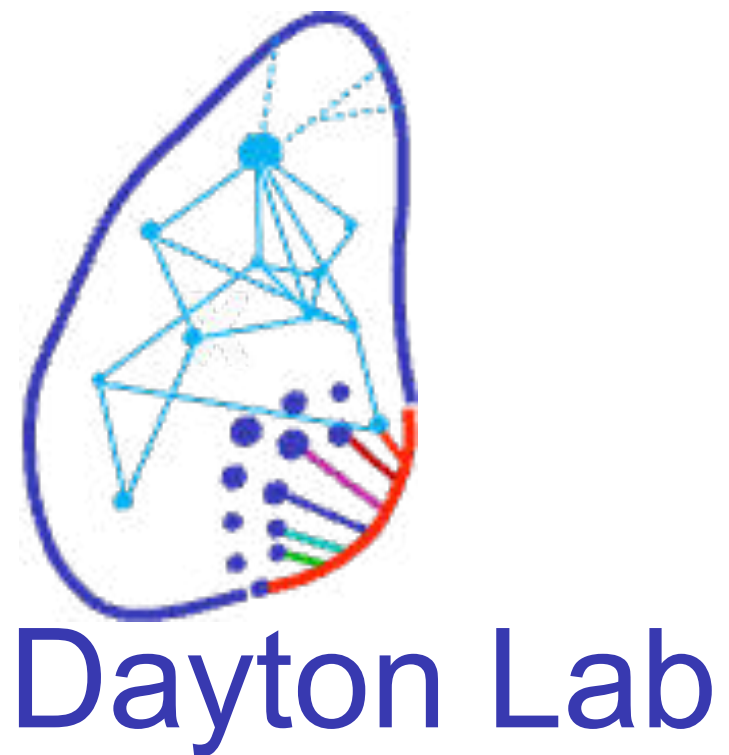
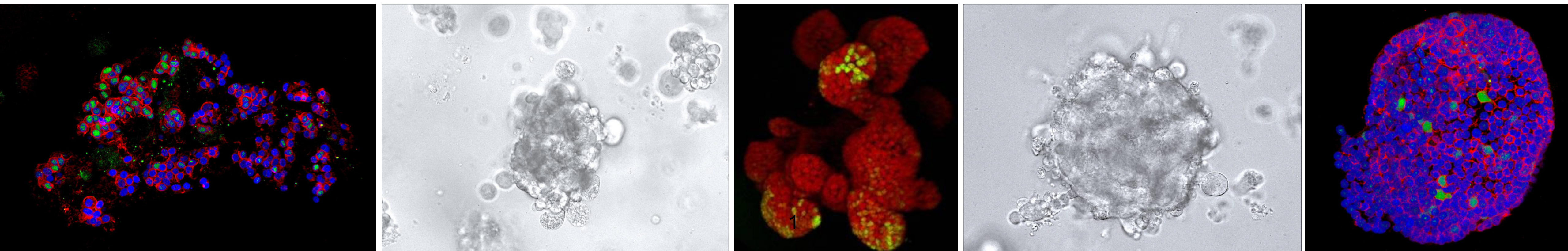


Modelling genomic profiles and cancer initiation in organoids

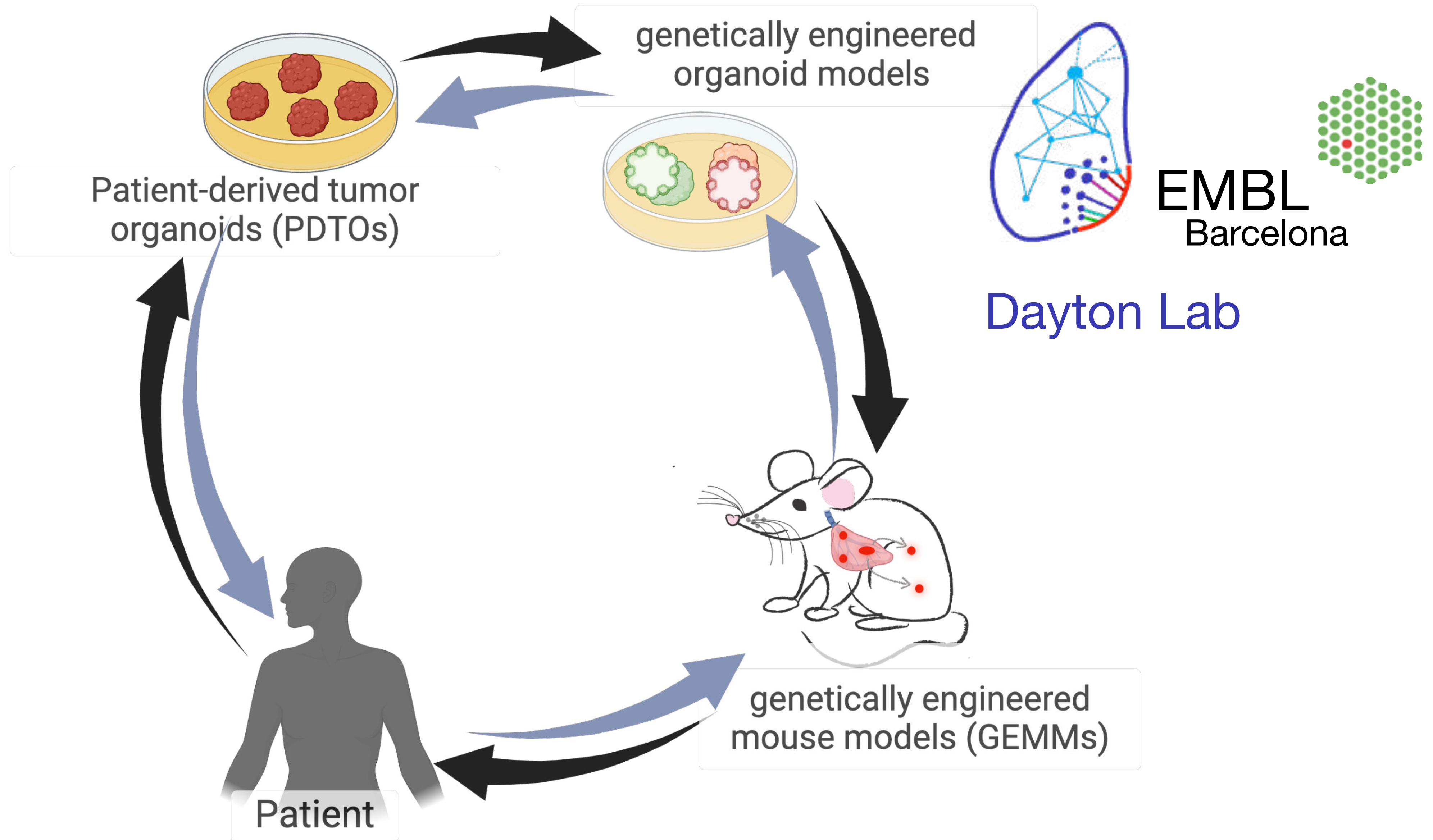


Talya Dayton, PhD
Group Leader | EMBL Barcelona

 @talyadayton.bsky.social



Crosstalk between model systems is needed in Cancer Research

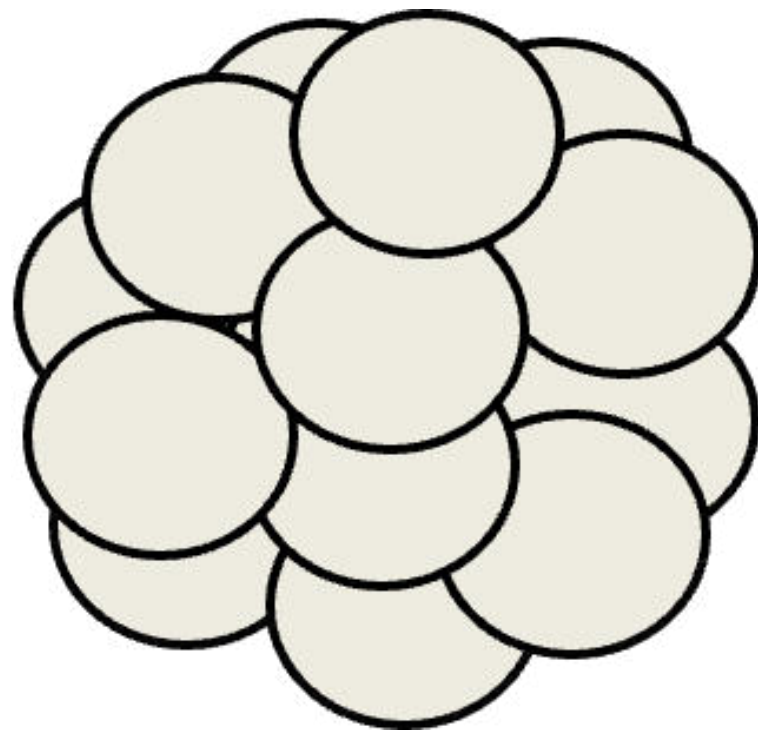


What are organoids?

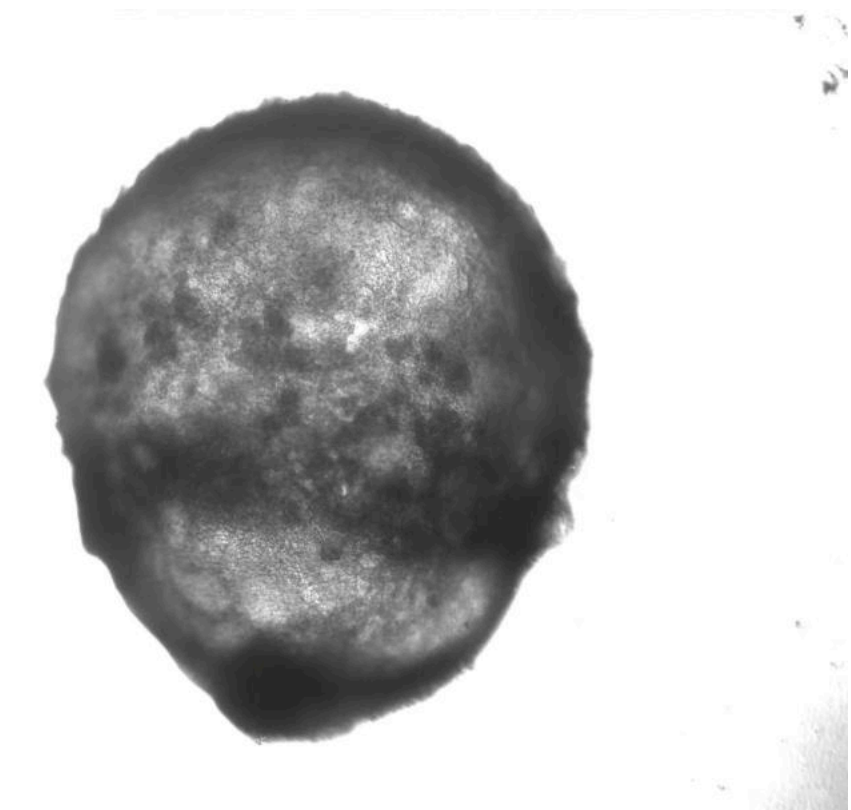
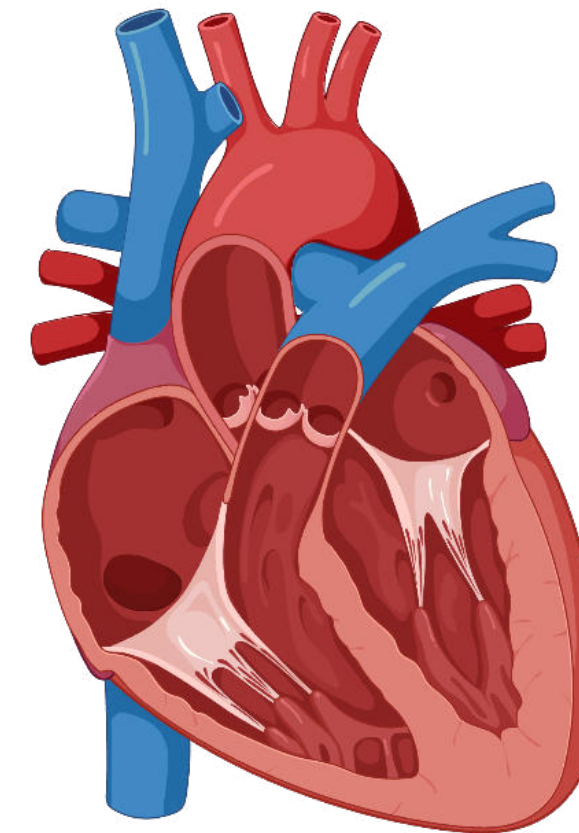
Organoids: Stem cell-derived *in vitro* 3D tissue cultures that mimic certain aspects of organ formation.

- Self organise to recapitulate spatial organization of the *in vivo* tissue
- Recapitulate tissue-specific features and function
- Contain multiple cell types

Aggregate of stem cells



Tissue like structures



Hofbauer et al, Cell (2021)

Organoids are **not** organs

Stem cells to organoids

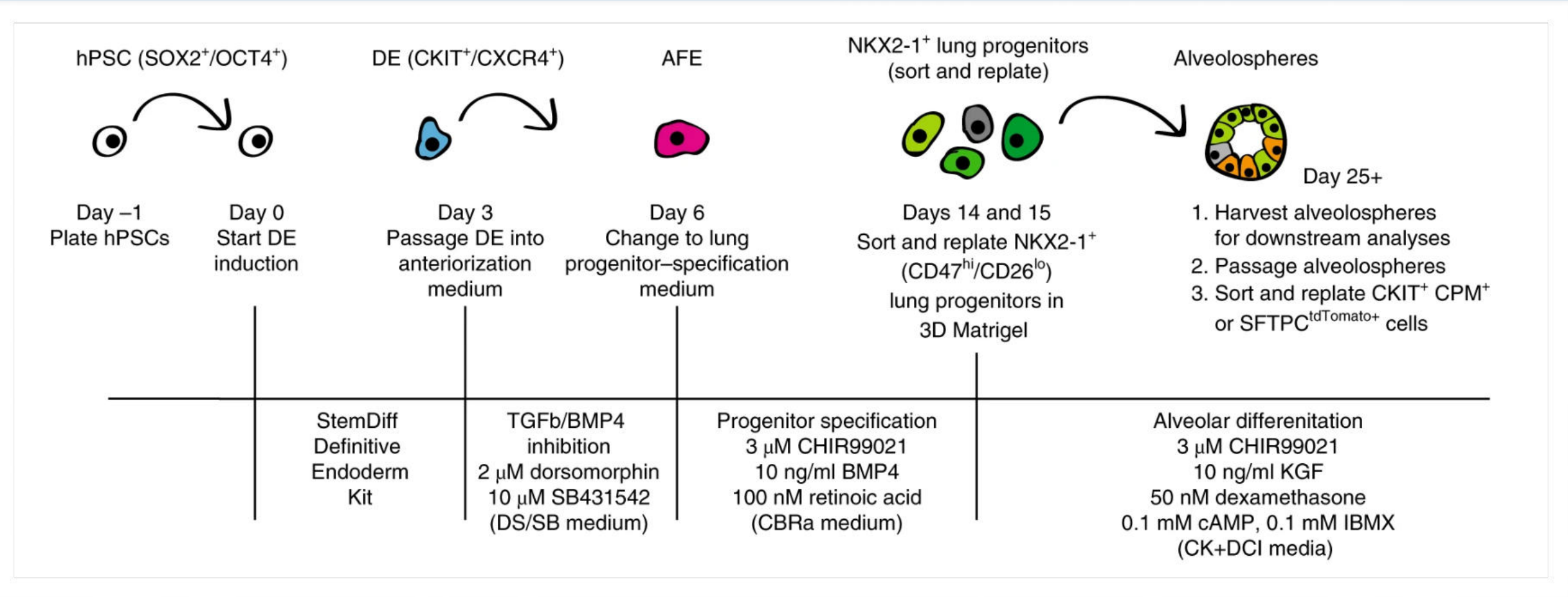
Stem cells: cells that can self-renew and differentiate into various cell types

Pluripotent stem cells:
Can become most cell types

Embryonic stem cells (ESCs): from embryo

Induced pluripotent stem cells (iPSCs):
from reprogrammed somatic cells

Organoid protocols recapitulate development



Tissue stem cells:

More restricted; can become most cell types of the corresponding tissue

Intestinal stem cells

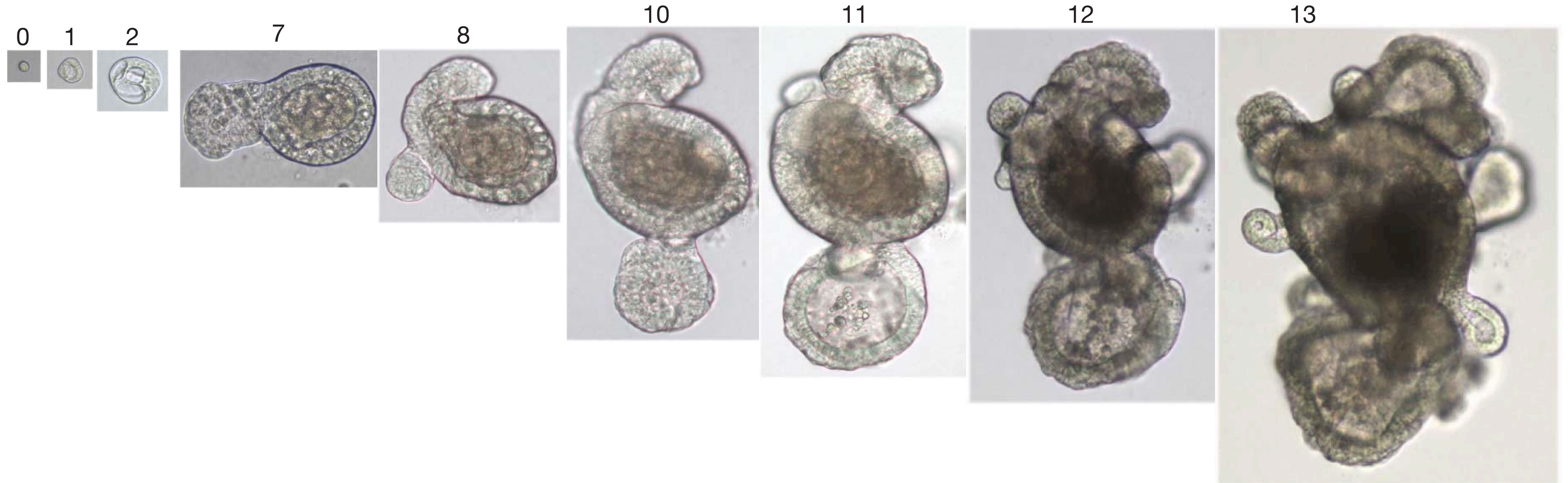
Pulmonary stem cells (airway, alveolar)

Hematopoietic stem cells

Neural stem cells

Organoid protocols recapitulate the *in vivo* stem cell niche and differentiation signals

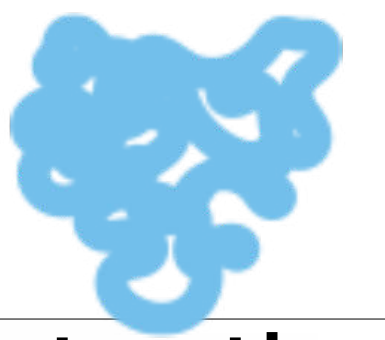
Tissue stem cell derived organoids



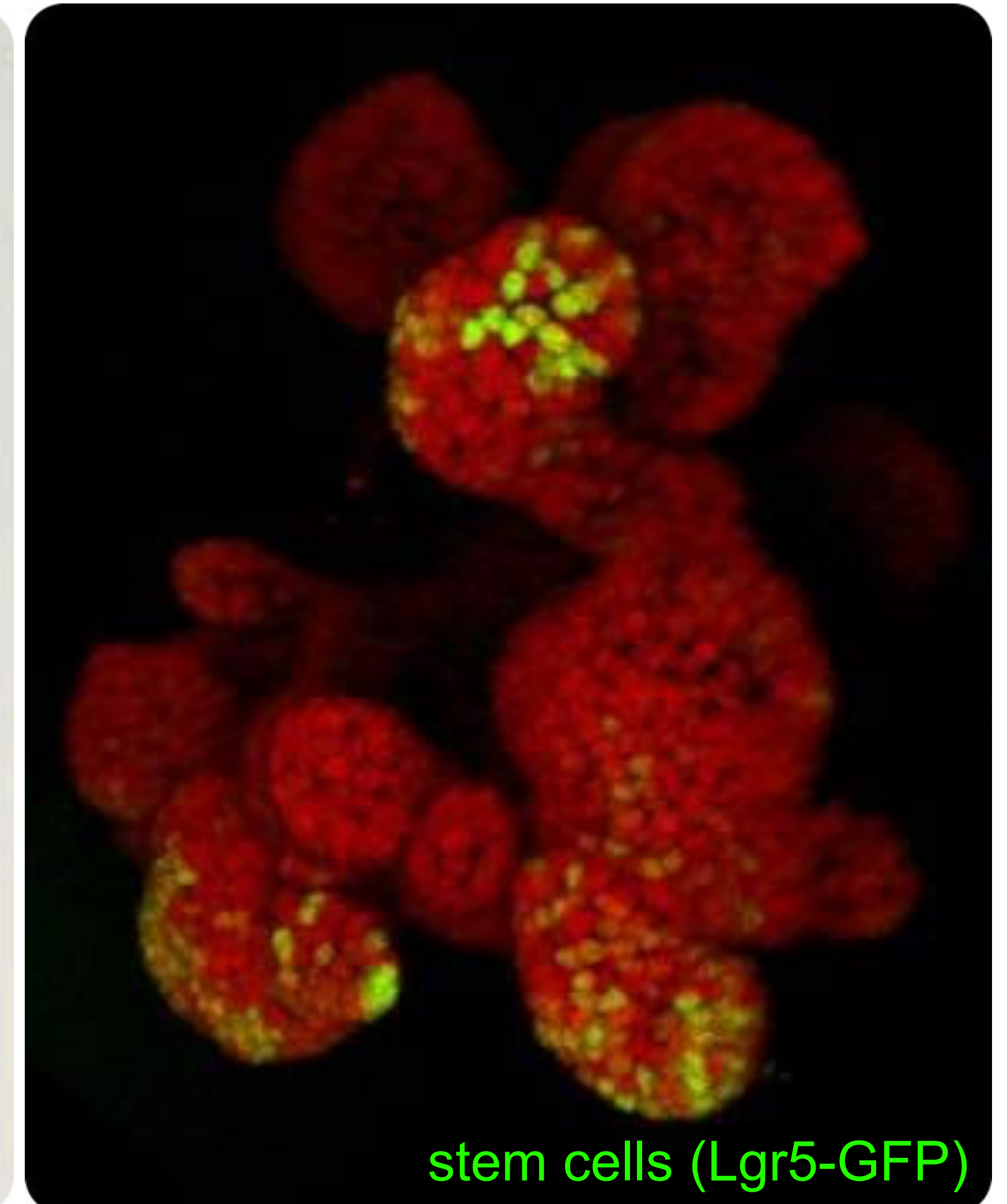
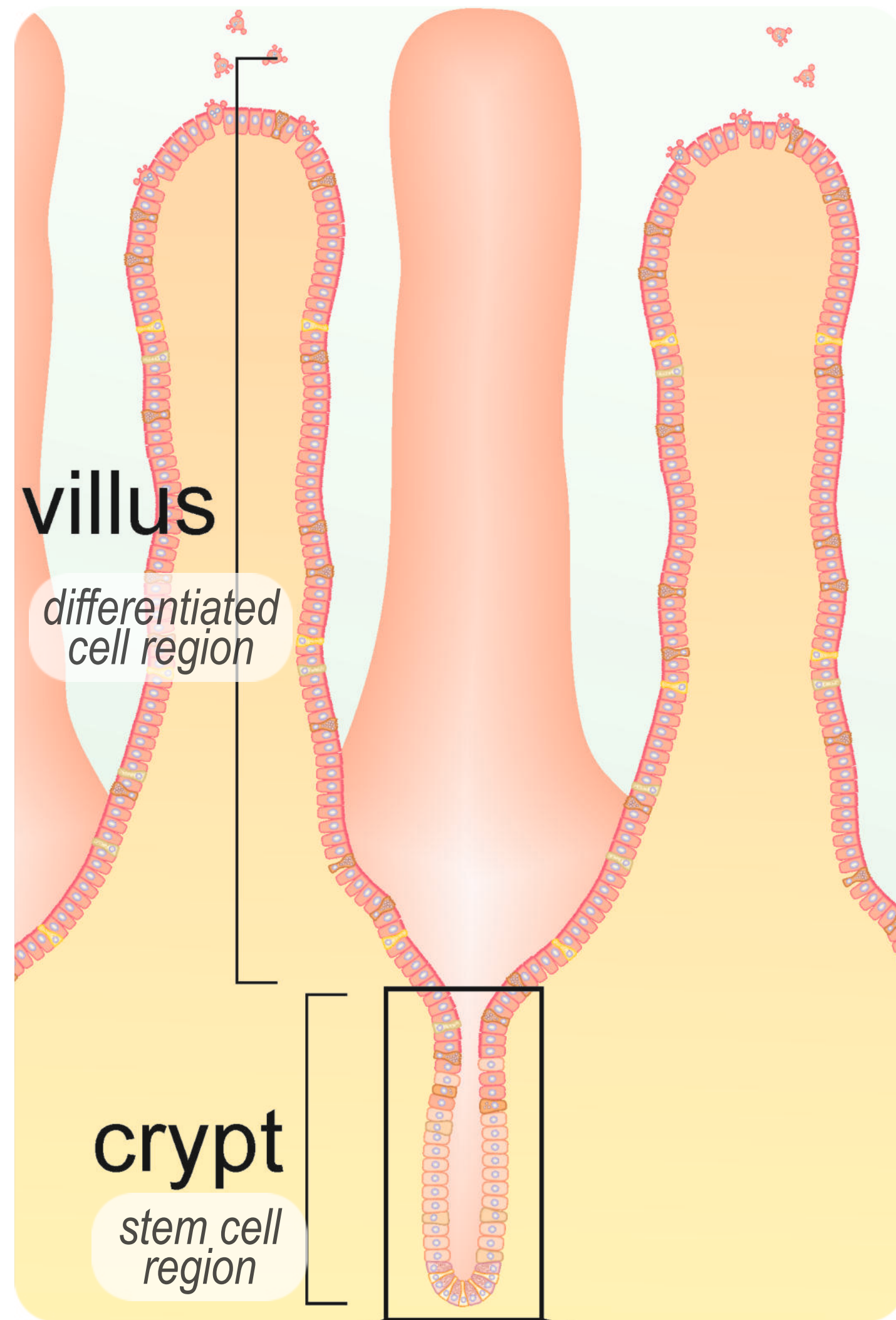
+EGF, Noggin, and Rspondin1 in matrigel

Sato et al, 2009

Tissue stem cell derived organoids

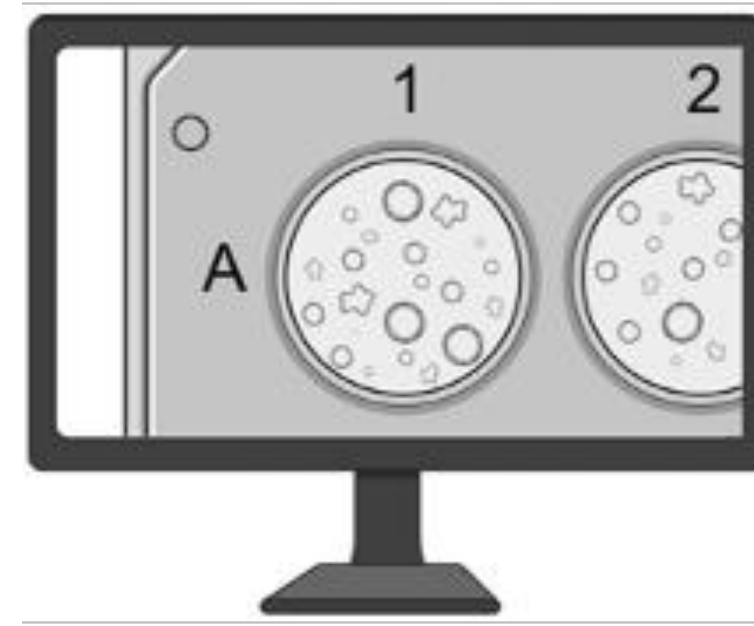


Intestine



Live imaging of airway organoid growth dynamics & morphologies

Incucyte - Sartorius



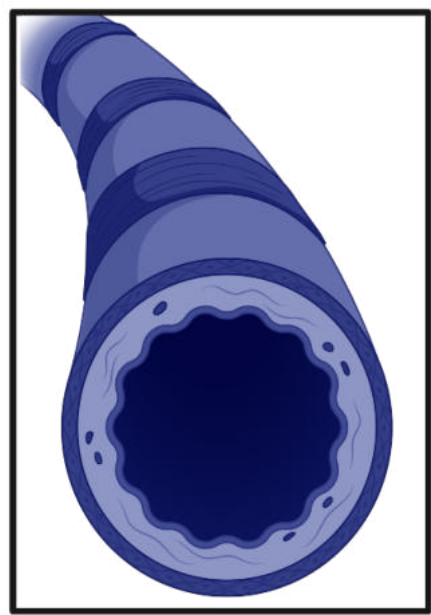
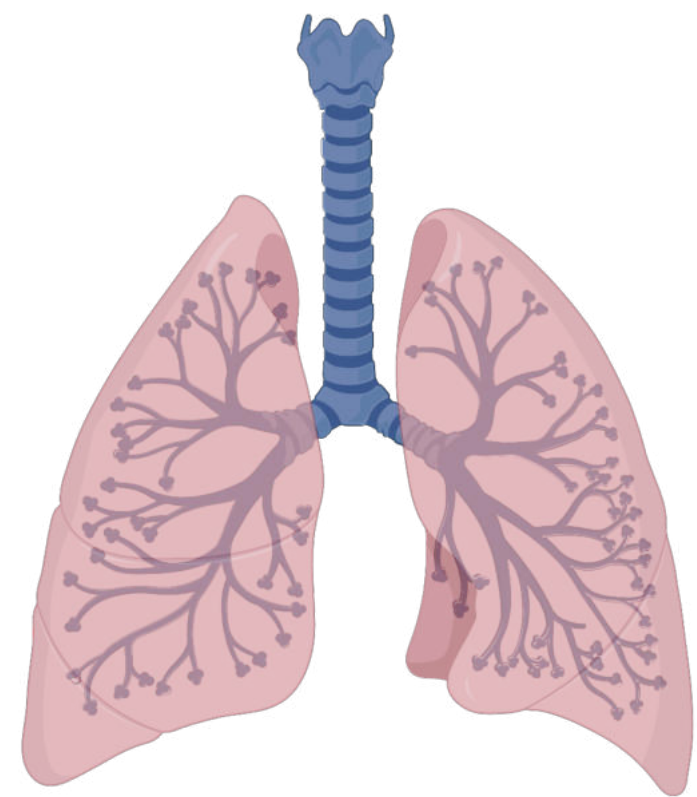
Ilastik



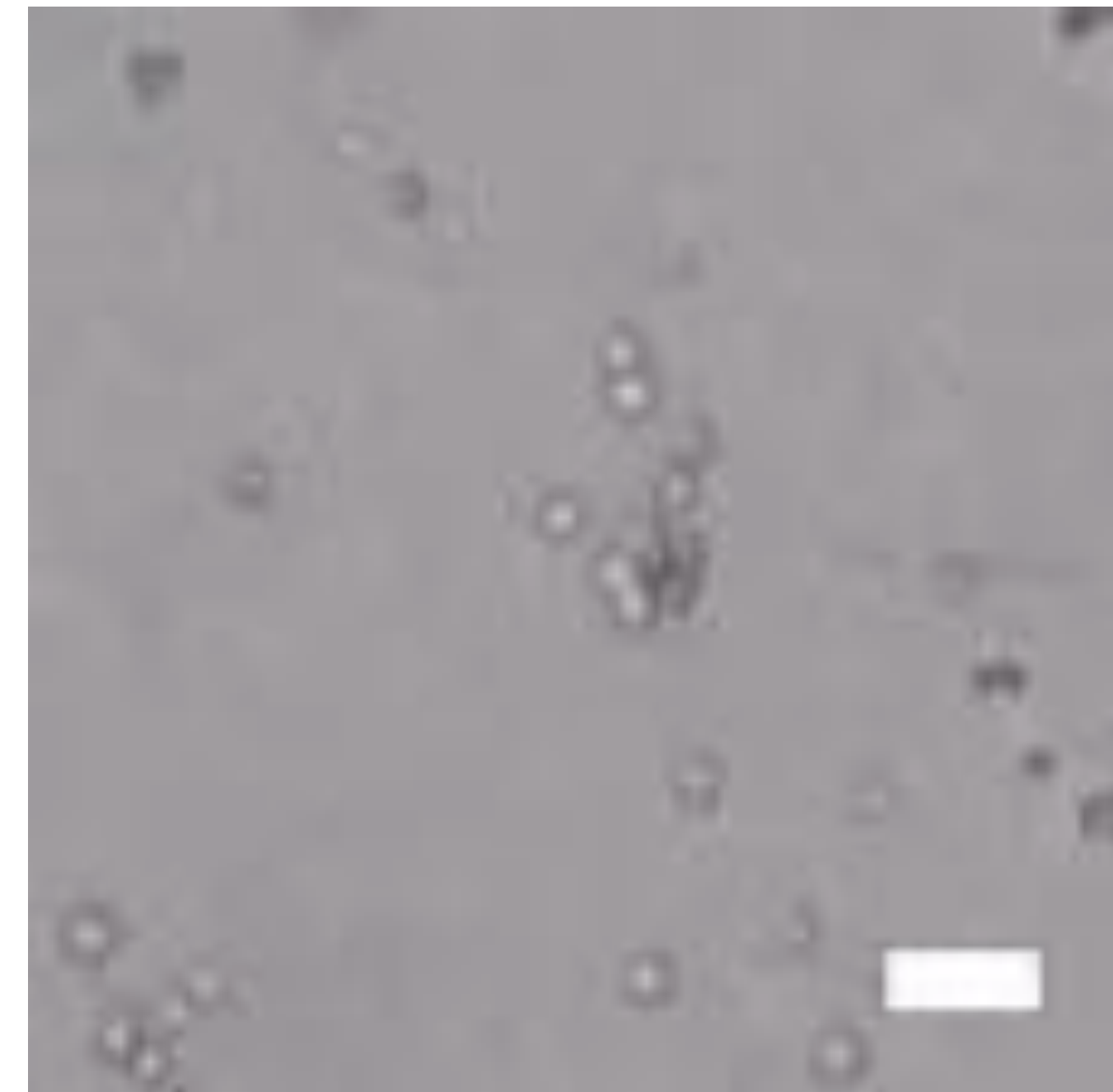
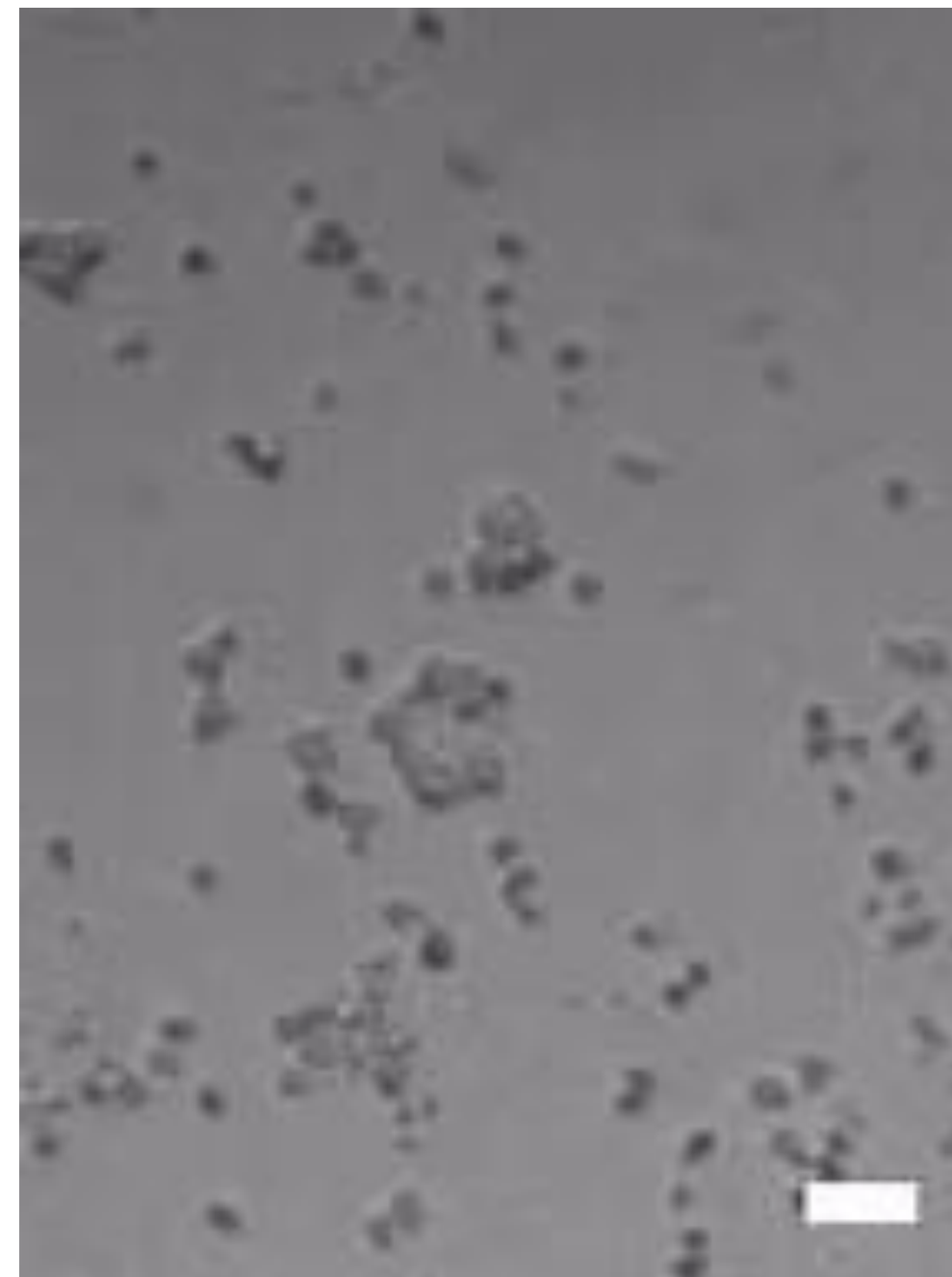
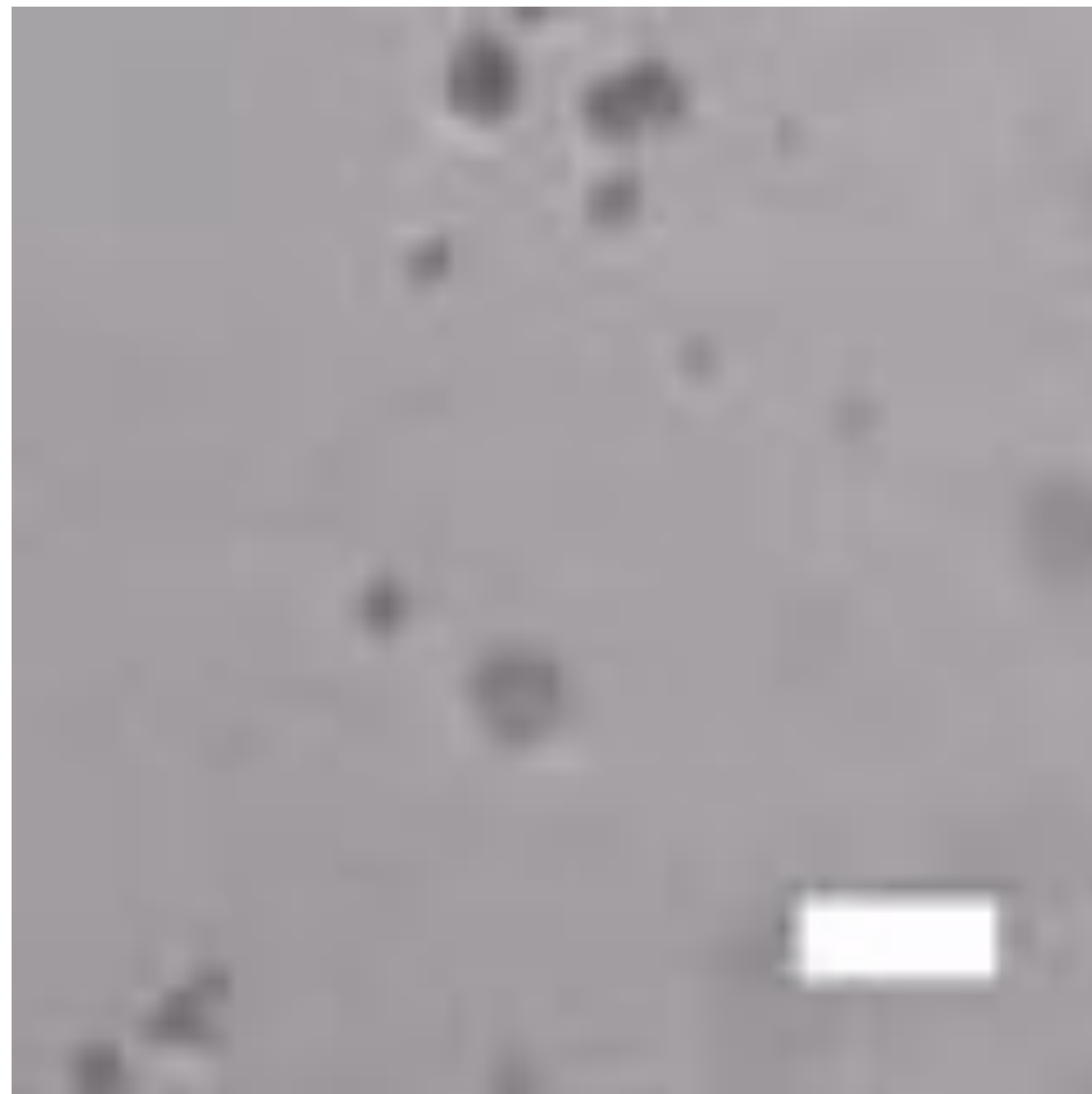
Andrés Marco

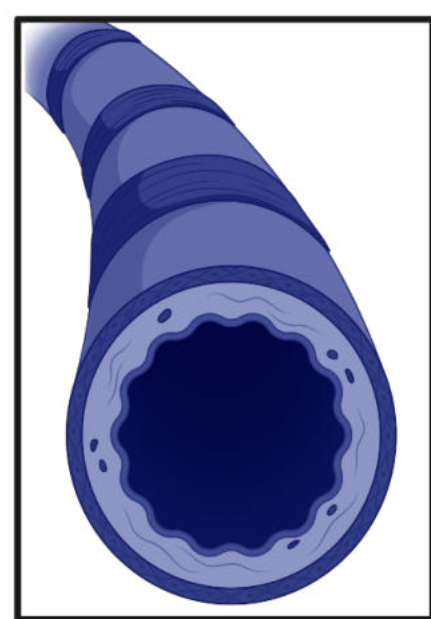
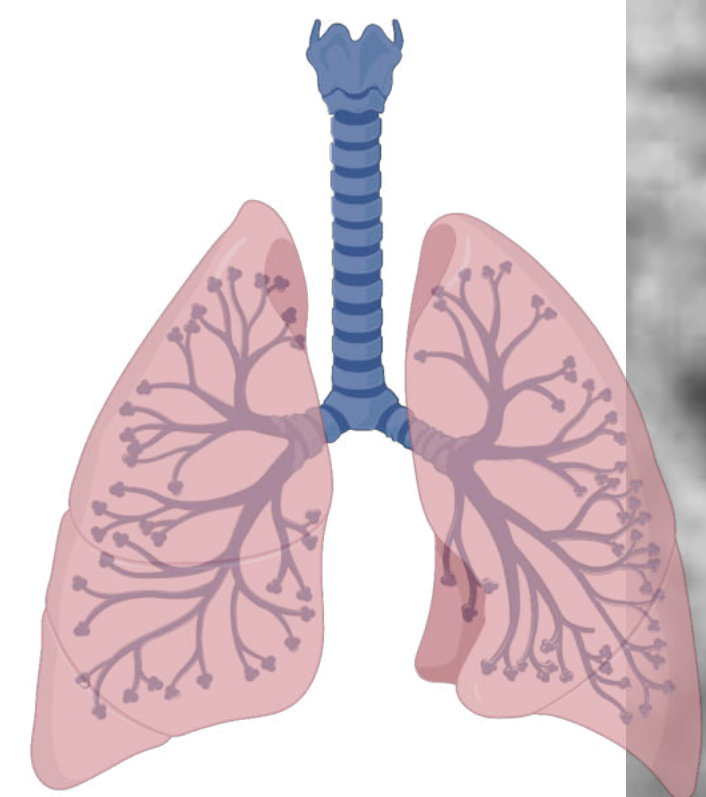


Andrea García

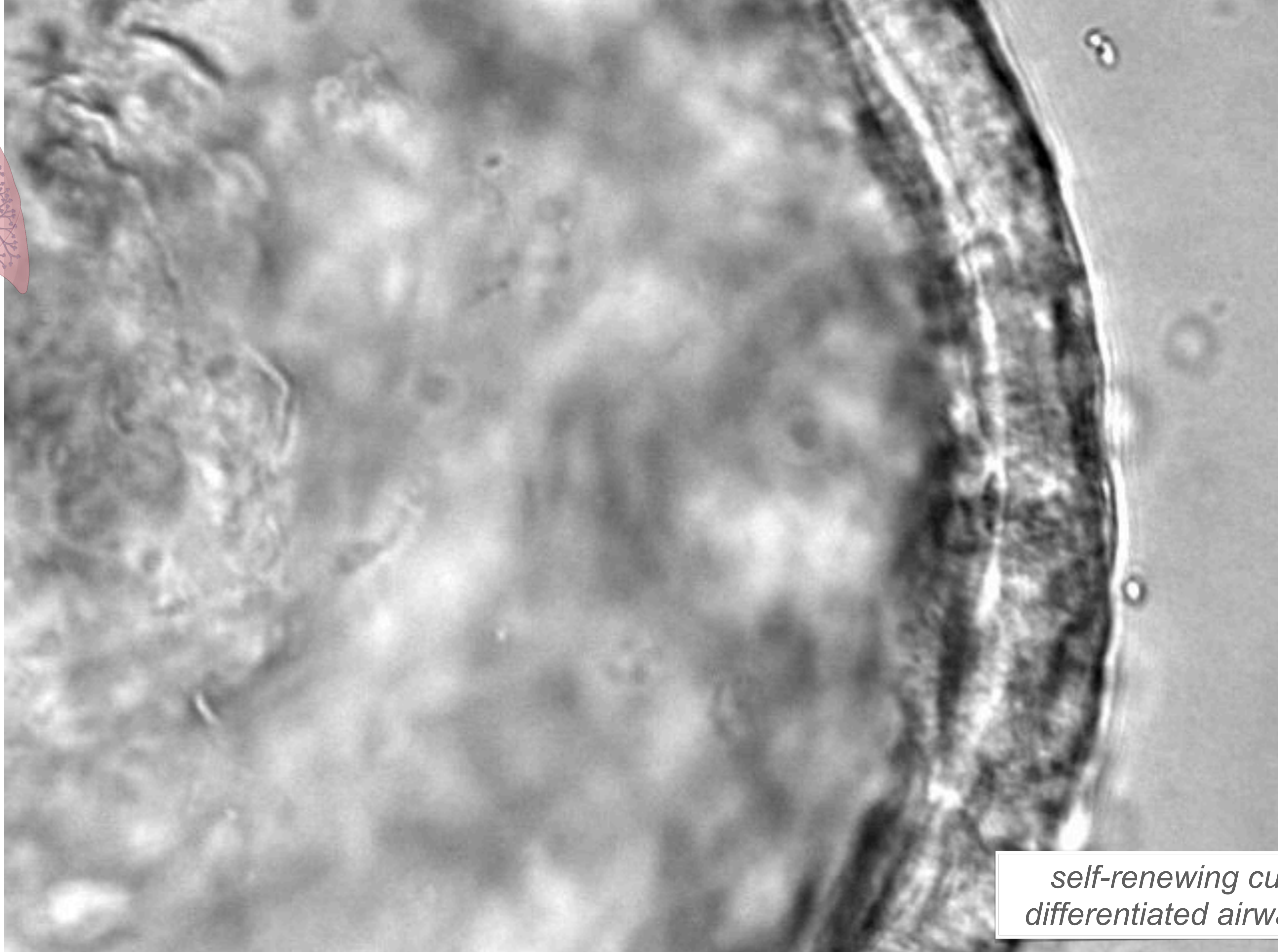


Airway





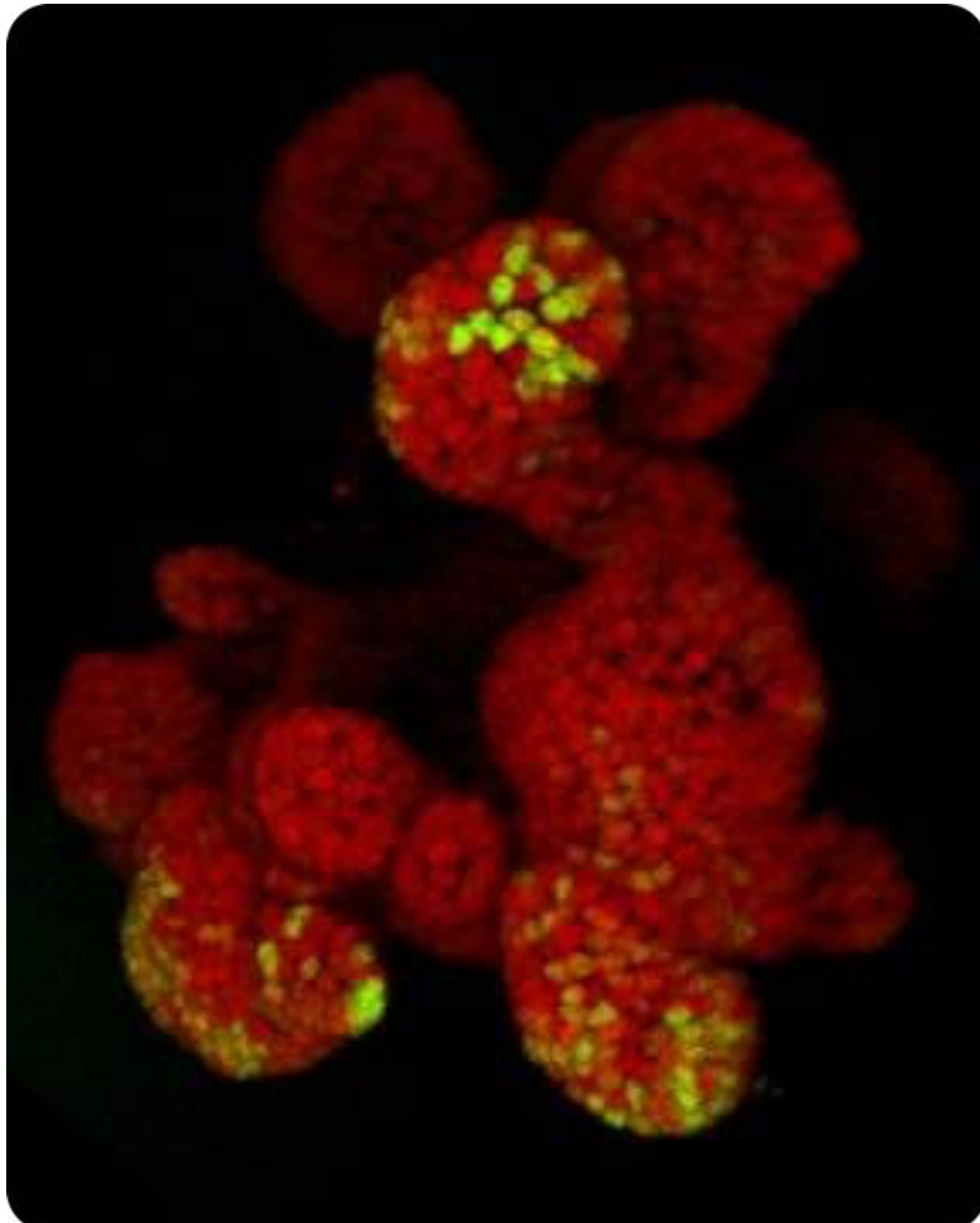
Airway



HEALTH

*self-renewing cultures with
differentiated airway cell types*

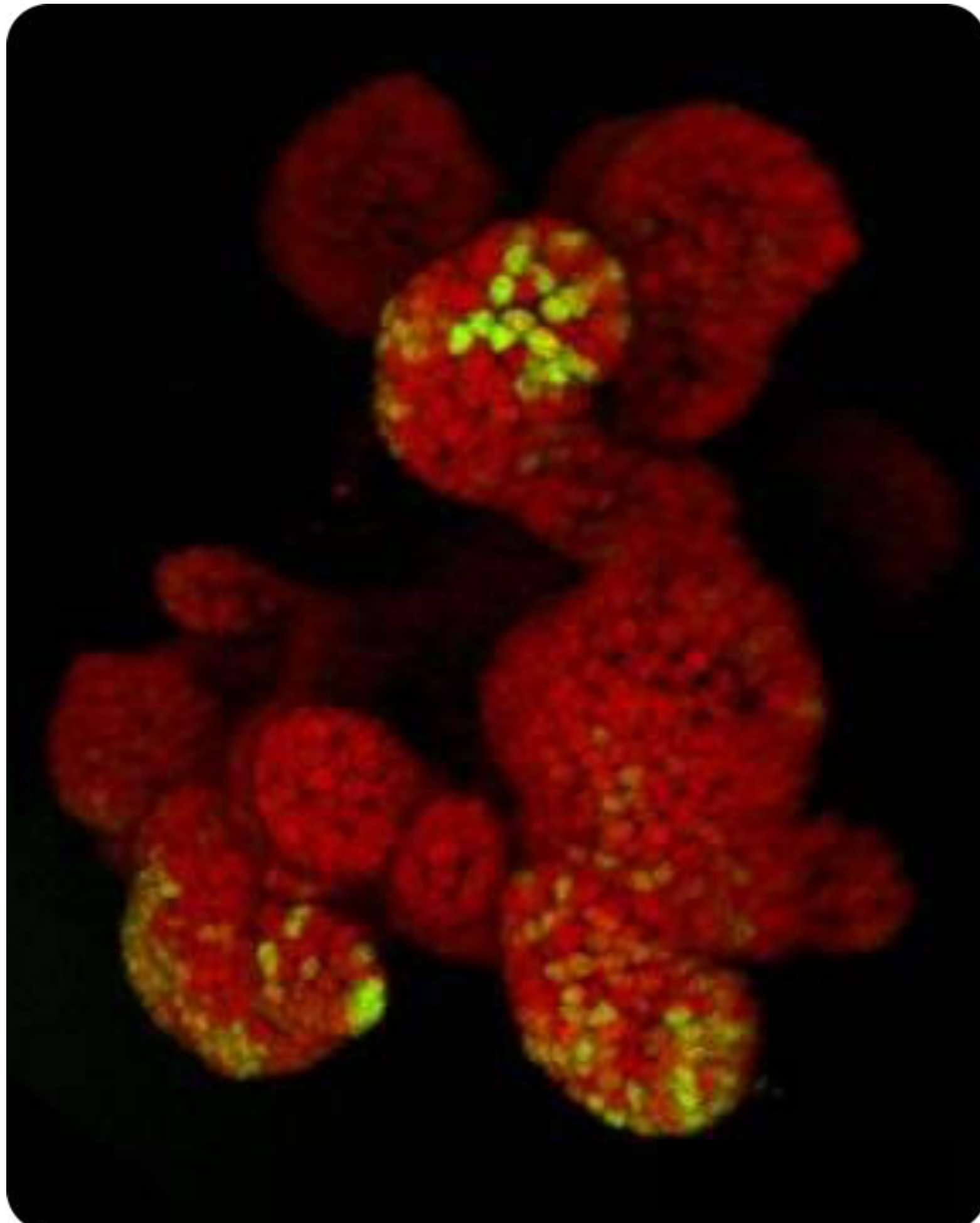
Tissue stem cell derived organoids



Advantages of organoids:

- Self-renewing, longterm cultures
- Defined *in vitro* system
- Can be grown from both healthy and diseased tissues
- Media mimics the *in vivo* niche and is easily modified
- Recapitulate stem cell differentiation dynamics

Tissue stem cell derived organoids

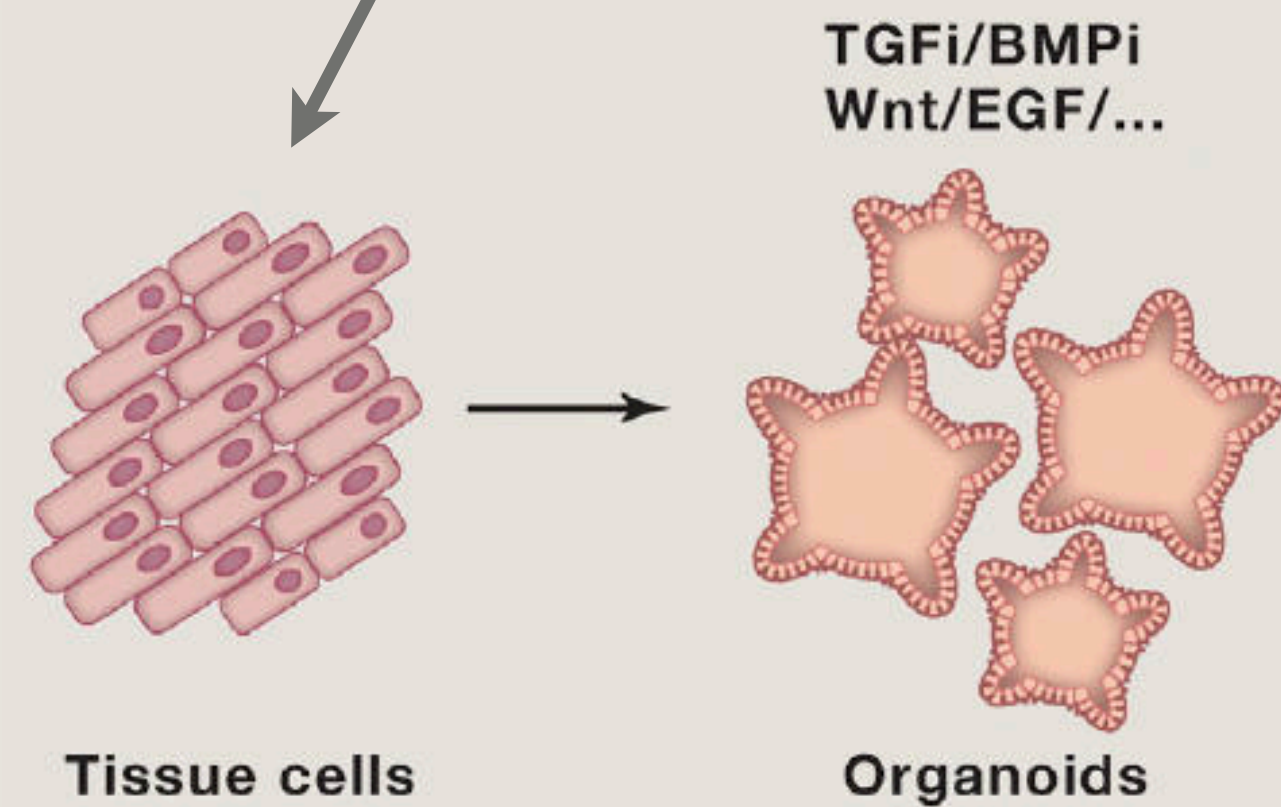
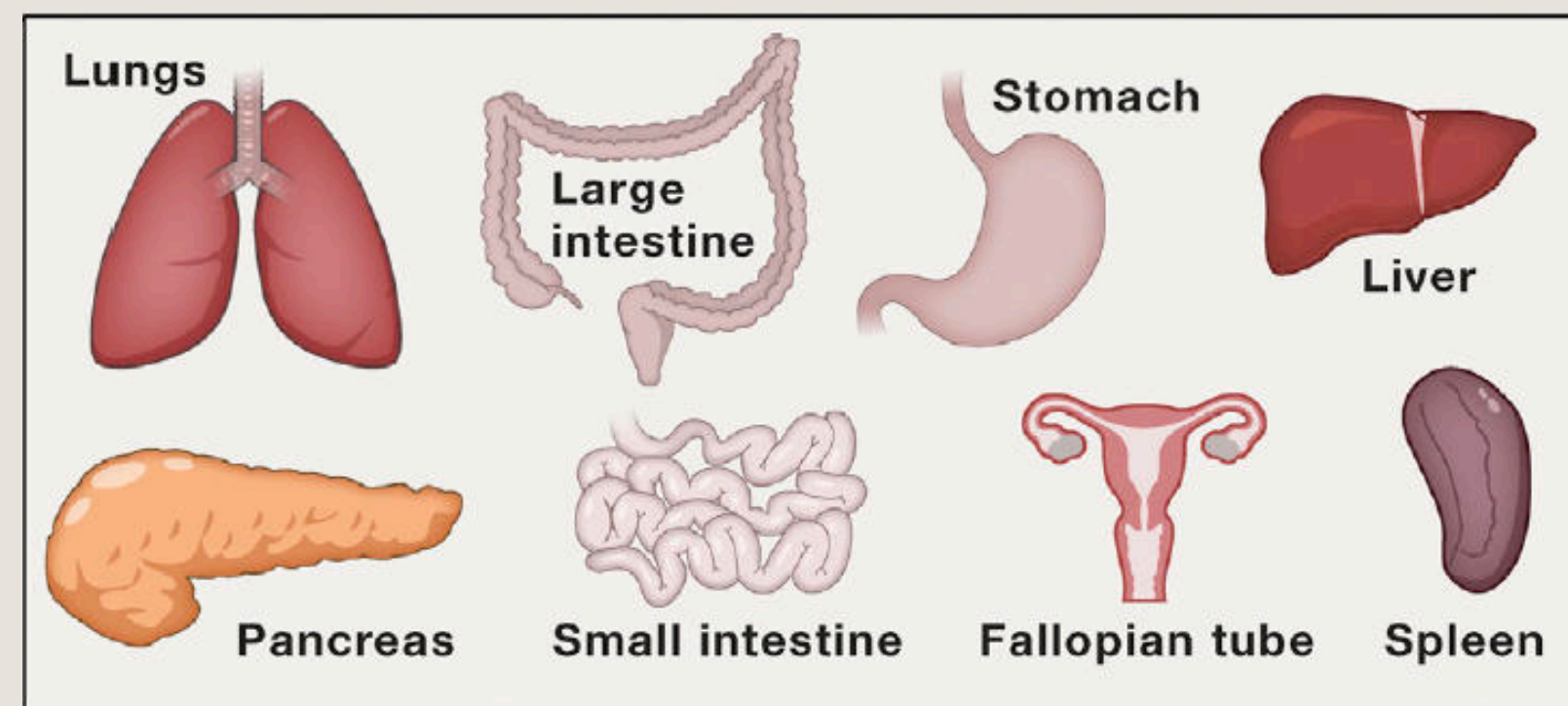


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- Recapitulate stem cell differentiation dynamics

Tissue stem cell derived organoids: an adaptable system

Growth: Organoids can be grown from multiple tissue types



modified from Clevers (2016) Cell

Differentiation: Media modifications to enrich for specific cell types

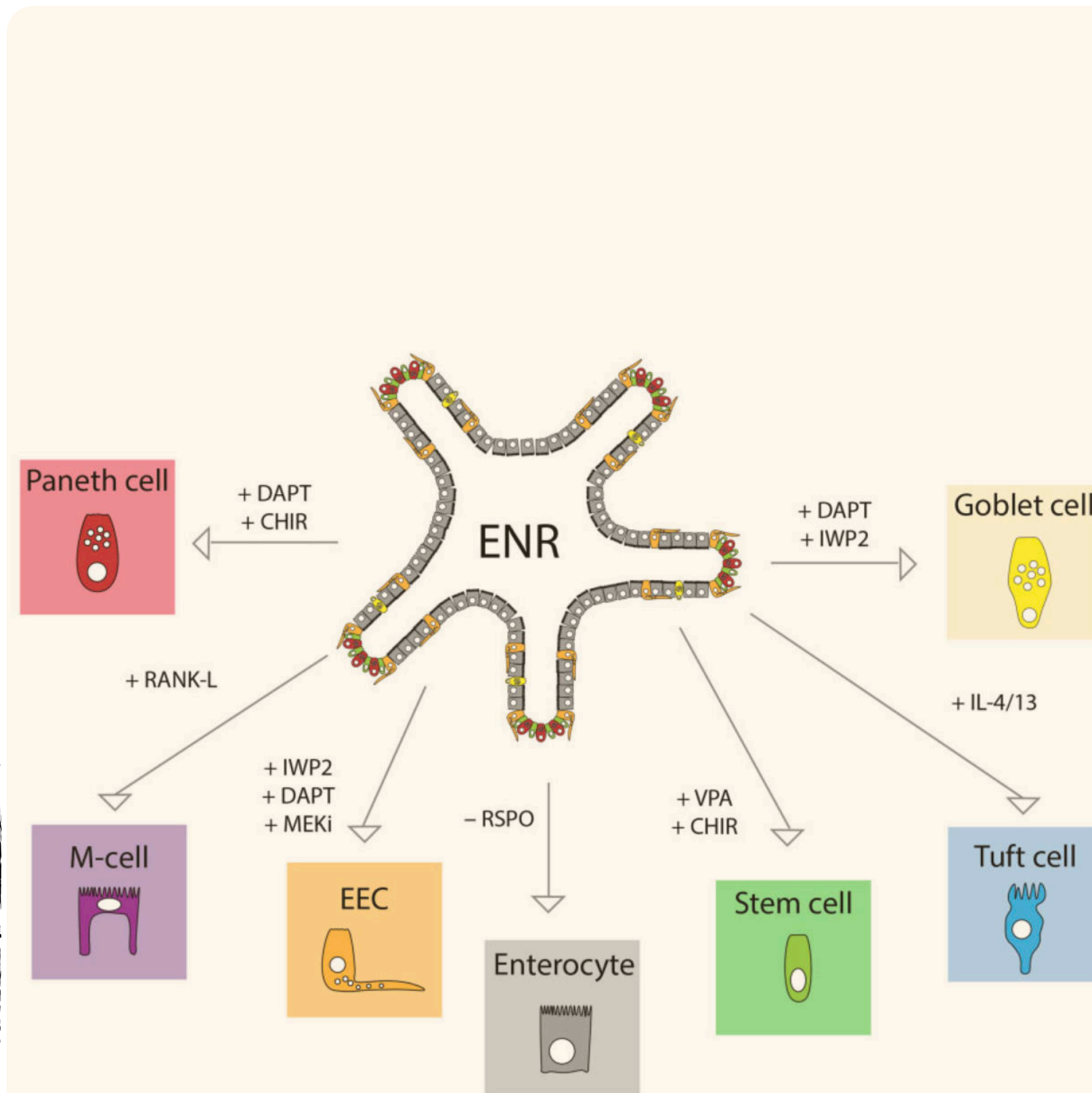
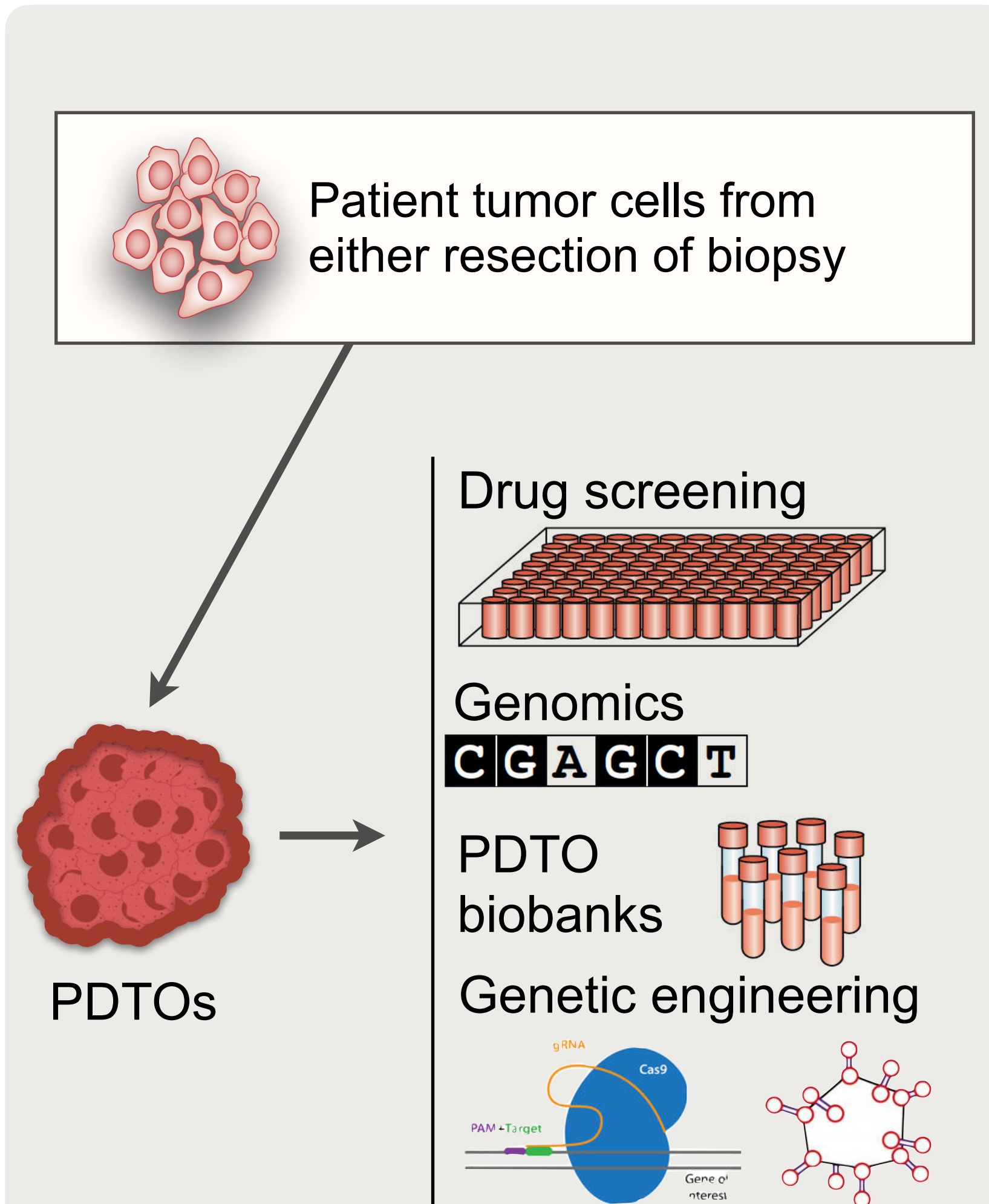


Image from Boonekamp, Dayton, & Clevers (2020)
J Mol Cell Biol.

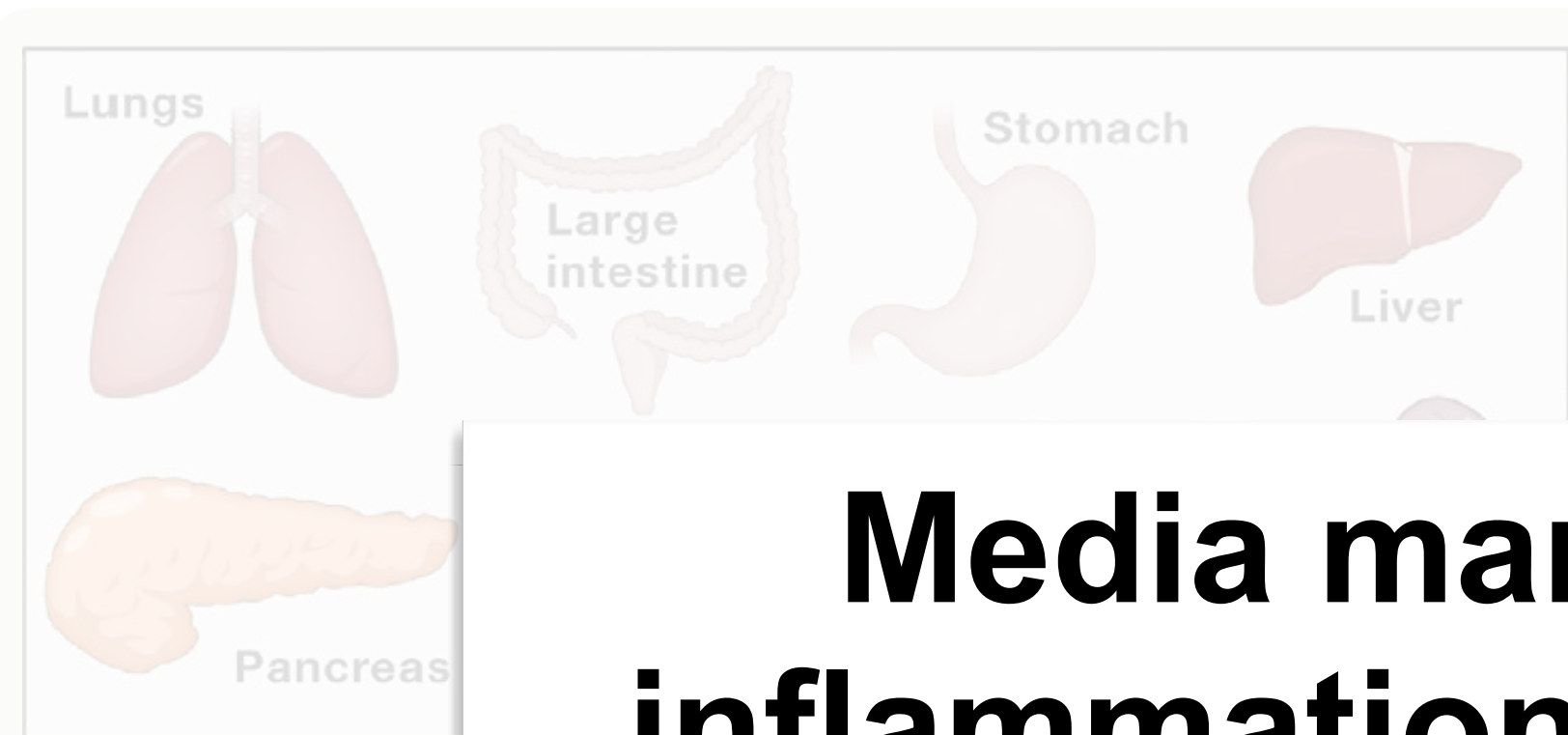
Transformation: Patient derived tumor organoids (PDTOs) can be grown from tumors



modified from Drost & Clevers (2018) Nat Rev Cancer

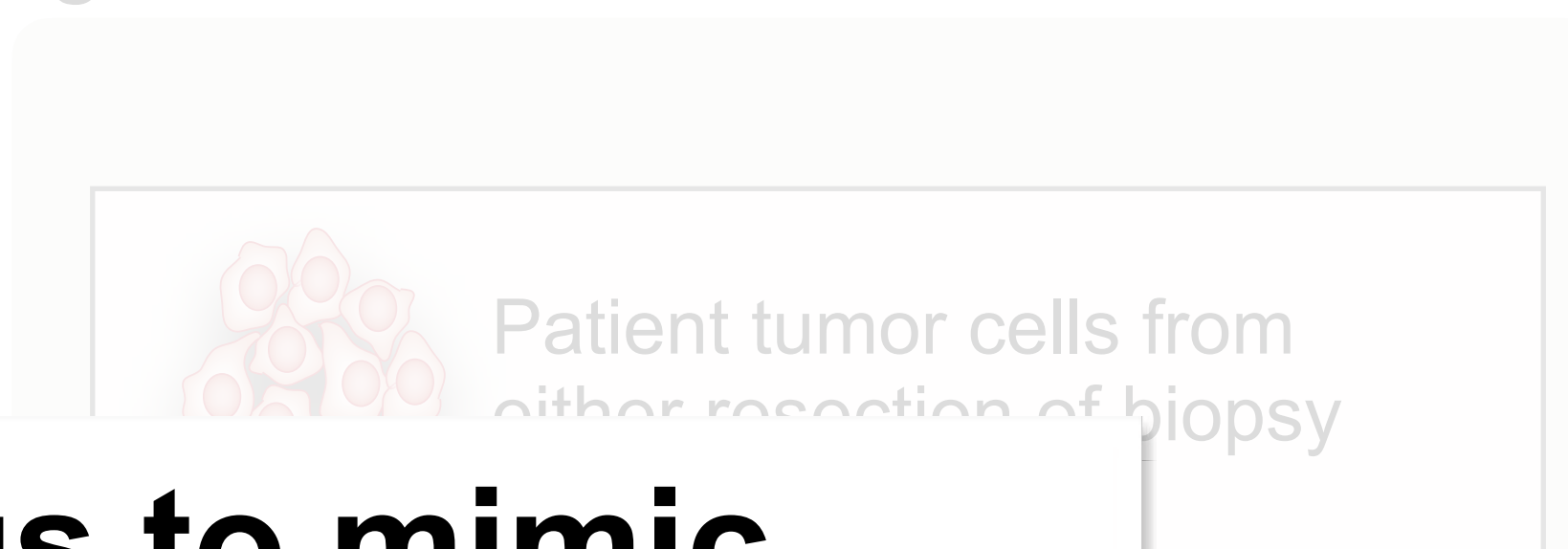
Tissue stem cell derived organoids: an adaptable system

***Growth:** Organoids can be grown from multiple tissue types*

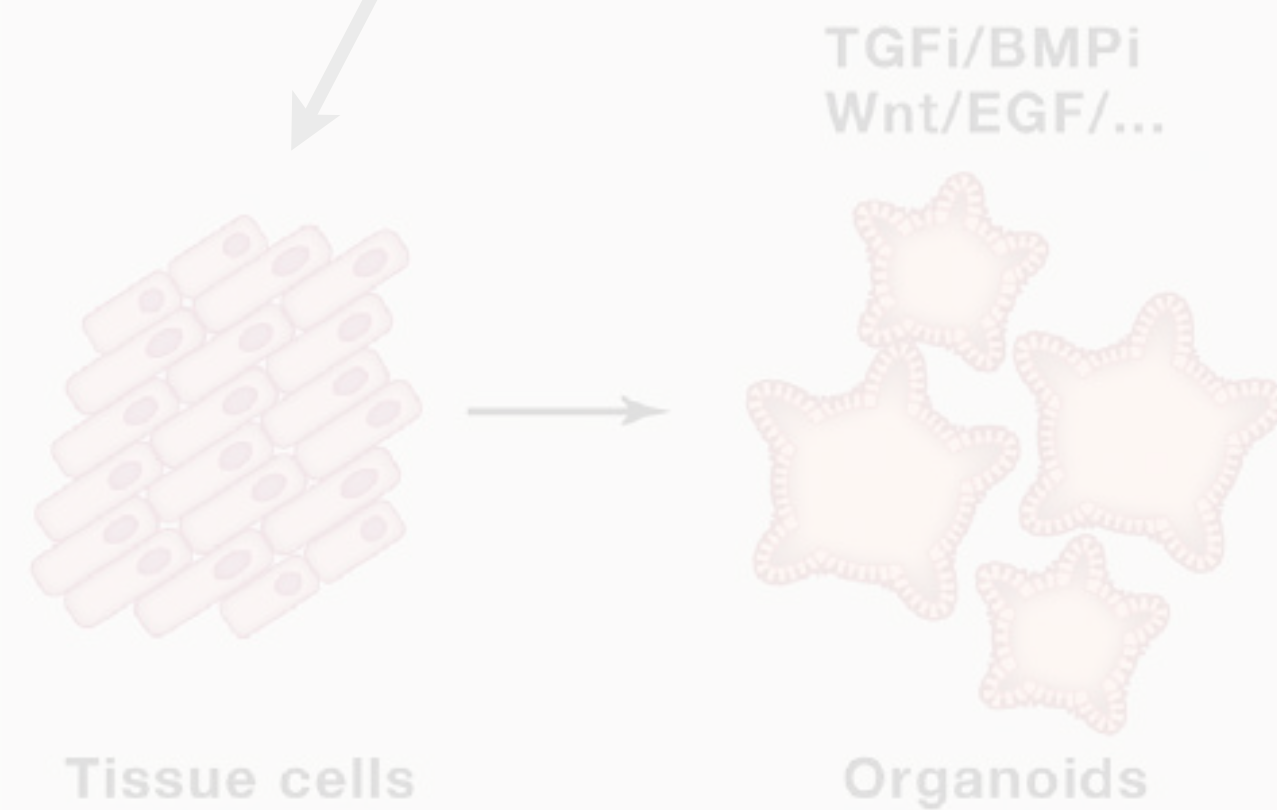


***Differentiation:** Media modifications to enrich for specific cell types*

***Transformation:** Patient derived tumor organoids (PDTOs) can be grown from tumors*



Media manipulations also allow us to mimic inflammation & some environmental exposures...



modified from Clevers (2016) Cell

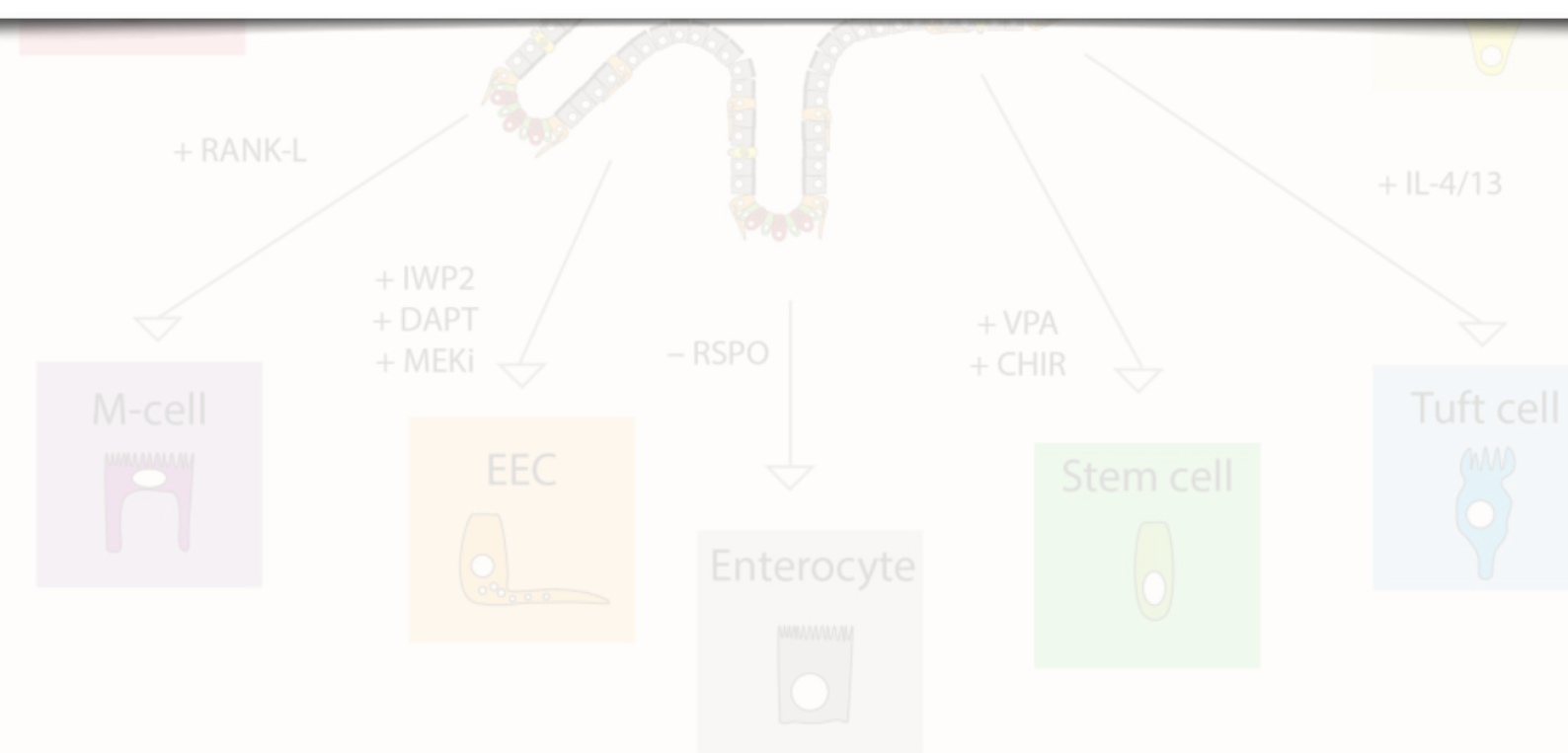
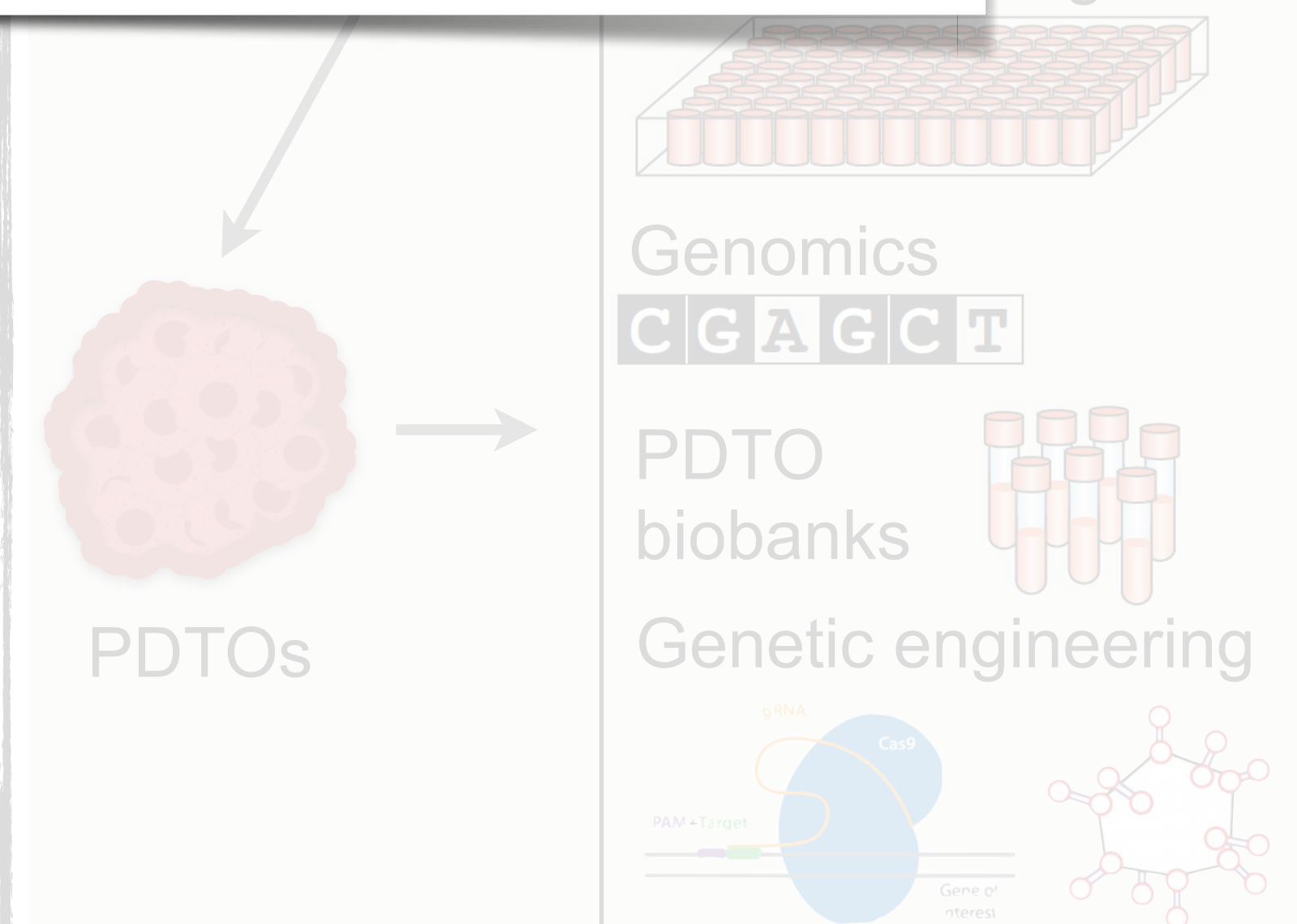
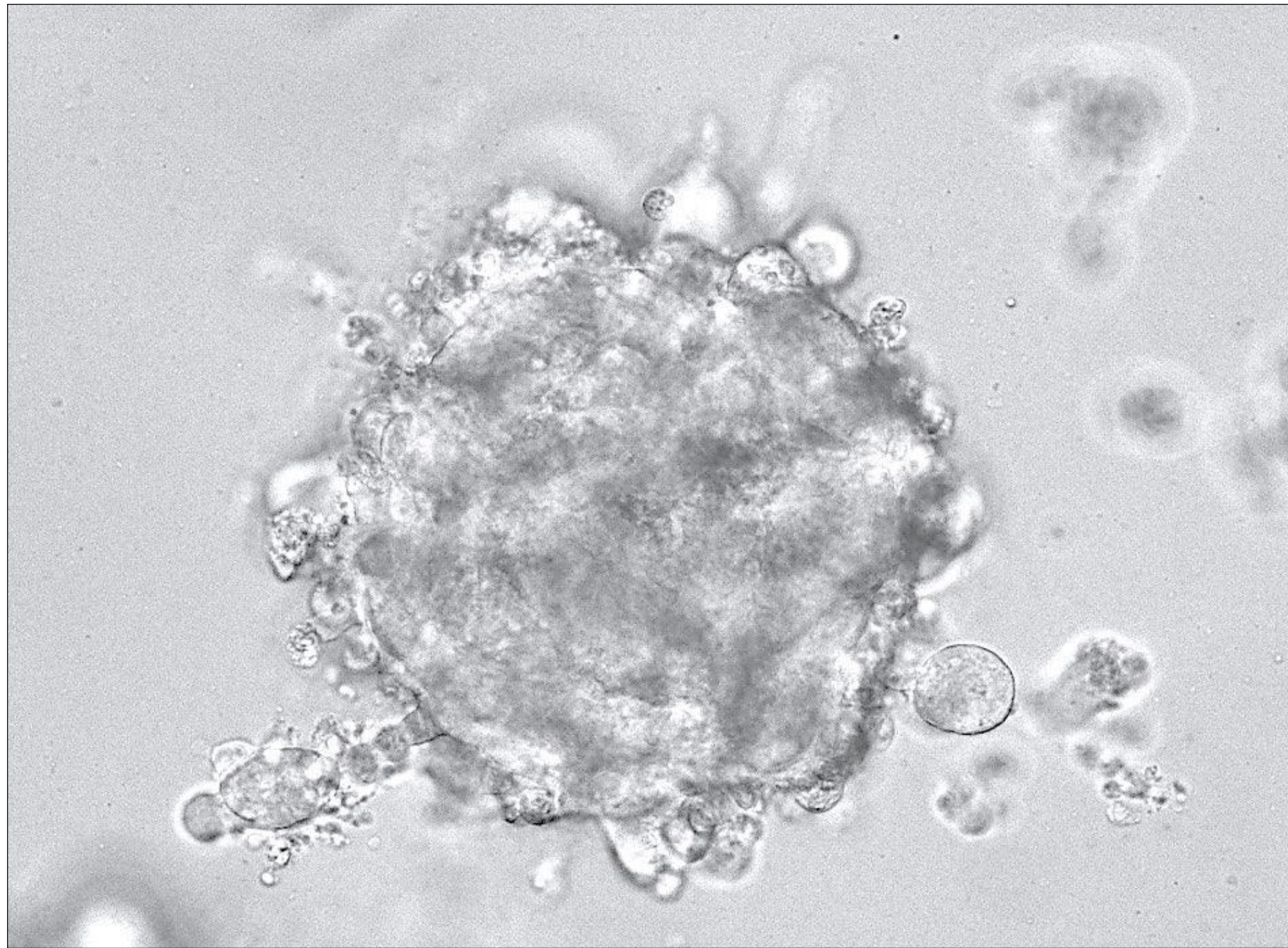


Image from Boonekamp, Dayton, & Clevers (2020) J Mol Cell Biol.



modified from Drost & Clevers (2018) Nat Rev Cancer

PDTOs: a preclinical model system

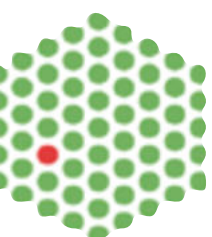


Consist entirely/primarily of tumors cells

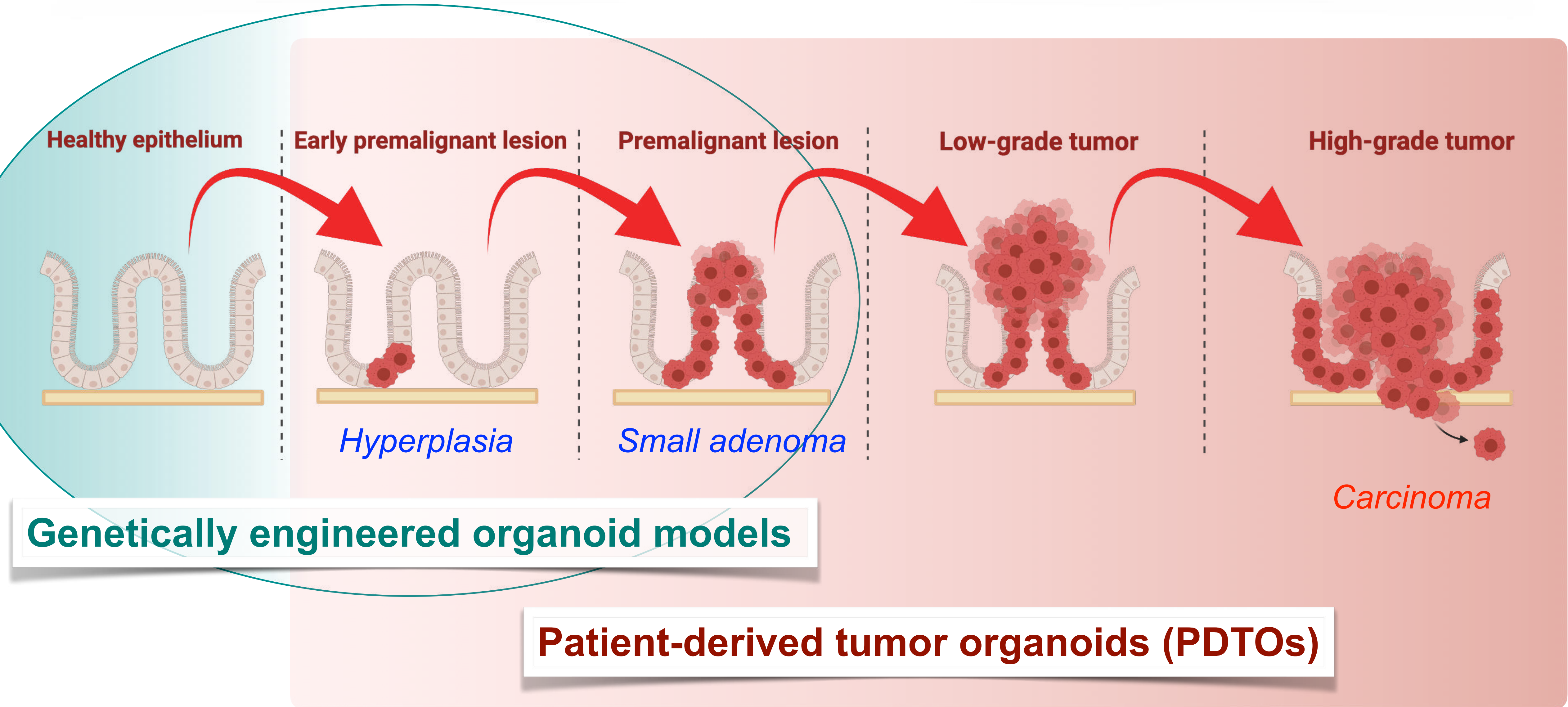
Advantages of Patient Derived Tumor Organoids (PDTOs):

- Can be expanded *in vitro* **indefinitely** - **unlimited supply**
- Can be frozen for long term storage and subsequently thawed for outgrowth
- Allow for drug sensitivity testing
- Can be genetically modified
- Culture conditions mimic the native tumor microenvironment** = **enables the growth of low grade tumors**

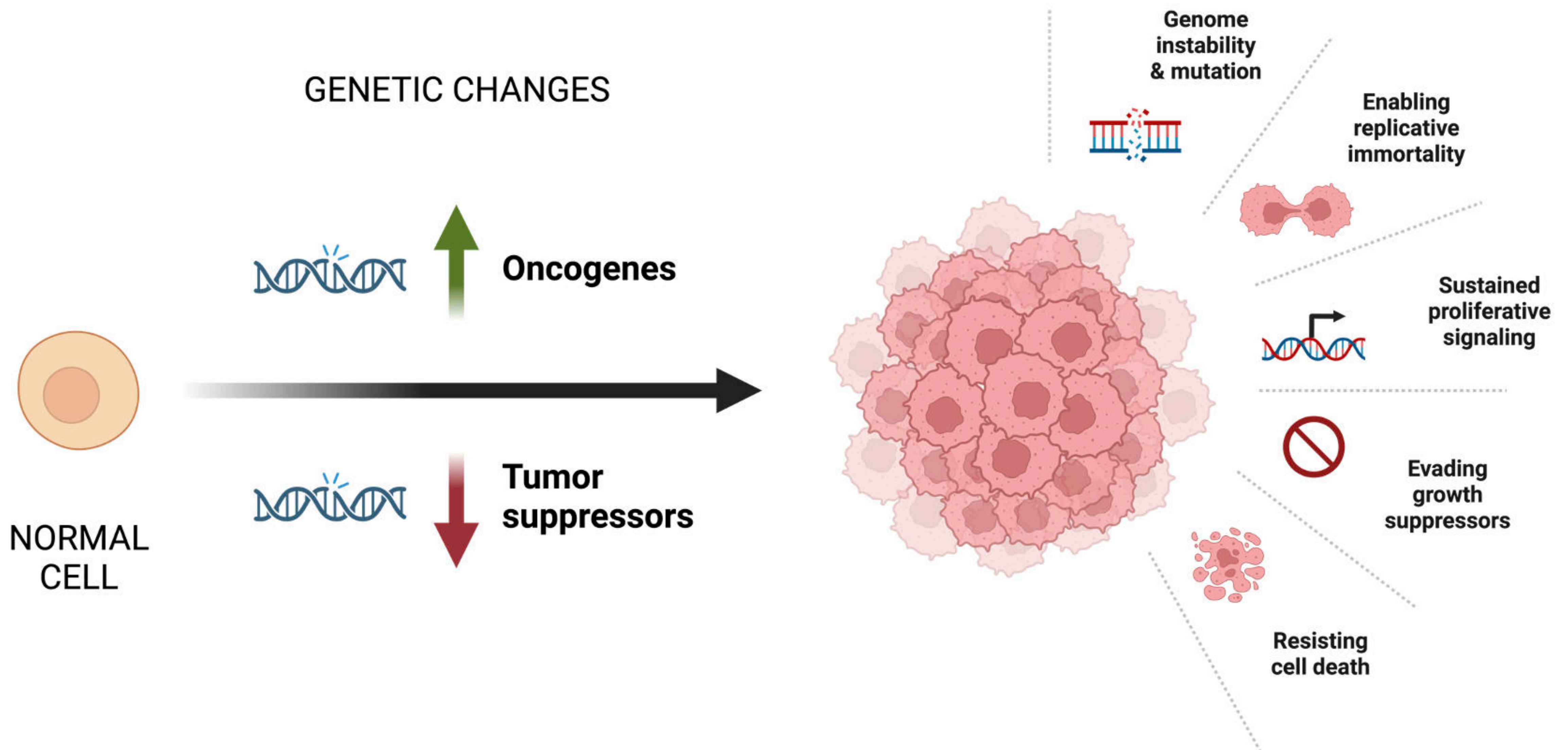
****Media is serum-free**



Human organoids: *a platform to study the full spectrum of cancer*



Cancer is a genetic disease

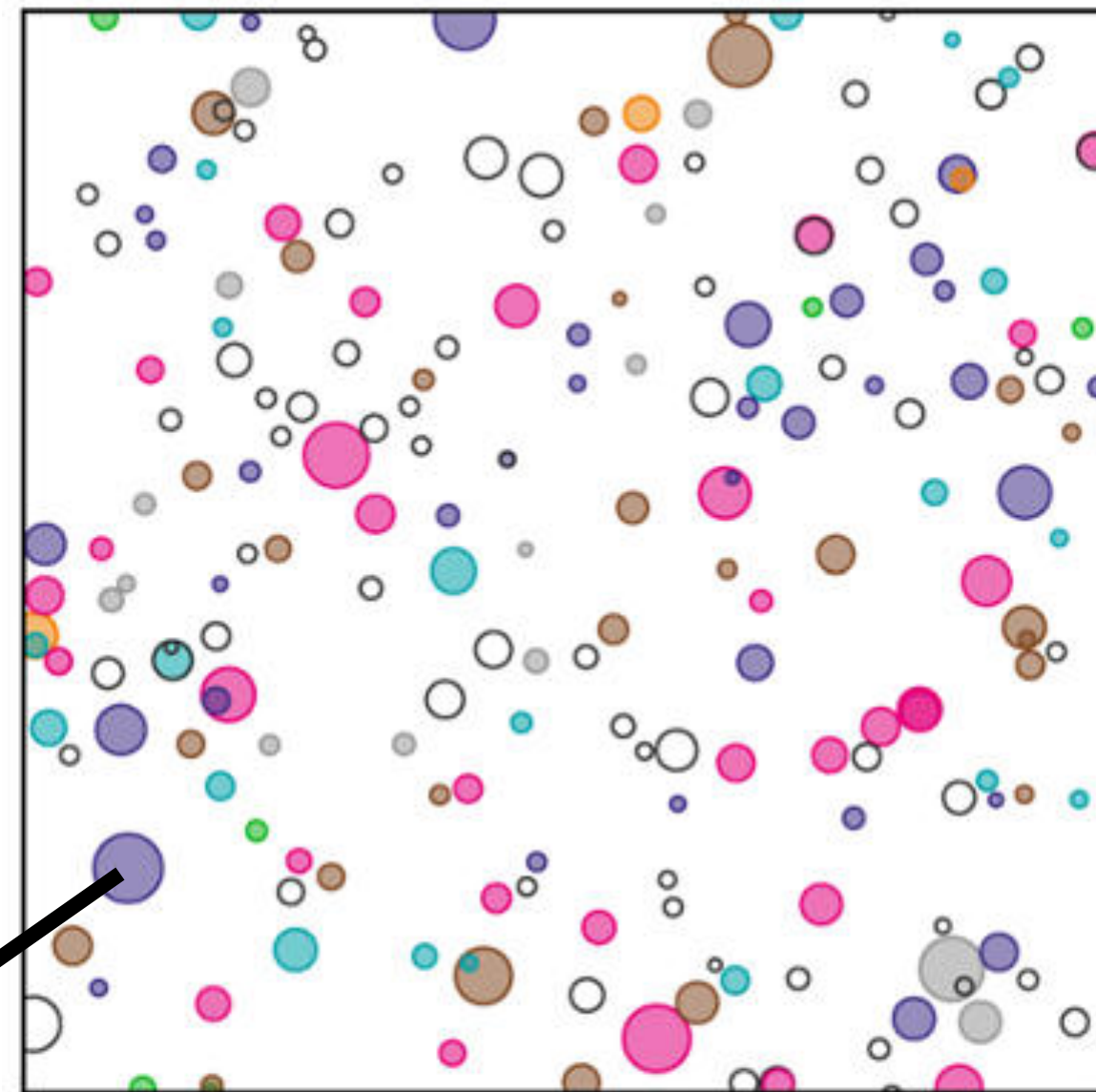


But mutation alone is not always sufficient to drive cancer...

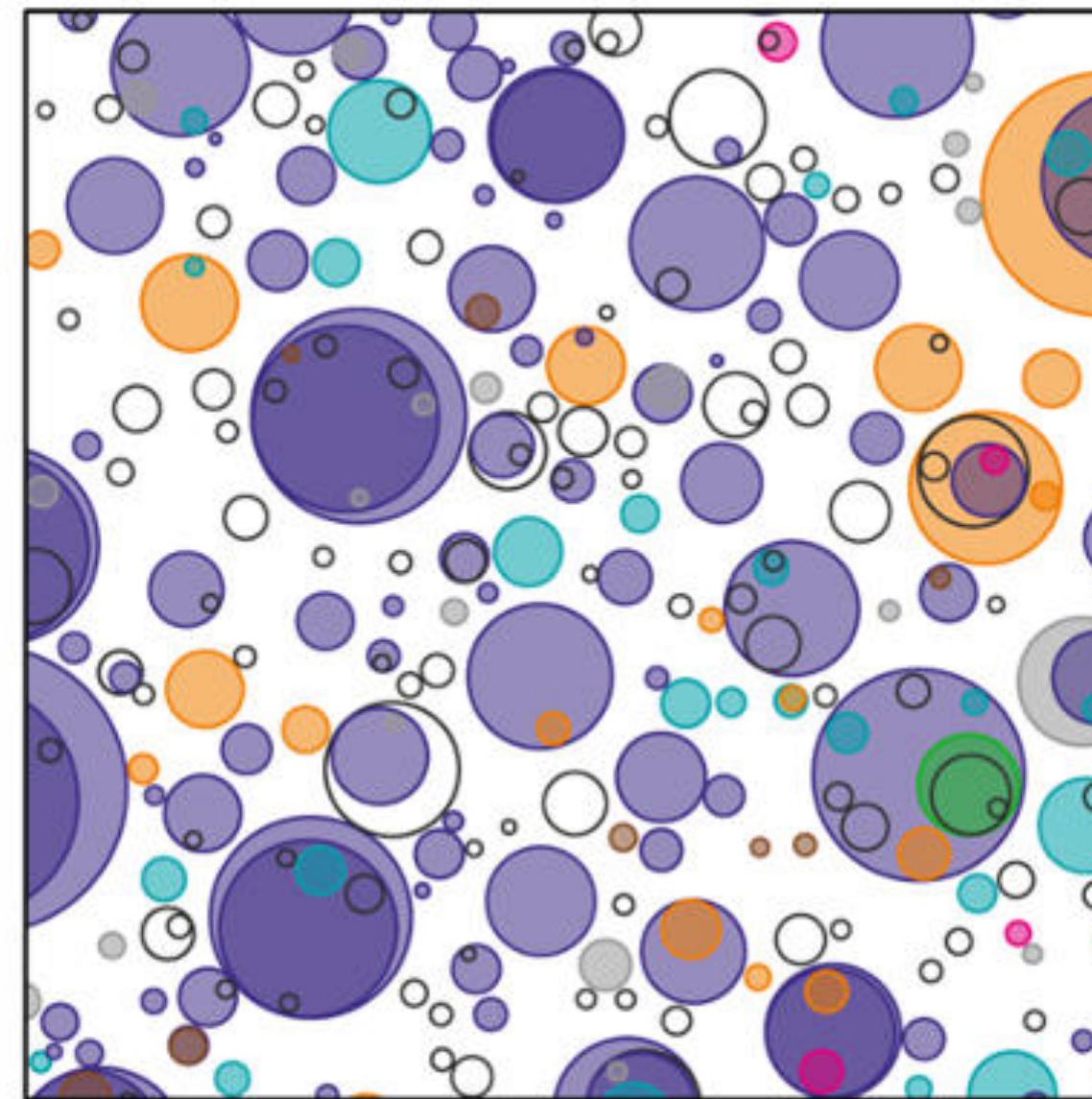
Genetic alterations alone are not always sufficient to drive cancer

Clonal expansion of mutant cell clones in healthy oesophageal epithelium with age

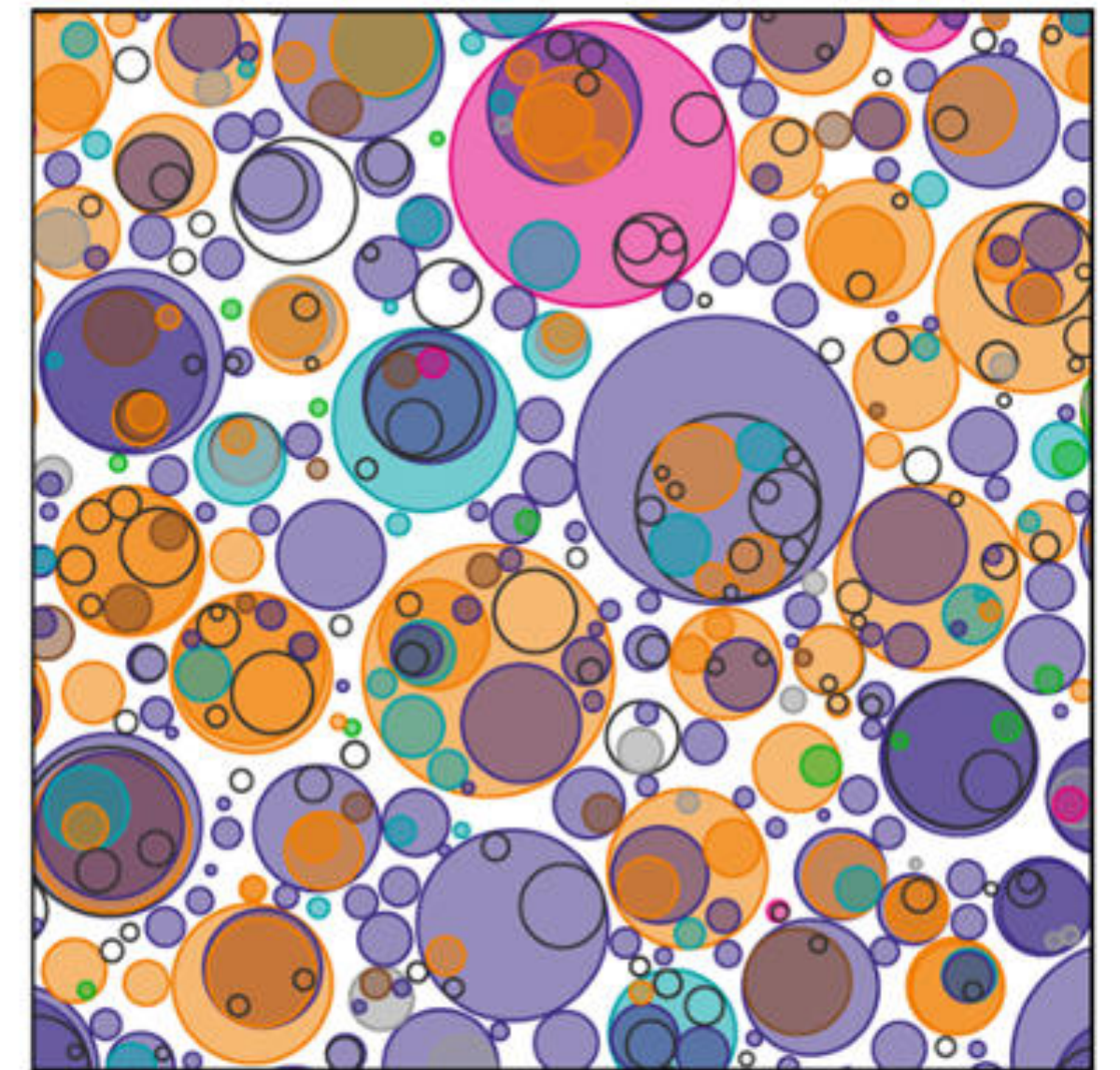
24–27 years old



52–55 years old



72–75 years old

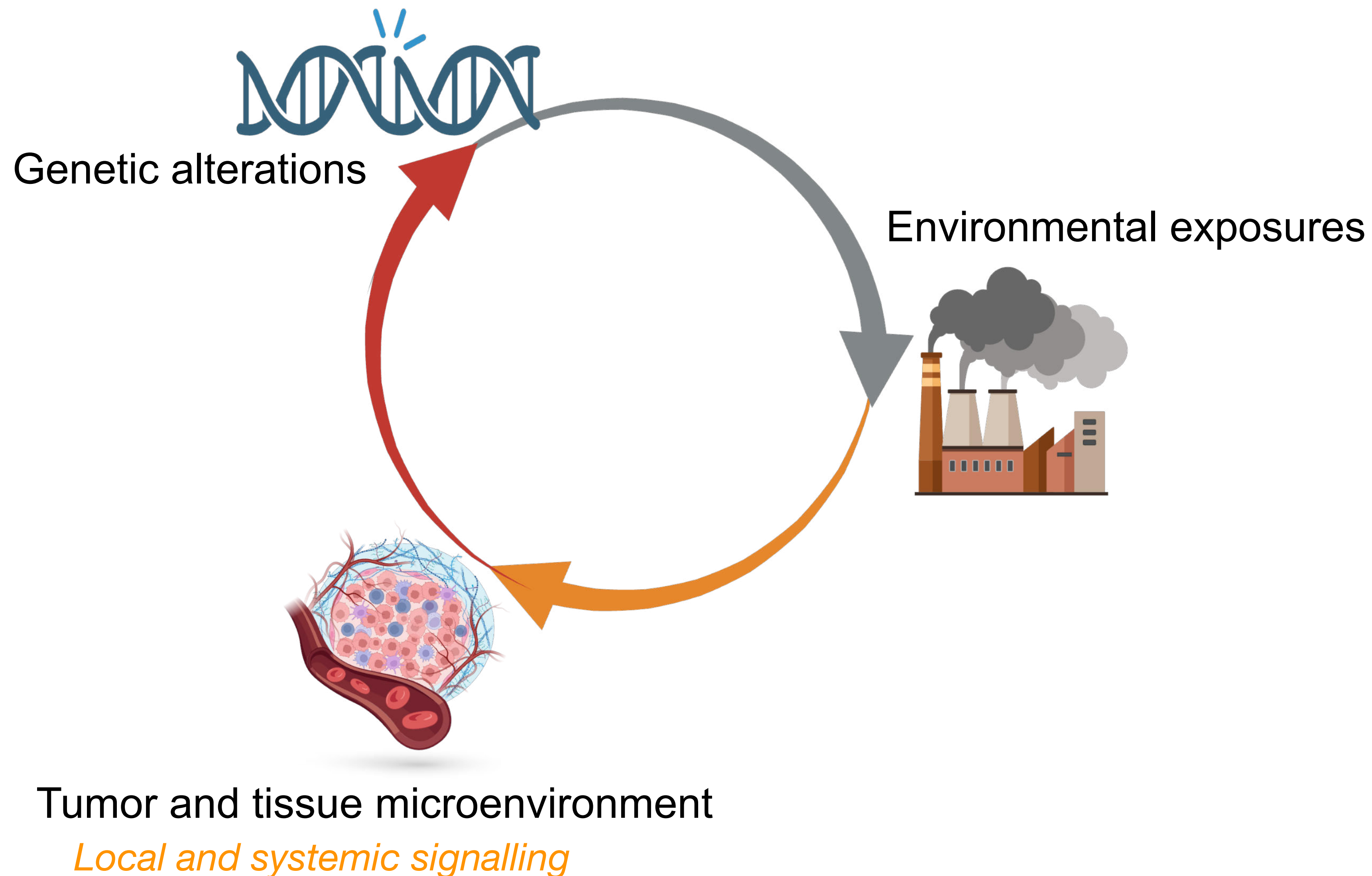


Time

Clonal abundance in 1 mm² of healthy tissue

Cancer driving mutations

Cancer genesis and progression: a dynamic interplay between genetics, environmental exposures, and the microenvironment



How have human stem cell and organoid models contributed to disentangling the GxE contribution to cancer?

Article

Tobacco smoking and somatic mutations in human bronchial epithelium

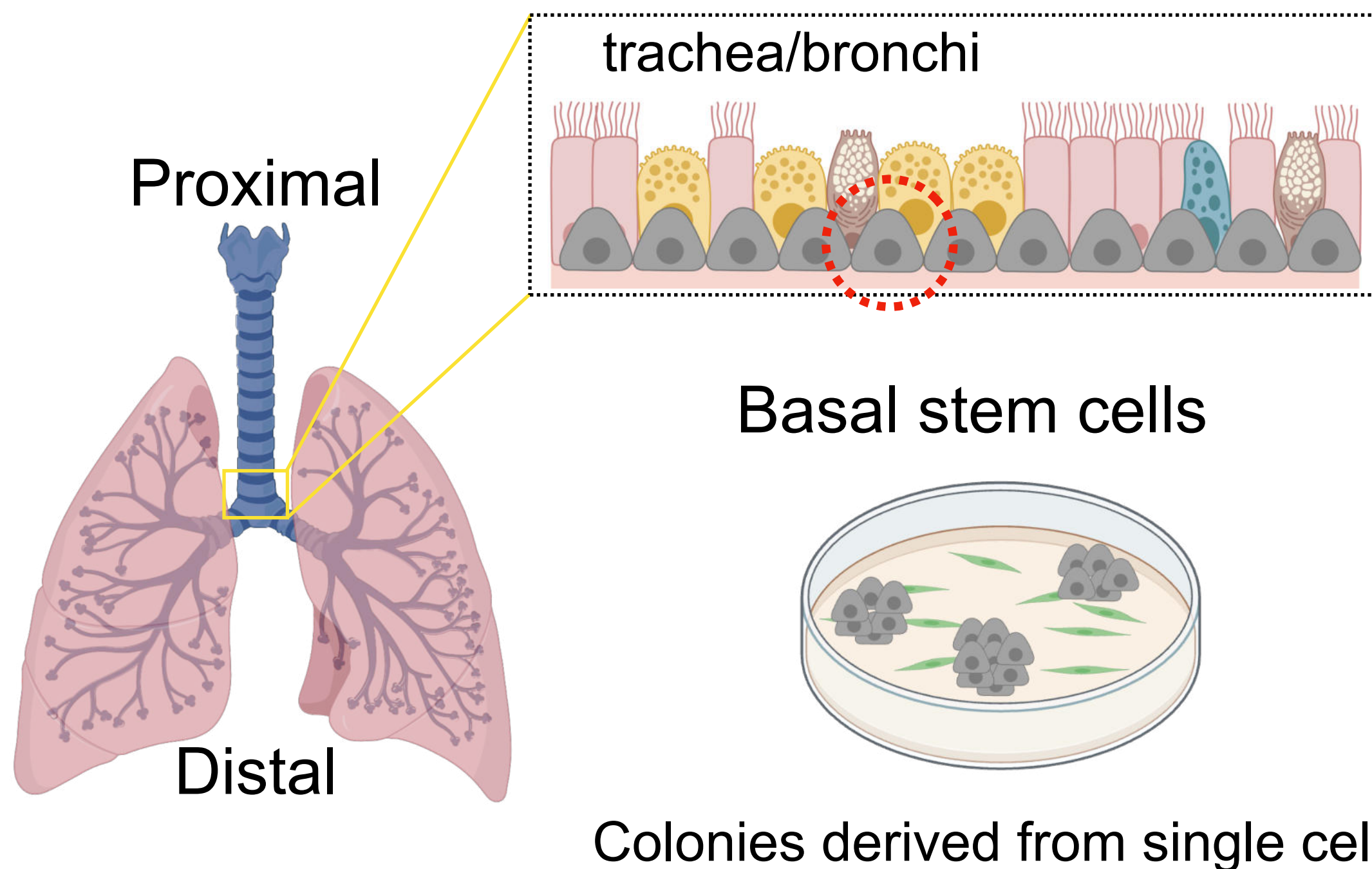
<https://doi.org/10.1038/s41586-020-1961-1>

Received: 7 June 2019

Accepted: 29 December 2019

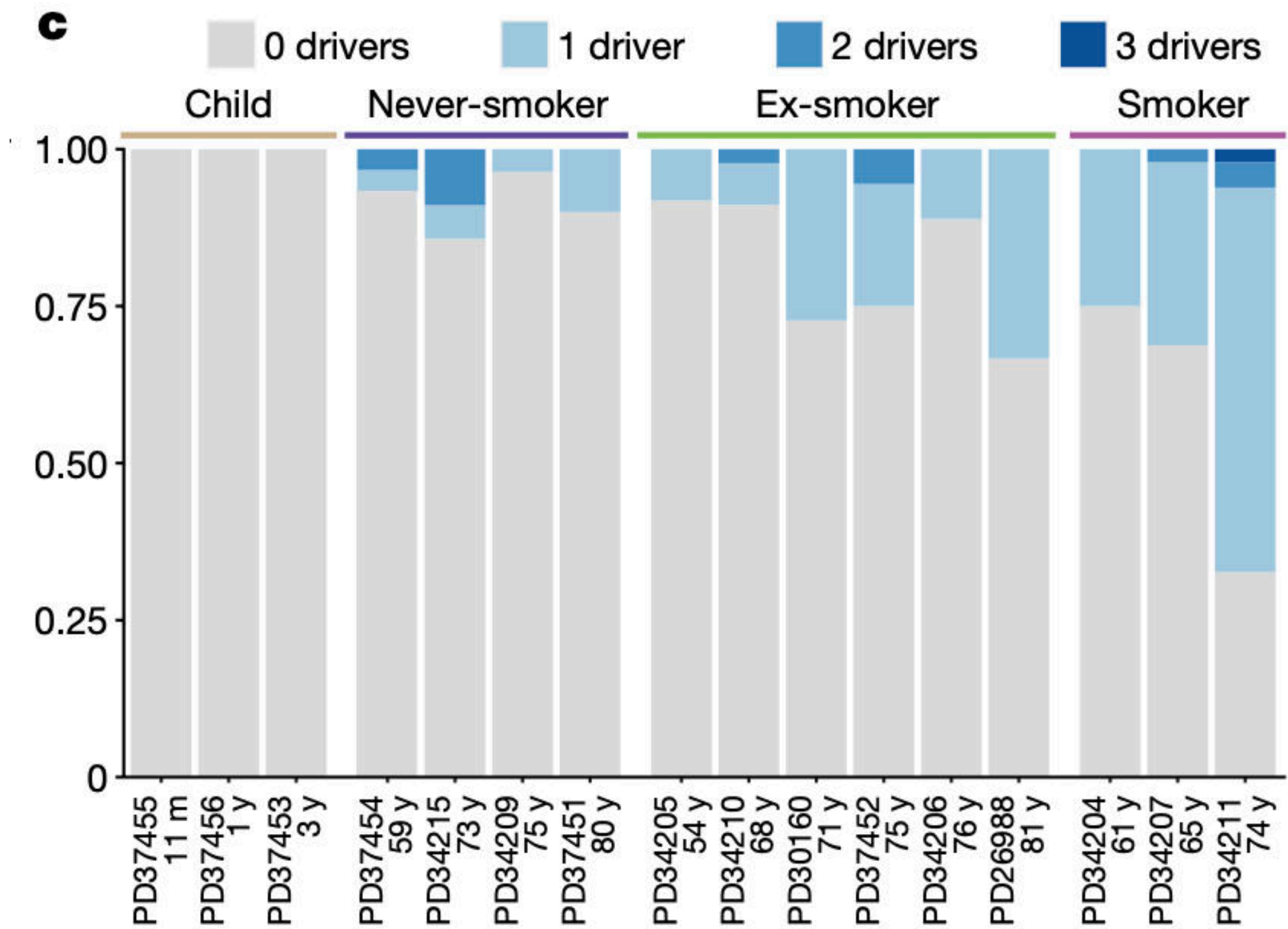
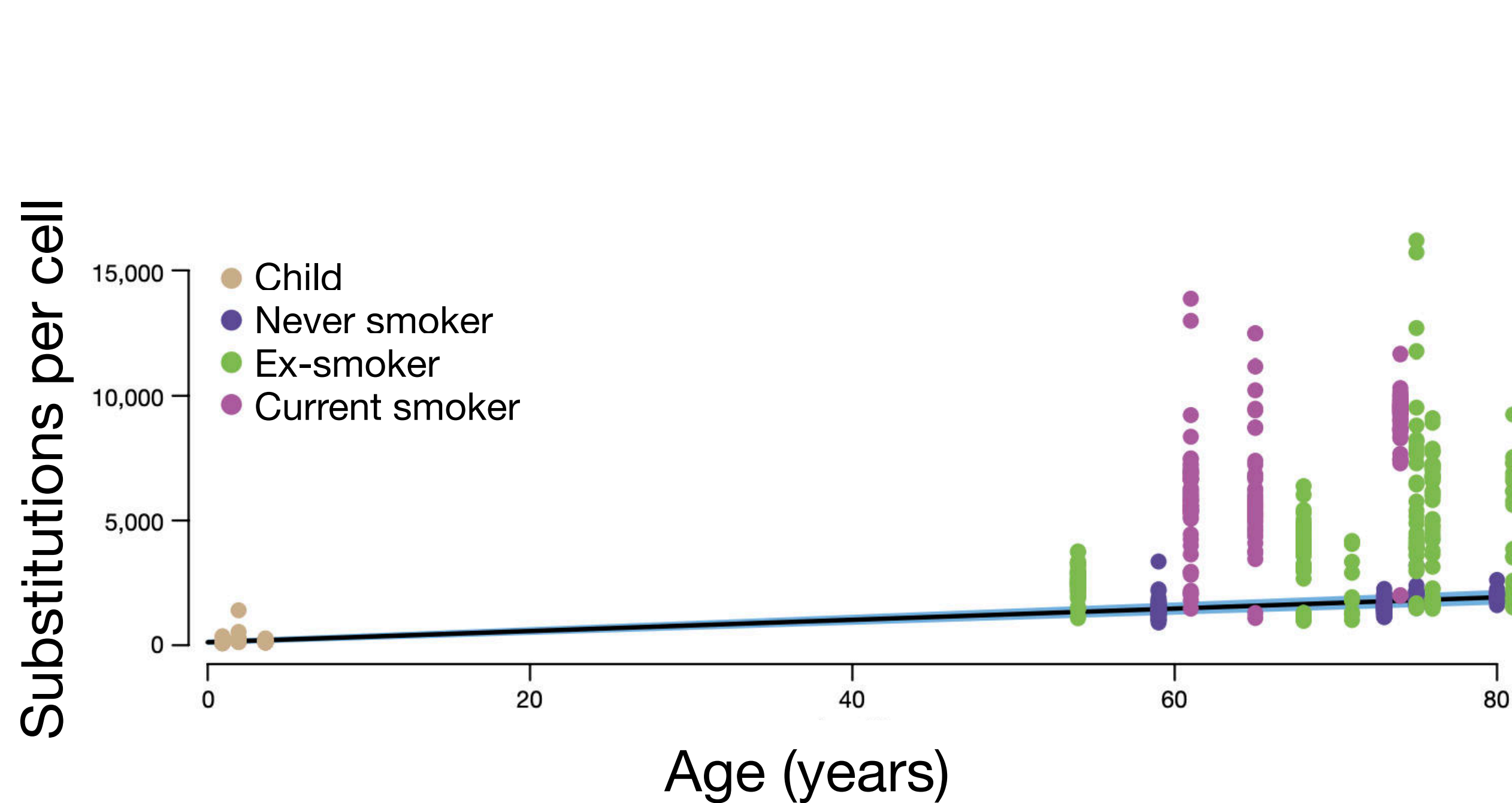
Published online: 29 January 2020

Kenichi Yoshida^{1,7}, Kate H. C. Gowers^{2,7}, Henry Lee-Six¹, Deepak P. Chandrasekharan², Tim Coorens¹, Elizabeth F. Maughan², Kathryn Beal¹, Andrew Menzies¹, Fraser R. Millar², Elizabeth Anderson¹, Sarah E. Clarke², Adam Pennycuick², Ricky M. Thakrar^{2,3}, Colin R. Butler^{2,3}, Nobuyuki Kakiuchi⁴, Tomonori Hirano⁴, Robert E. Hynds^{2,5}, Michael R. Stratton¹, Iñigo Martincorena¹, Sam M. Janes^{2,3,8*} & Peter J. Campbell^{1,6,8*}

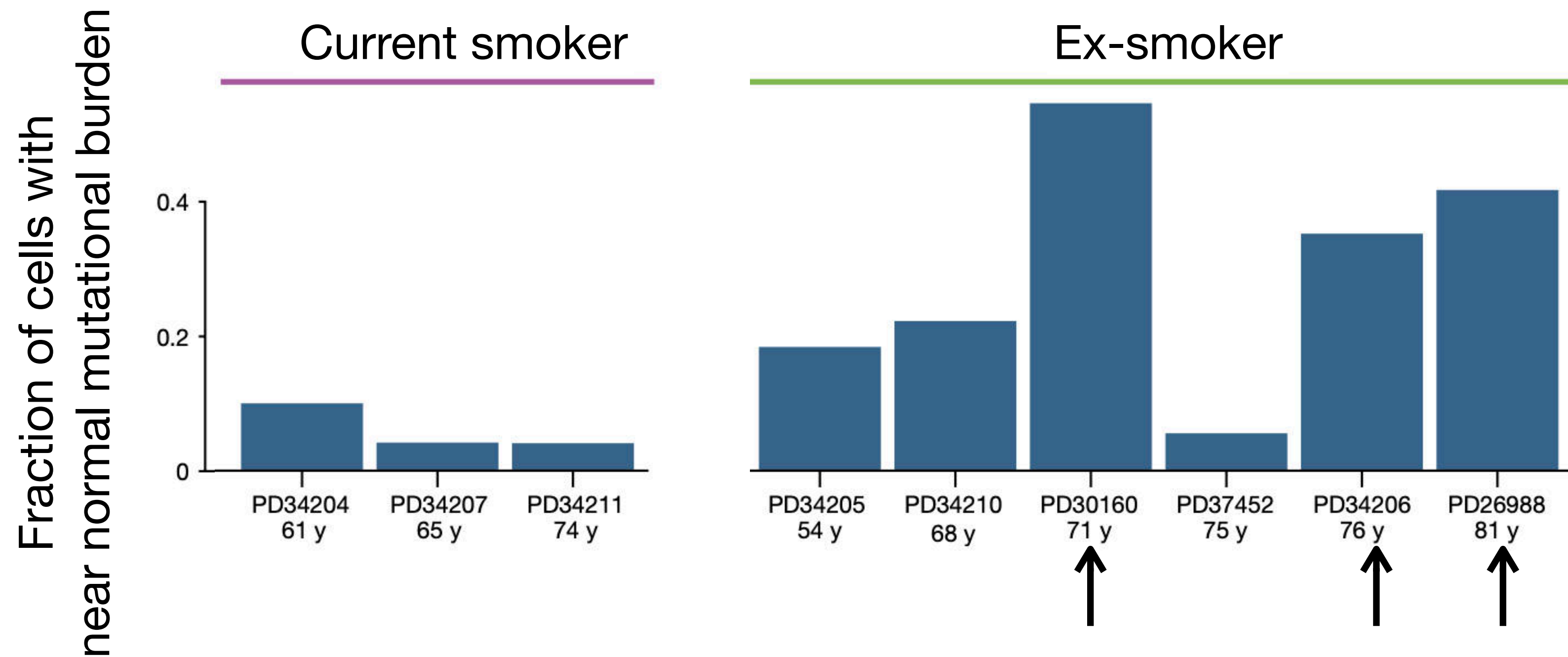


WGS = whole genome sequencing

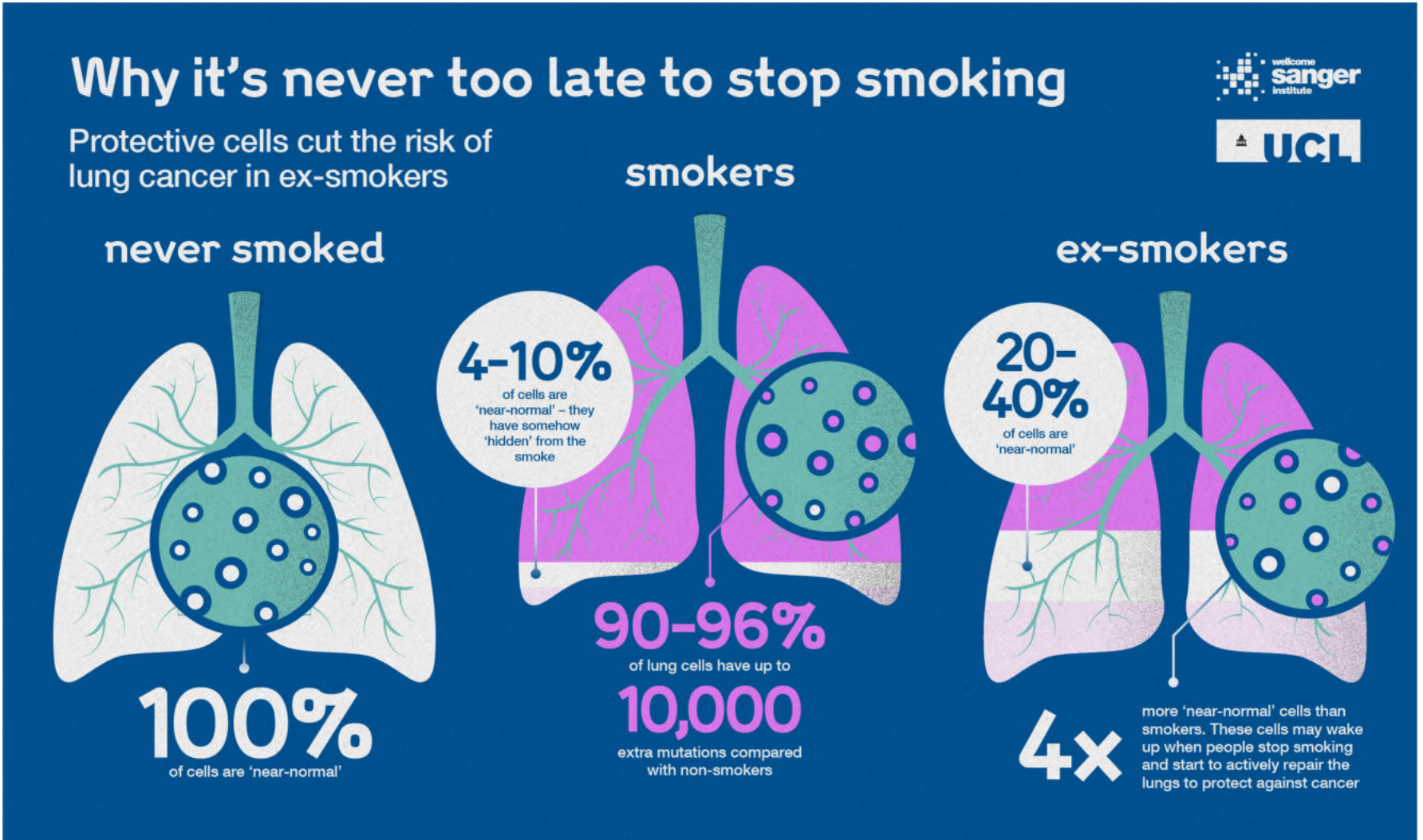
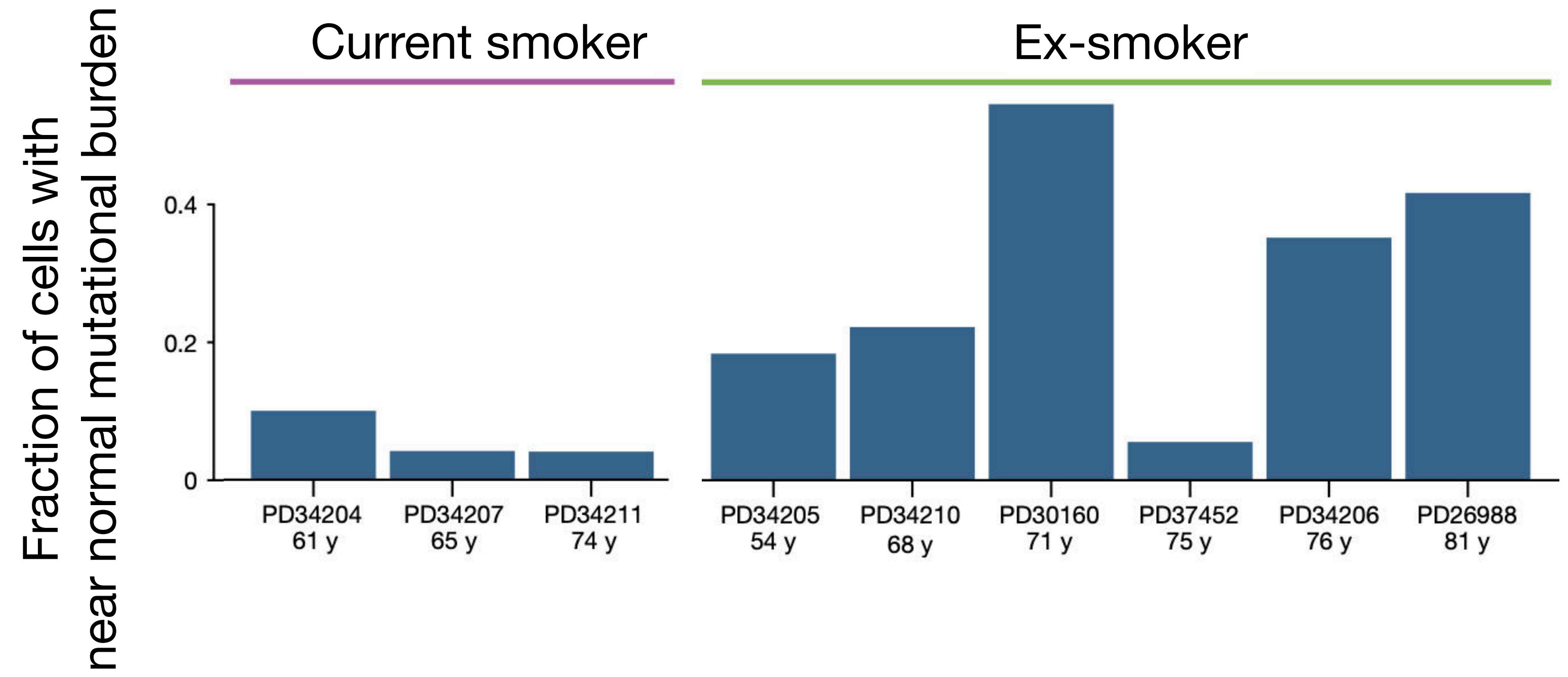
Basal cell colonies from smokers and ex-smokers have more mutations and more cells with at least 1 driver mutation



Ex-smokers have a higher proportion of “near normal” basal cells than current smokers



Ex-smokers have a higher proportion of “near normal” basal cells than current smokers



How did these 'near normal' cells avoid mutation?
Did non-mutant cells expand after smoking-cessation?

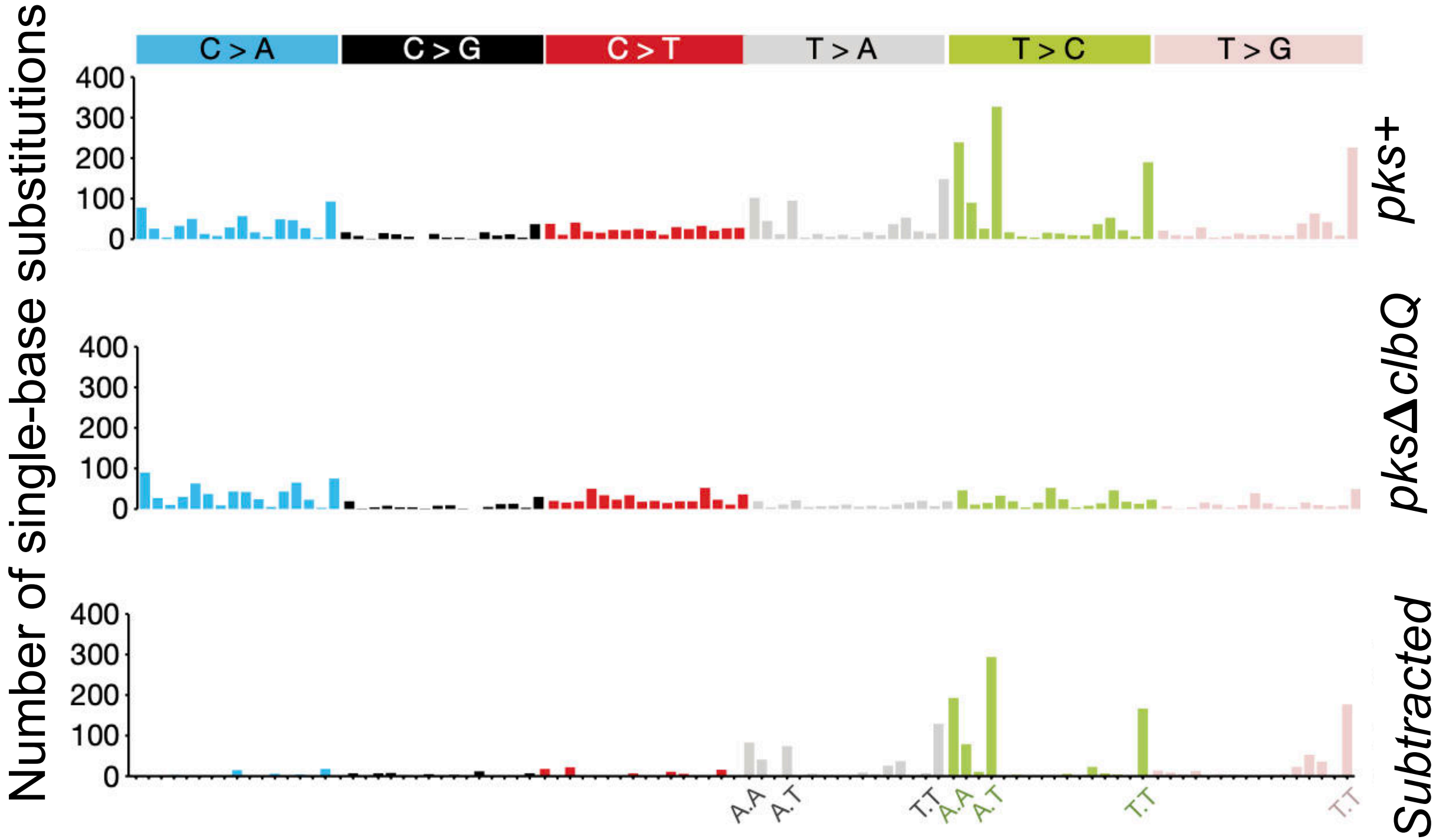
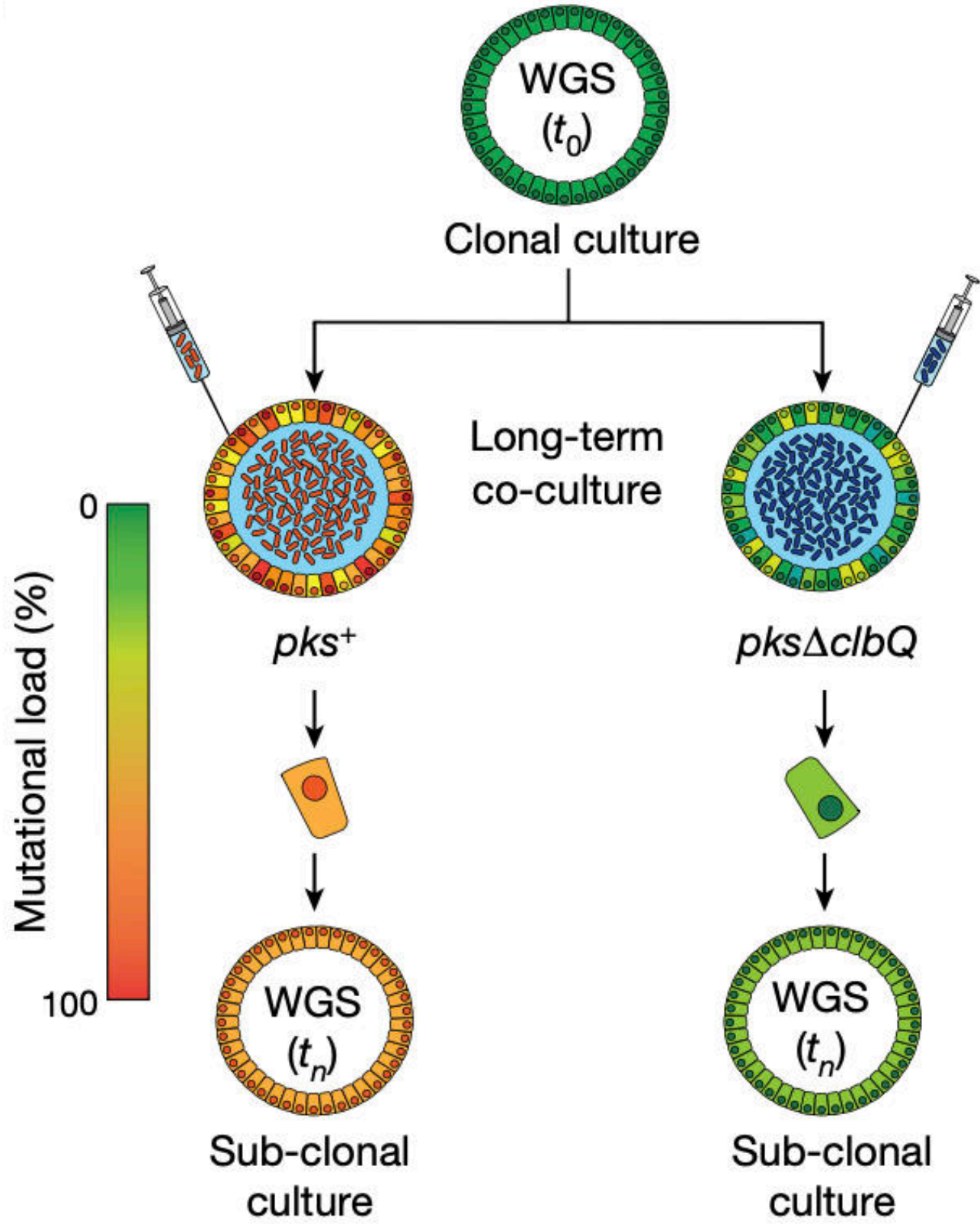
Article

Mutational signature in colorectal cancer caused by genotoxic *pks*⁺ *E. coli*

<https://doi.org/10.1038/s41586-020-2080-8>
Received: 14 September 2019
Accepted: 17 February 2020
Published online: 27 February 2020
 Check for updates

Cayetano Pleguezuelos-Manzano^{1,2,21}, Jens Puschhof^{1,2,21}, Axel Rosendahl Huber^{2,3,21}, Arne van Hoeck^{2,4}, Henry M. Wood⁵, Jason Nomburg^{6,7,8}, Carino Gurjao^{7,8}, Freek Manders^{2,3}, Guillaume Dalmasso⁹, Paul B. Stege¹⁰, Fernanda L. Paganelli¹⁰, Maarten H. Geurts^{1,2}, Joep Beumer¹, Tomohiro Mizutani^{1,2}, Yi Miao^{11,12,13}, Reinier van der Linden¹, Stefan van der Elst¹, Genomics England Research Consortium*, K. Christopher Garcia^{11,12,13}, Janetta Top¹⁰, Rob J. L. Willems¹⁰, Marios Giannakis^{7,8}, Richard Bonnet^{9,14}, Phil Quirke⁵, Matthew Meyerson^{7,8,15,16}, Edwin Cuppen^{2,4,17,18}, Ruben van Boxtel^{2,3} & Hans Clevers^{1,2,3}

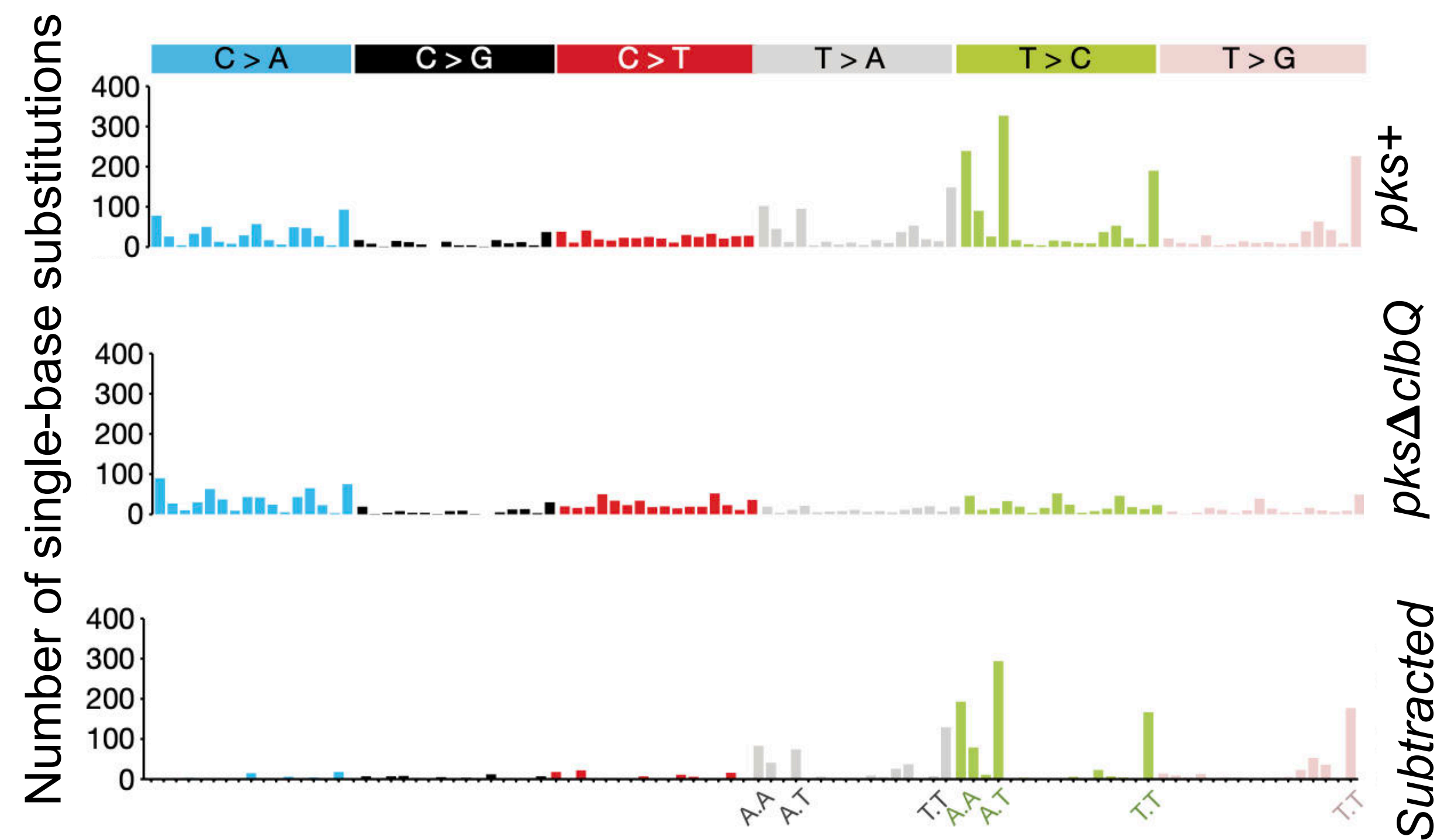
Identified a mutational signature associated with *pks*⁺ *E.coli* (colibactin)



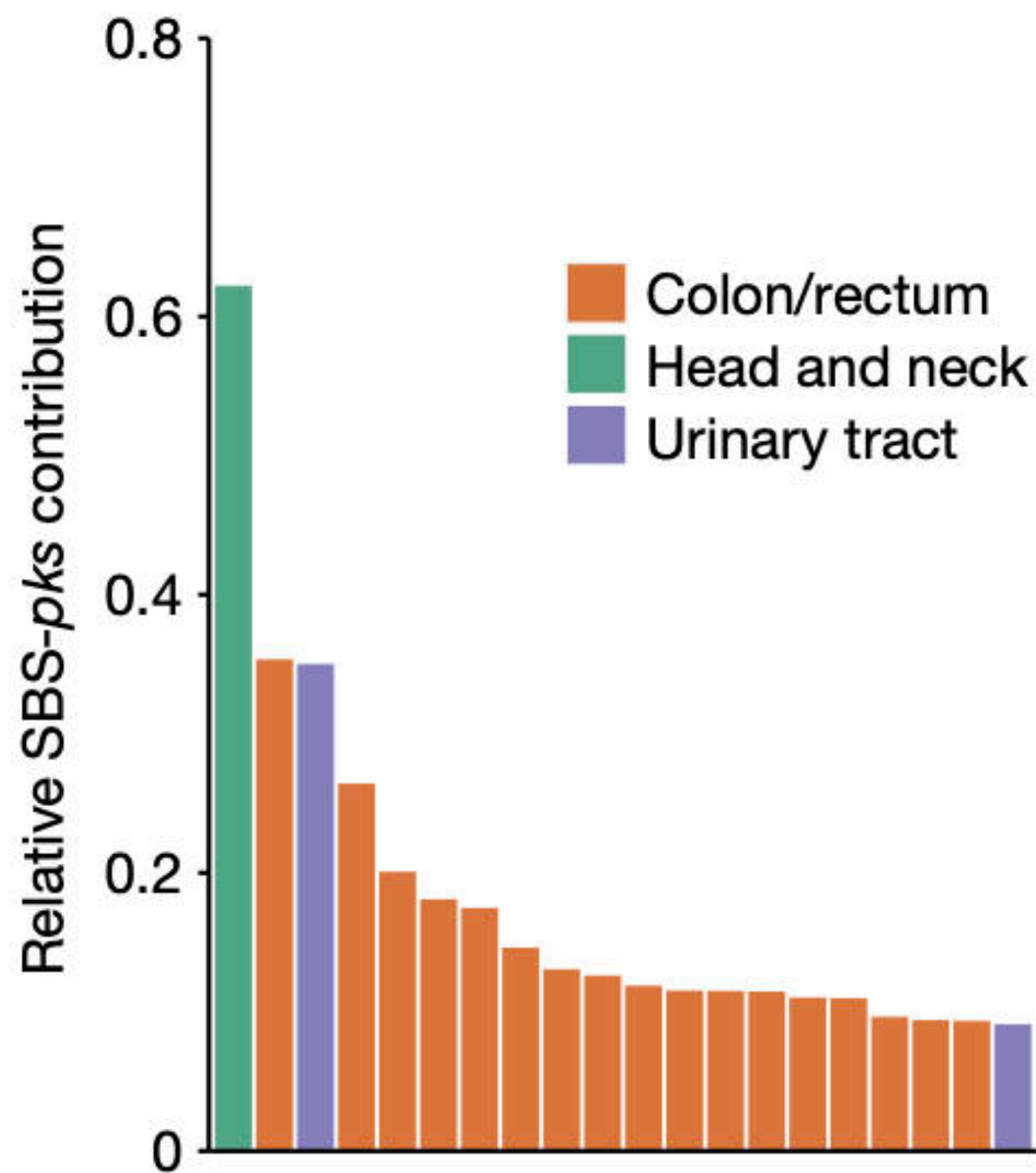
pks⁺ *E.coli* produce the genotoxic compound colibactin

Mutational signature, *SBS-pks* (*SBS88*), identified in cancer samples

Identified a mutational signature associated with *pks*+
E.coli (colibactin)



20/3668 metastasis samples show *SBS-pks*



Mutational signature, *SBS-pks* (*SBS88*), identified in cancer samples

Identified a mutational signature

Article

Geographic and age variations in mutational processes in colorectal cancer

<https://doi.org/10.1038/s41586-025-09025-8>
Received: 25 August 2024
Accepted: 15 April 2025
Published online: 23 April 2025
Open access
Check for updates

Marcos Díaz-Gay^{1,2,3,4,5†}, Wellington dos Santos^{5,5†}, Sarah Moody^{6,5†}, Mariya Kazachkova^{1,3,7}, Ammal Abbasi^{1,2,3}, Christopher D. Steele^{1,2,3}, Raviteja Vangara^{1,2,3}, Sergey Senkin⁵, Jingwei Wang⁶, Stephen Fitzgerald⁶, Erik N. Bergstrom^{1,2,3}, Azhar Khandekar^{1,2,3,8}, Burçak Otlu^{1,2,3,9}, Behnoush Abedi-Ardekani⁵, Ana Carolina de Carvalho⁵, Thomas Cattiaux⁵, Ricardo Cortez Cardoso Penha⁵, Valérie Gaborieau⁵, Priscilia Chopard⁵, Christine Carreira¹⁰, Saamin Cheema⁶, Calli Latimer⁶, Jon W. Teague⁶, Anush Mukeriyar¹¹, David Zaridze¹¹, Riley Cox¹², Monique Albert^{12,13}, Larry Phouthavongsy¹², Steven Gallinger¹⁴, Reza Malekzadeh¹⁵, Ahmadreza Niavarani¹⁵, Marko Miladinov¹⁶, Katarina Erić¹⁷, Sasa Milosavljević¹⁸, Suleeporn Sangrajrang¹⁹, Maria Paula Curado²⁰, Samuel Aguiar²¹, Rui Manuel Reis^{22,23}, Monise Tadin Reis²⁴, Luis Gustavo Romagnolo²⁵, Denise Peixoto Guimarães²⁶, Ivana Holcatova^{27,28}, Jaroslav Kalvach^{29,30,31,32}, Carlos Alberto Vaccaro³³, Tamara Alejandra Piñero³³, Beata Świątkowska³⁴, Jolanta Lissowska³⁵, Katarzyna Roszkowska-Purska³⁶, Antonio Huertas-Salgado³⁷, Tatsuhiro Shibata^{38,39}, Satoshi Shiba³⁹, Surasak Sangkhathat^{40,41,42}, Taned Chitapanarux⁴³, Gholamreza Roshandel⁴⁴, Patricia Ashton-Prolla^{45,46}, Daniel C. Damin⁴⁷, Francine Hehn de Oliveira⁴⁸, Laura Humphreys⁶, Trevor D. Lawley⁴⁹, Sandra Perdomo⁵, Michael R. Stratton⁶, Paul Brennan⁵ & Ludmil B. Alexandrov^{1,2,3,50,51}

Samples show SBS-pks

SBS88

$P\text{-trend} = 1.3 \times 10^{-4}$

9/27

16/70

27/136

28/279

29/290

0-39

40-49

50-59

60-69

70+

Colon/rectum

Head and neck

Urinary tract

Number of single-base substitutions

C > A

C > G

400

300

200

100

0

But can we say pks+ E.coli cause cancer?

Subtracted

400

300

200

100

0

AA

A.T

T.AA

A.T

TT

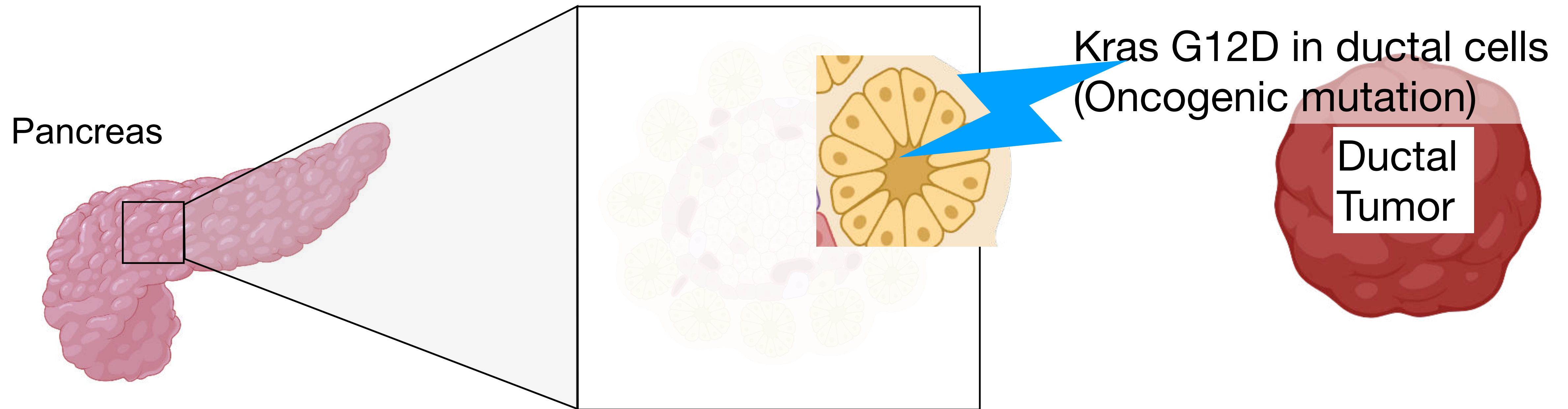
TT

What could explain why not all cells that have oncogenic mutations become tumors?

What is different about the cells that become tumors? The cells that don't become tumors?

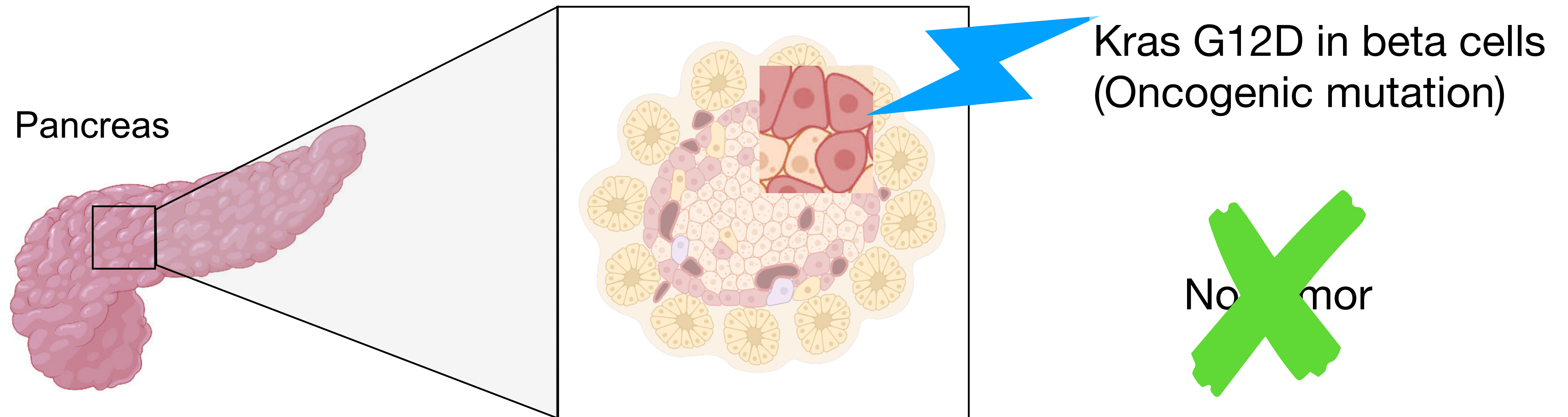
Injury influences susceptibility to cancer (via **plasticity**?)

We are used to defining cancers according to the tissues they arise in, but cell of origin also matters...



Injury influences susceptibility to cancer (via plasticity?)

We are used to defining cancers according to the tissues they arise in, but cell of origin also matters...

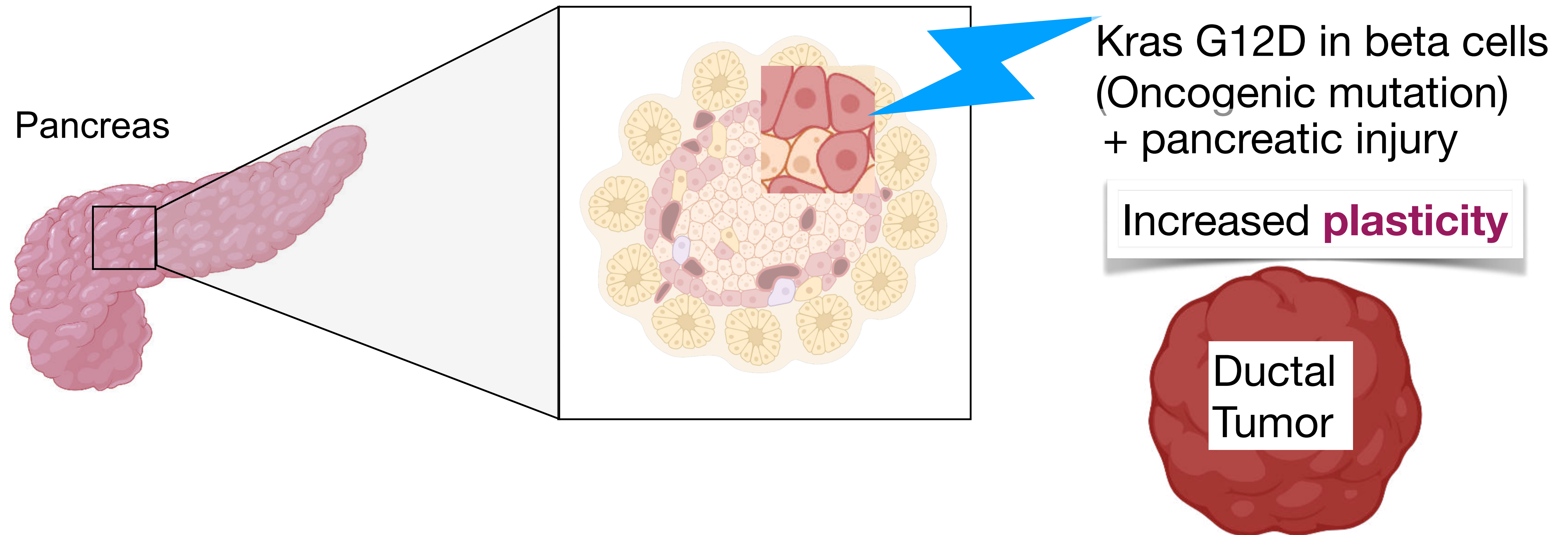


Burdziak and Alonso-Curbelo et al (2023) Science

Gidekel et al (2009) Cancer Cell

Injury influences susceptibility to cancer (via plasticity?)

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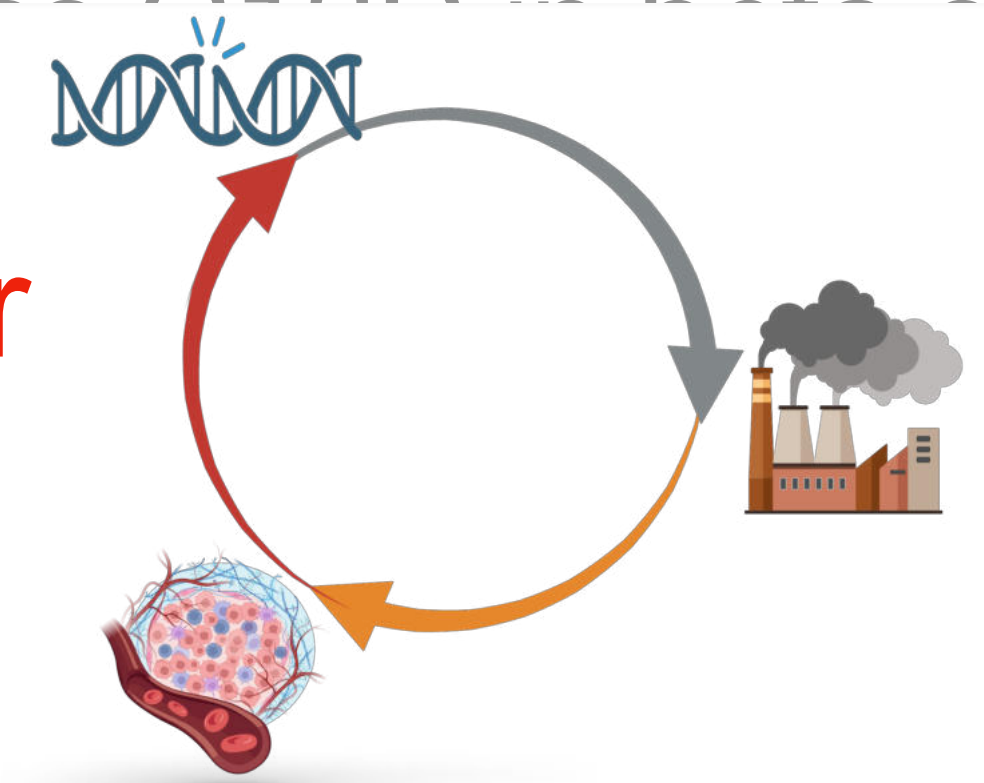
Burdziak and Alonso-Curbelo et al (2023) Science

Gidekel et al (2009) Cancer Cell

Injury influences susceptibility to cancer (via plasticity?)

We are used to defining cancers according to the tissues they arise in, but cell of origin also matters...

In this model, injury-induced plasticity and cancer susceptibility has been linked to expression of the inflammatory cytokine IL33

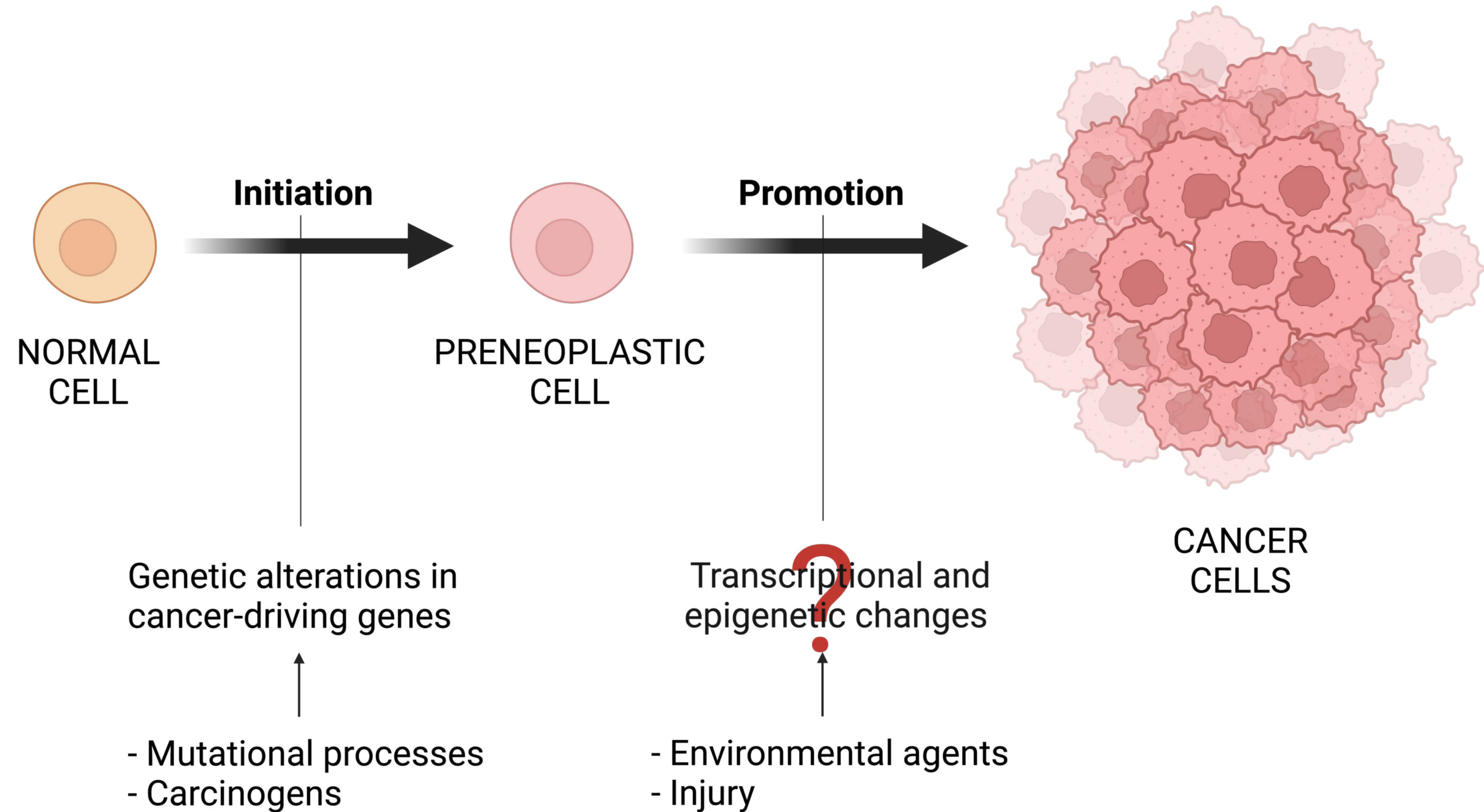


Ductal
Tumor

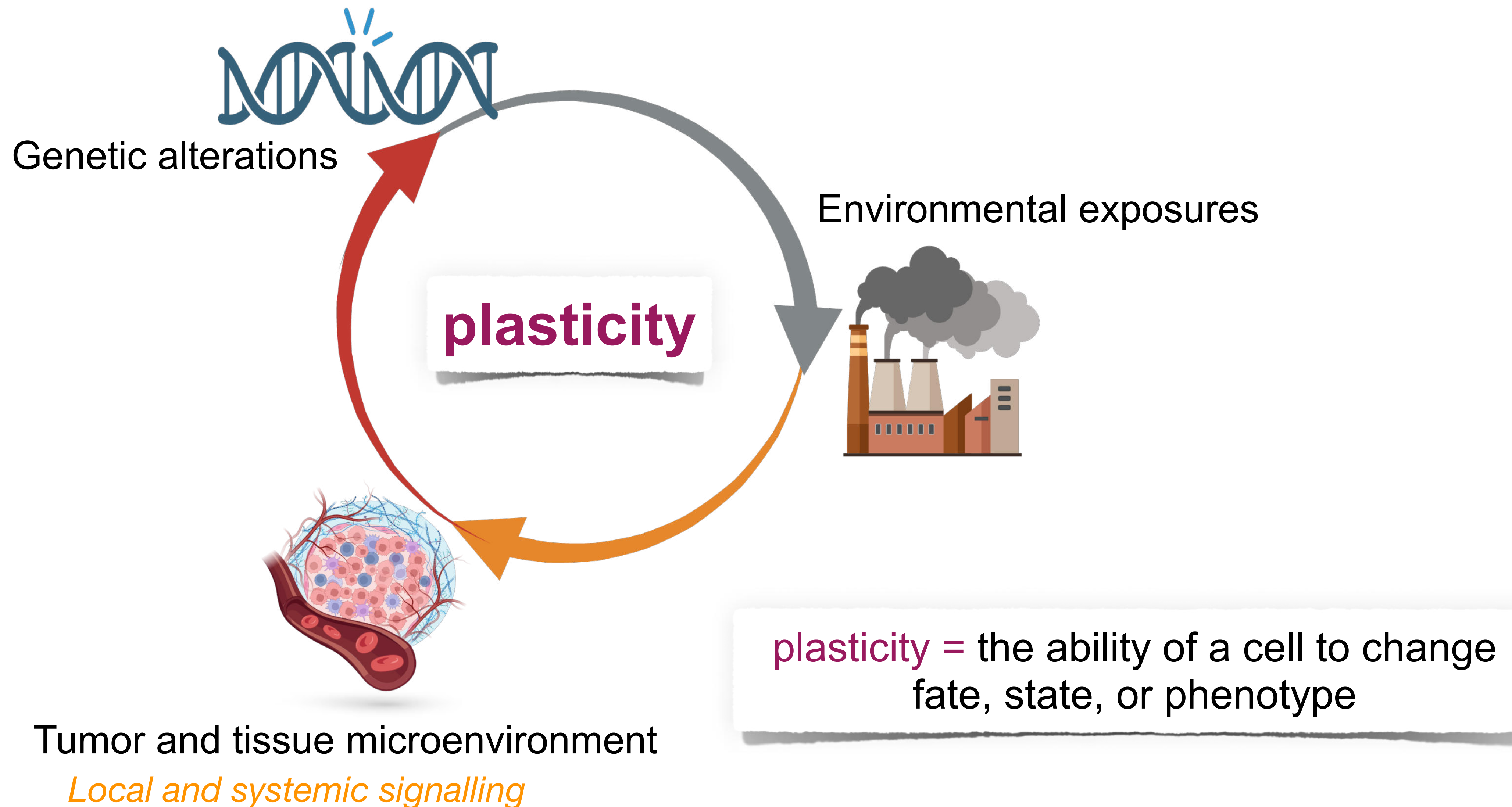
Burdziak and Alonso-Curbelo et al (2023) Science

Gidekel et al (2009) Cancer Cell

Genetic alterations alone are not always sufficient to drive cancer



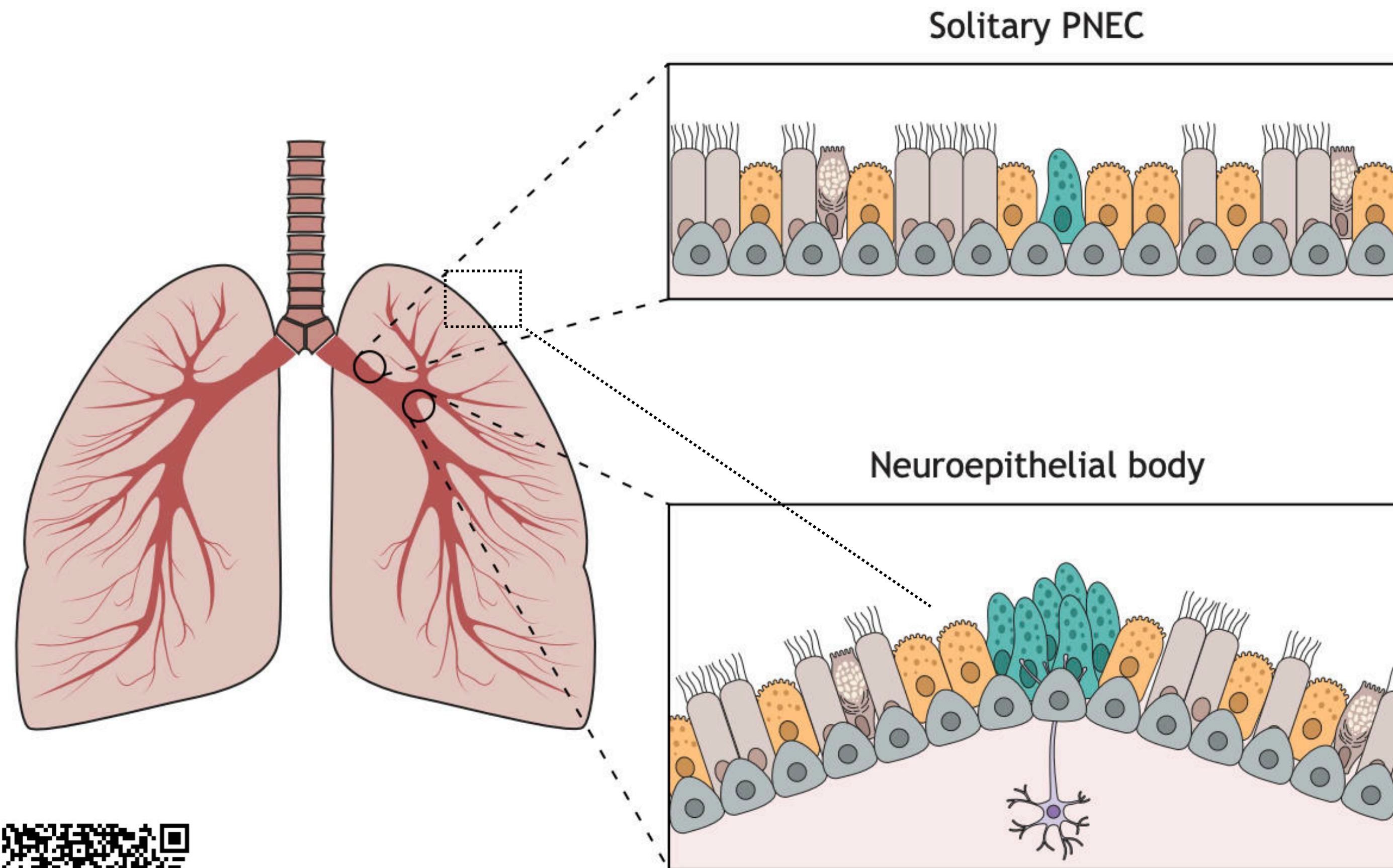
Cancer genesis and progression: a dynamic interplay between genetics, environmental exposures, and the microenvironment



Pulmonary neuroendocrine (NE) cells: rare chemosensory cells

Pulmonary NE cells account for 0.4% of lung epithelial cells

Pulmonary NE cell markers: ASCL1, UCHL1, CHGA



Pulmonary NE cell functions:

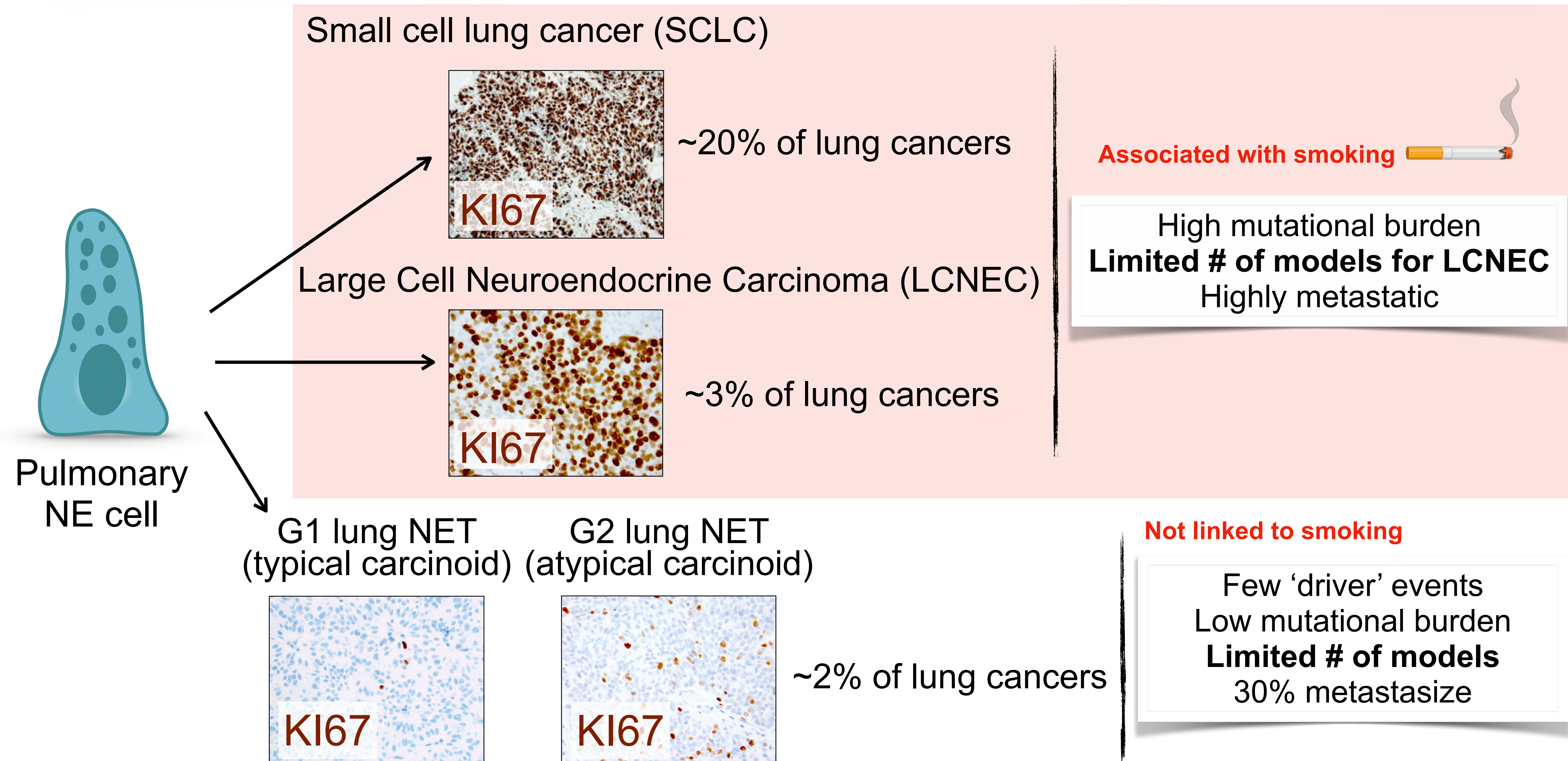
- Oxygen sensing
- Recruitment and stimulation of immune cells
- Reserve stem cells (NE^{stem})

Pulmonary NE cells in disease:

- COPD
- asthma
- cystic fibrosis
- pneumonia
- SIDS
- NE cancer

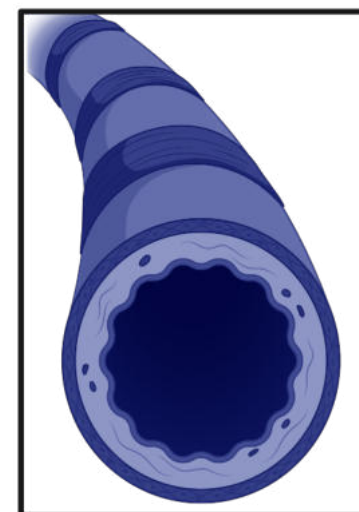
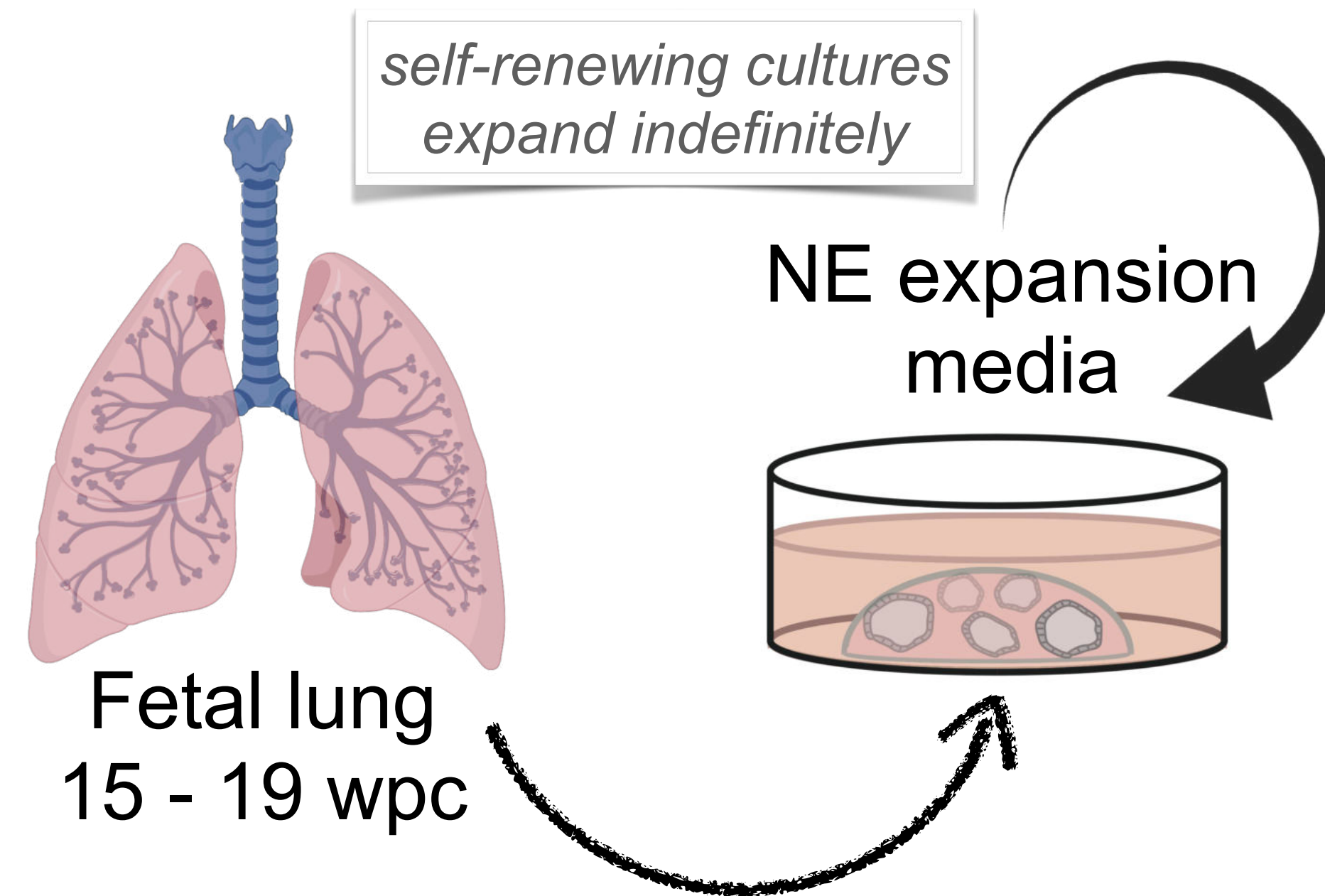


Pulmonary NE cells: cells of origin for NE cancers

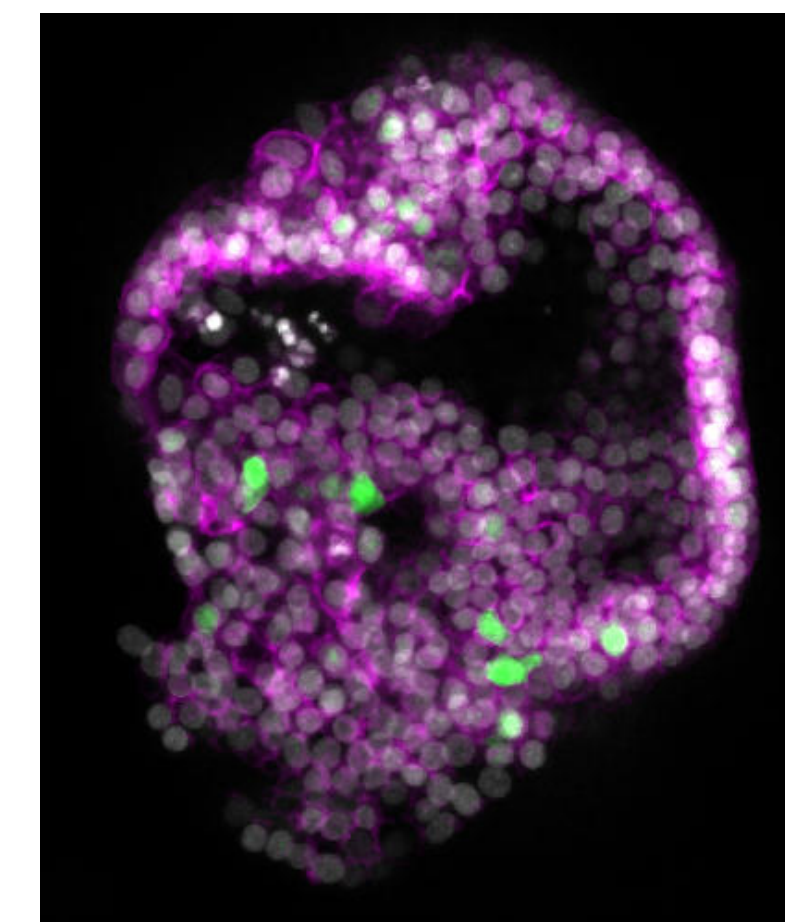
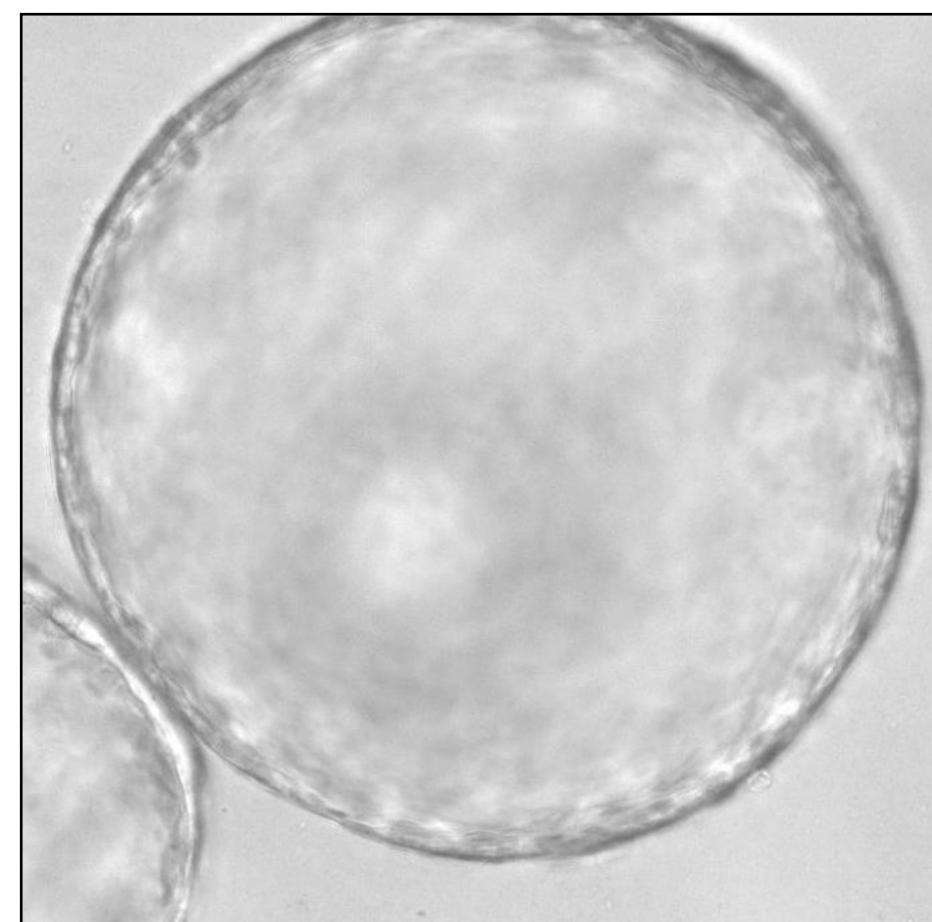


NE-enriched fetal Airway Organoids (NEr-fAOs)

Candeli et al., 2026, biorxiv



Airways



NE marker ACTIN DAPI

HEALTH

NE-enriched fetal Airway Organoids (NEr-fAOs)

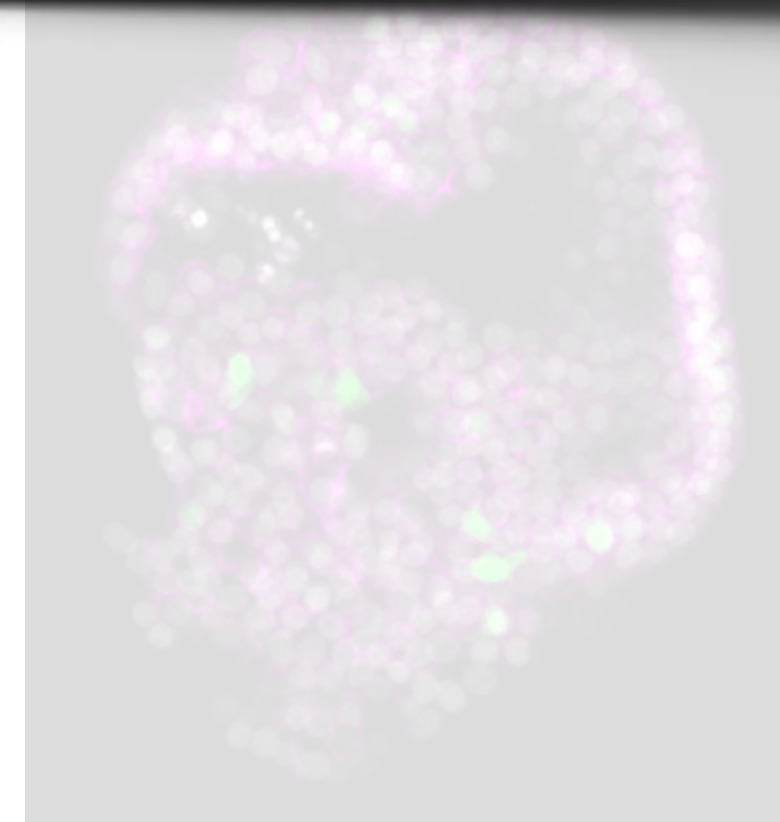
Candeli et al., 2026, biorxiv



These cultures can be expanded indefinitely but they can also be differentiated further or enriched for specific cell types by adding a differentiation media (remove growth factors & add signalling molecules)

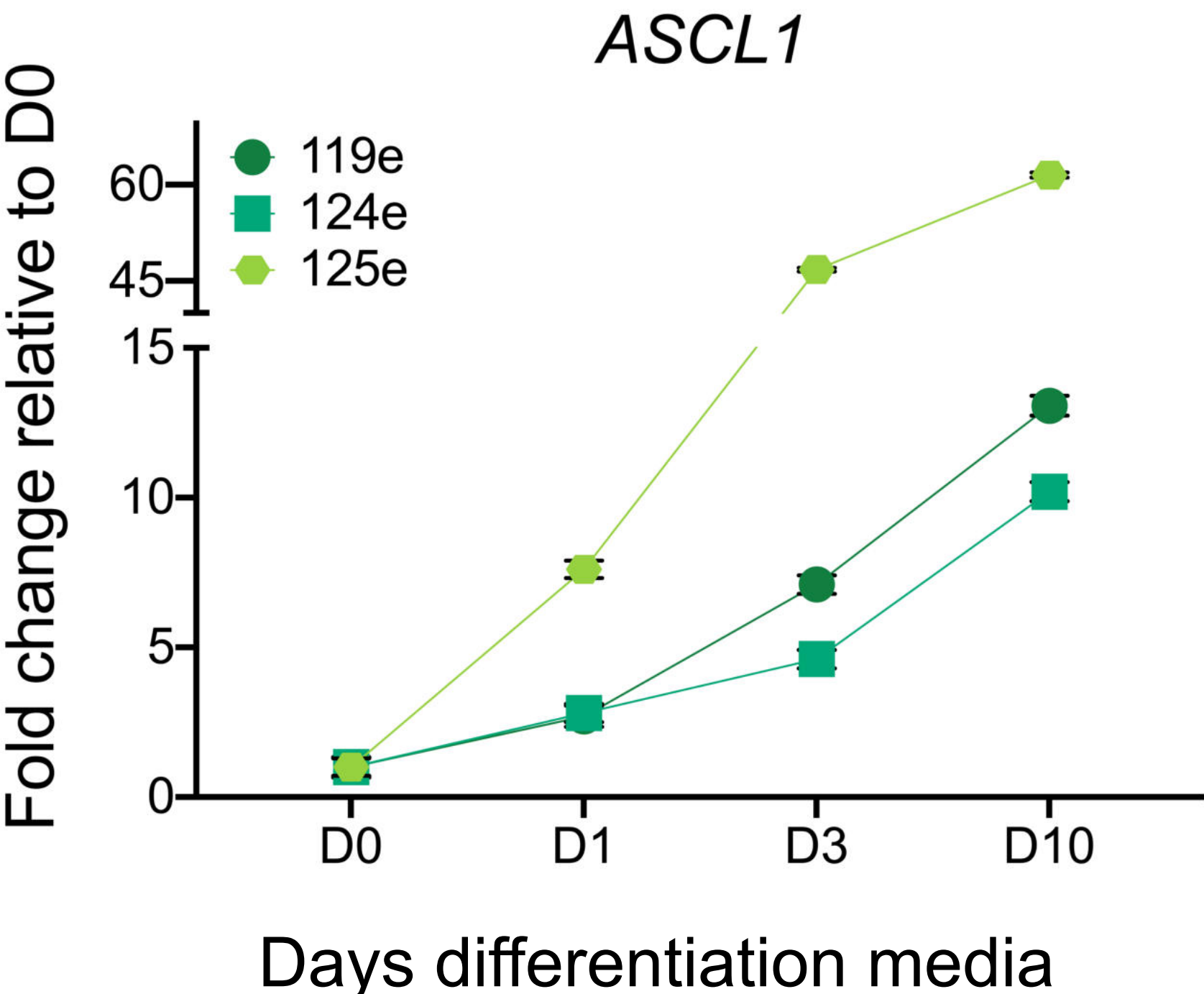
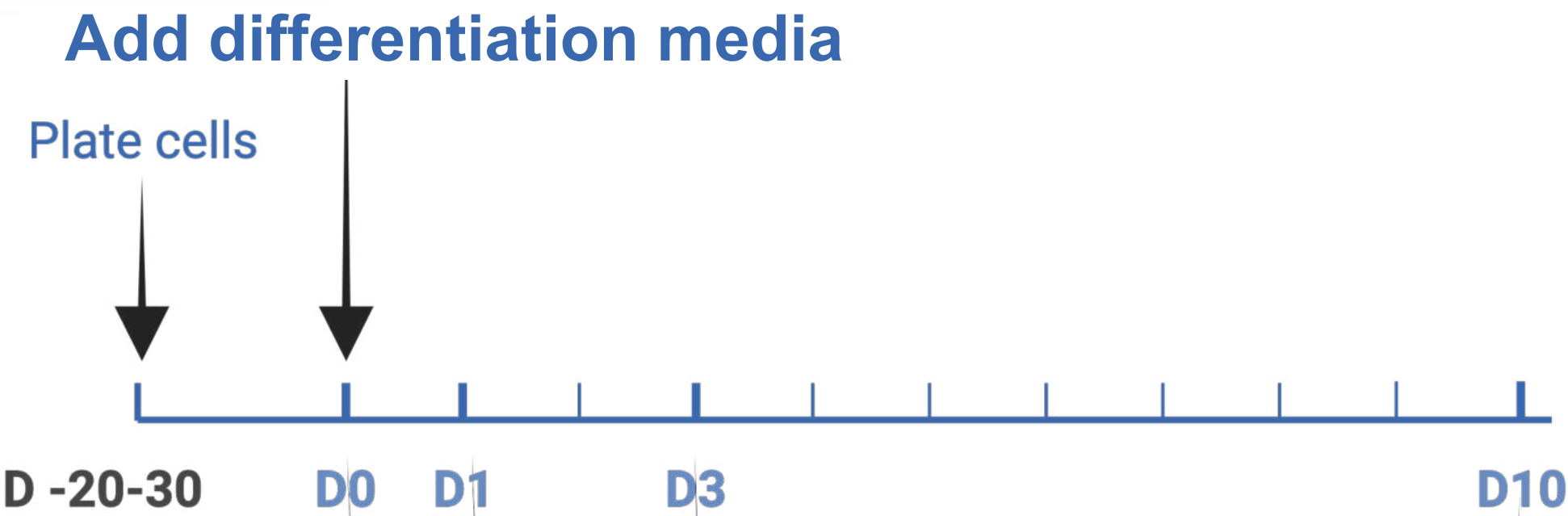


Airways

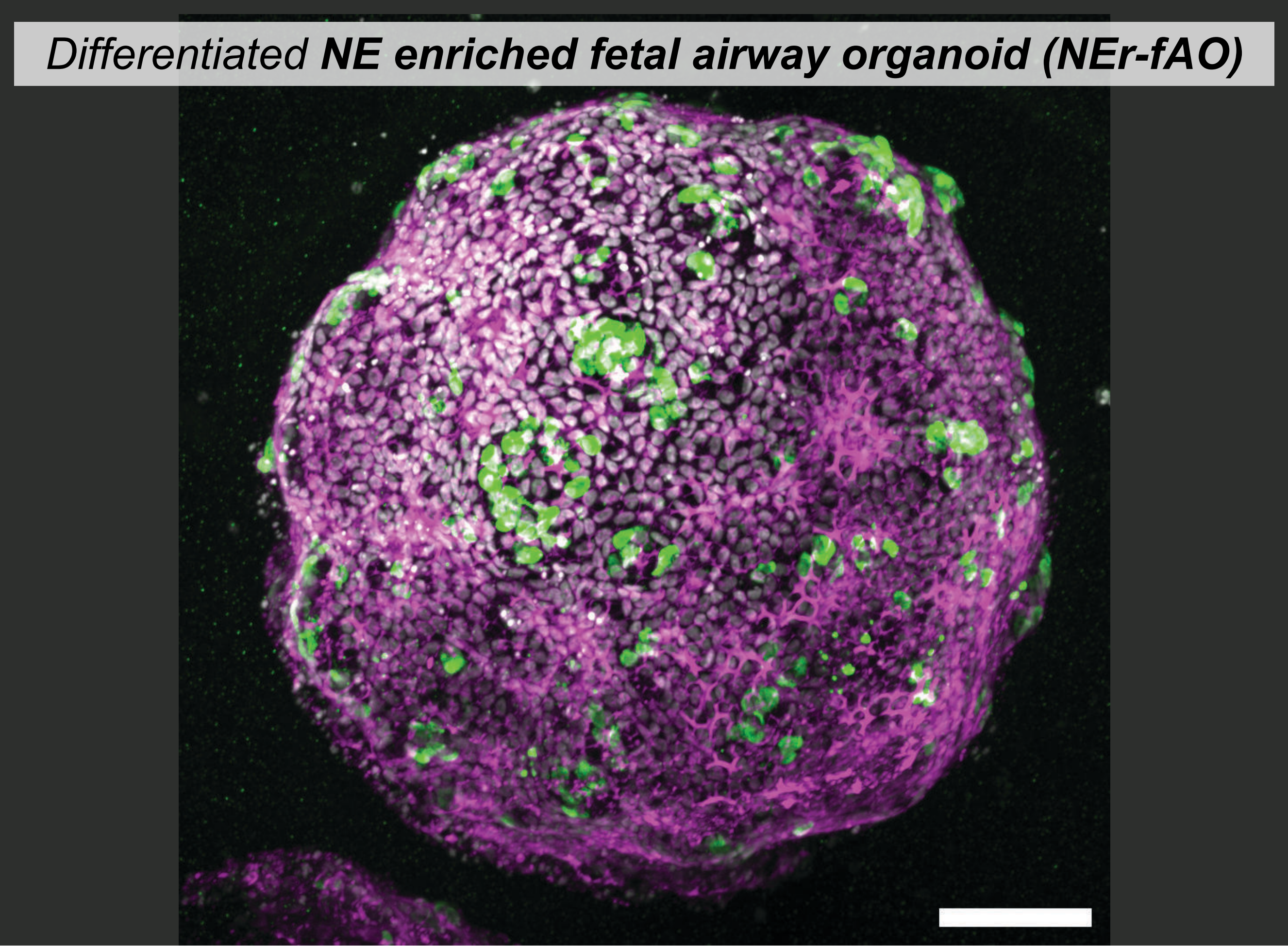


NE marker ACTIN DAPI

Pulmonary NE cell differentiation media promotes NE fate in NEr-fAOs



Addition of NE differentiation media leads to a 10 to 60 fold increase in NE marker expression after 10 days



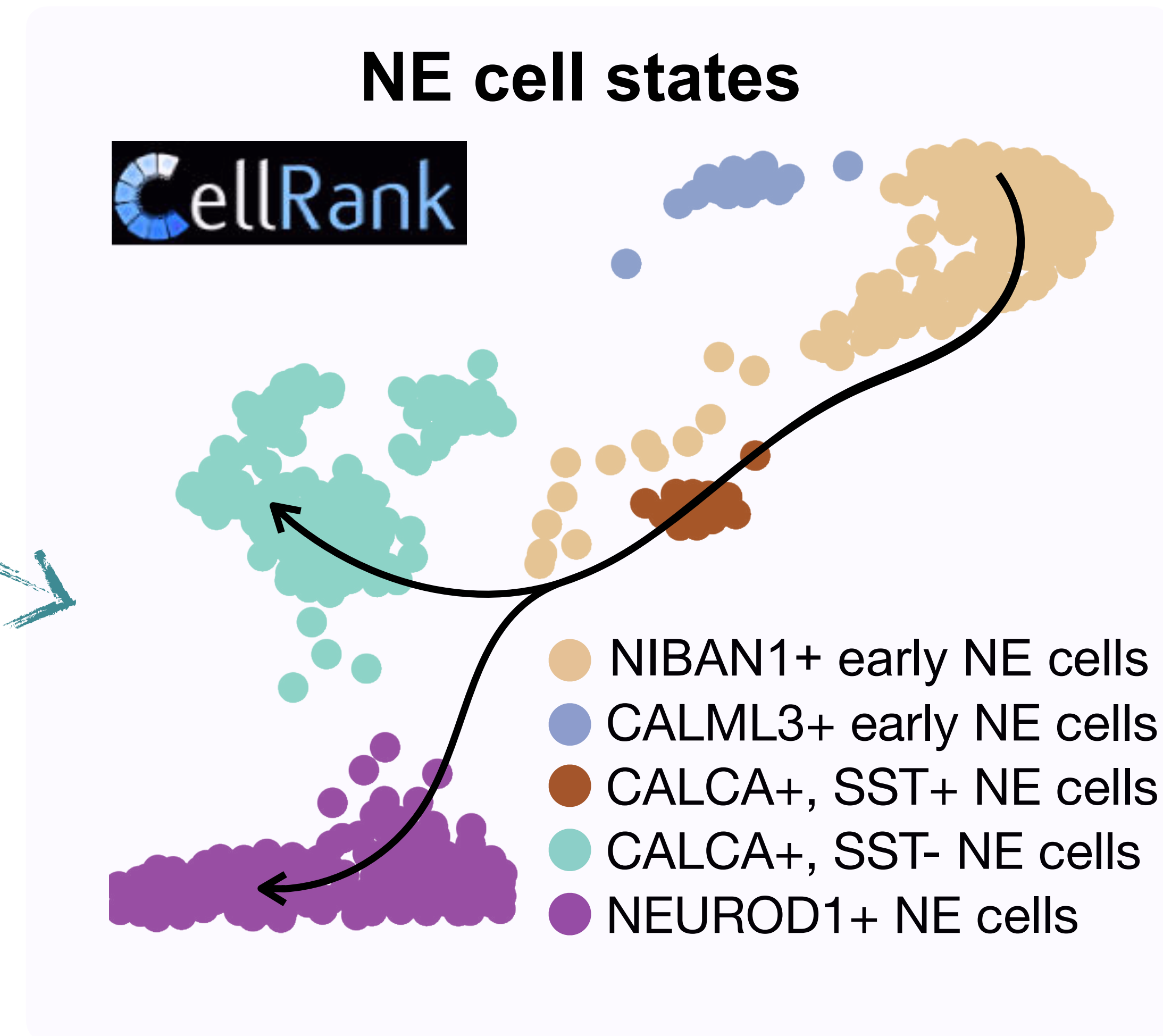
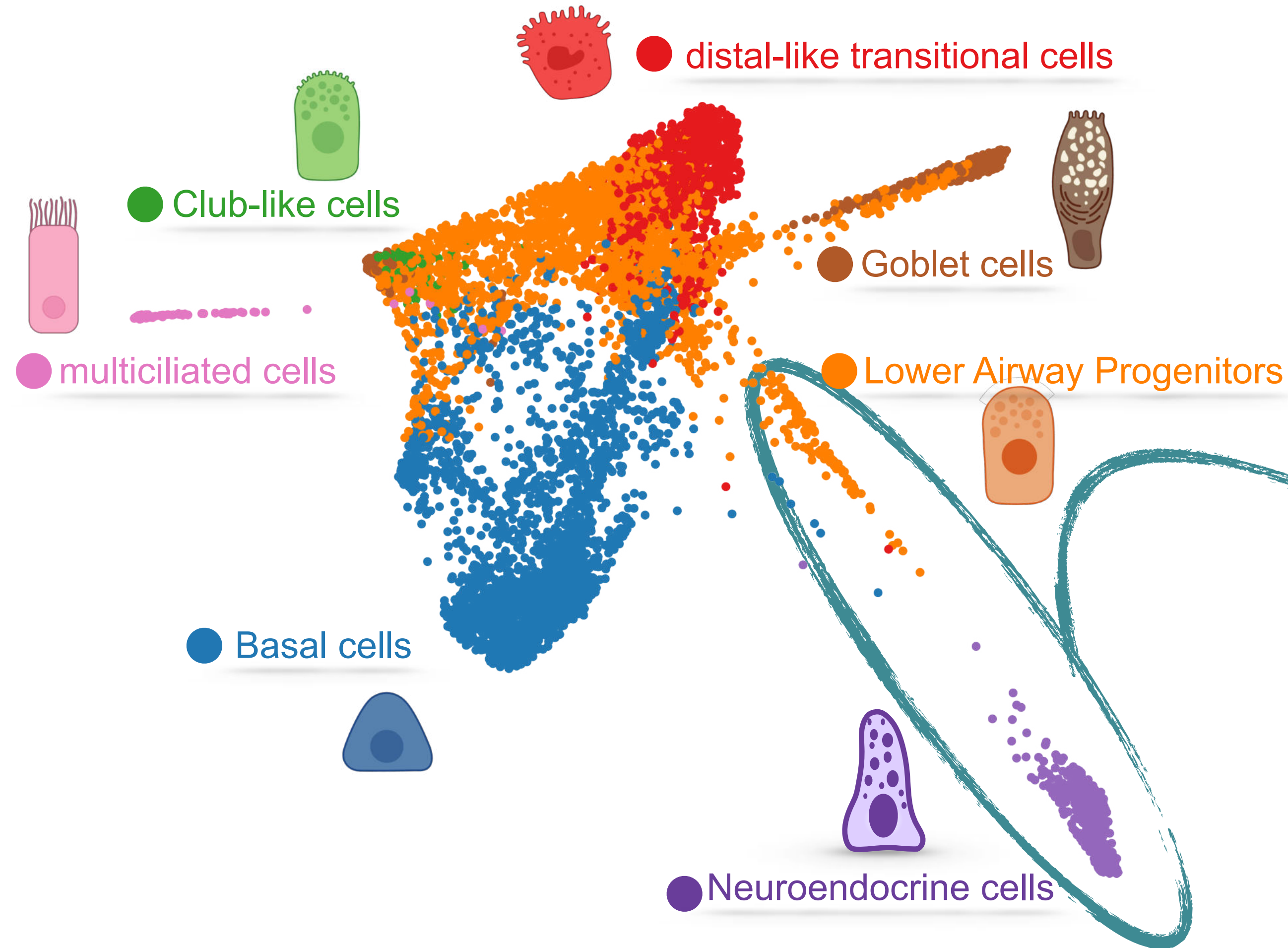
CHGA ACTIN DAPI
(NE cells)

scRNA-seq reveals pulmonary NE cell heterogeneity



Noah Candeli

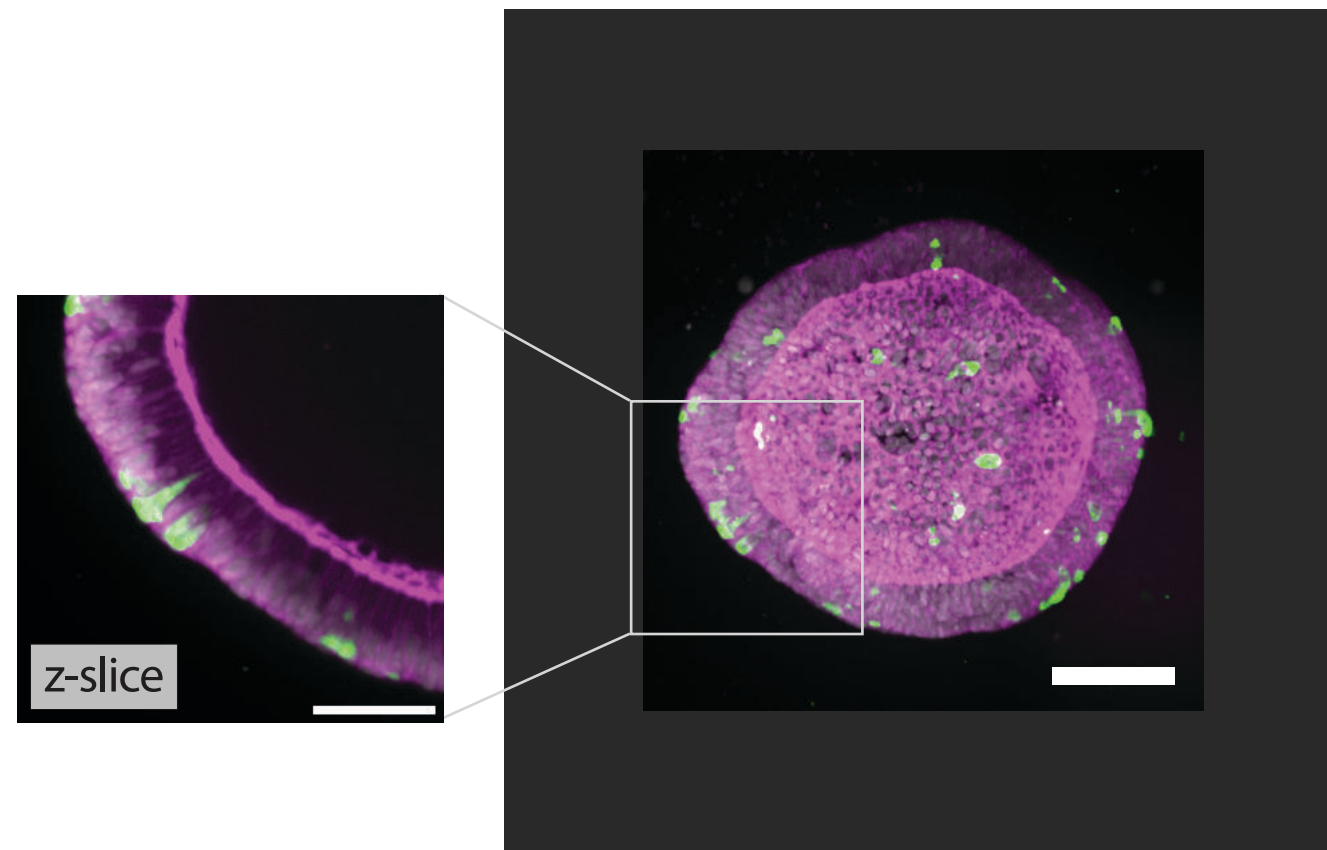
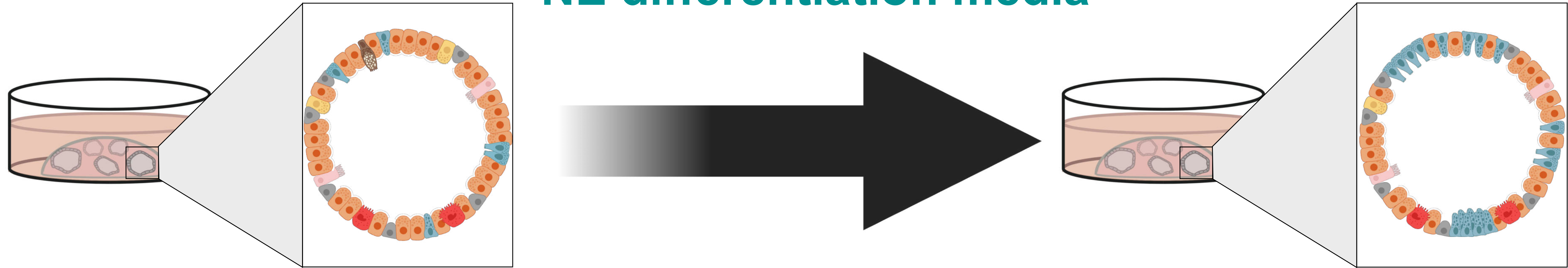
Standard, NE expansion, & D1, D3, and D10 of NE cell differentiation



An organoid system for the study of human pulmonary NE cells

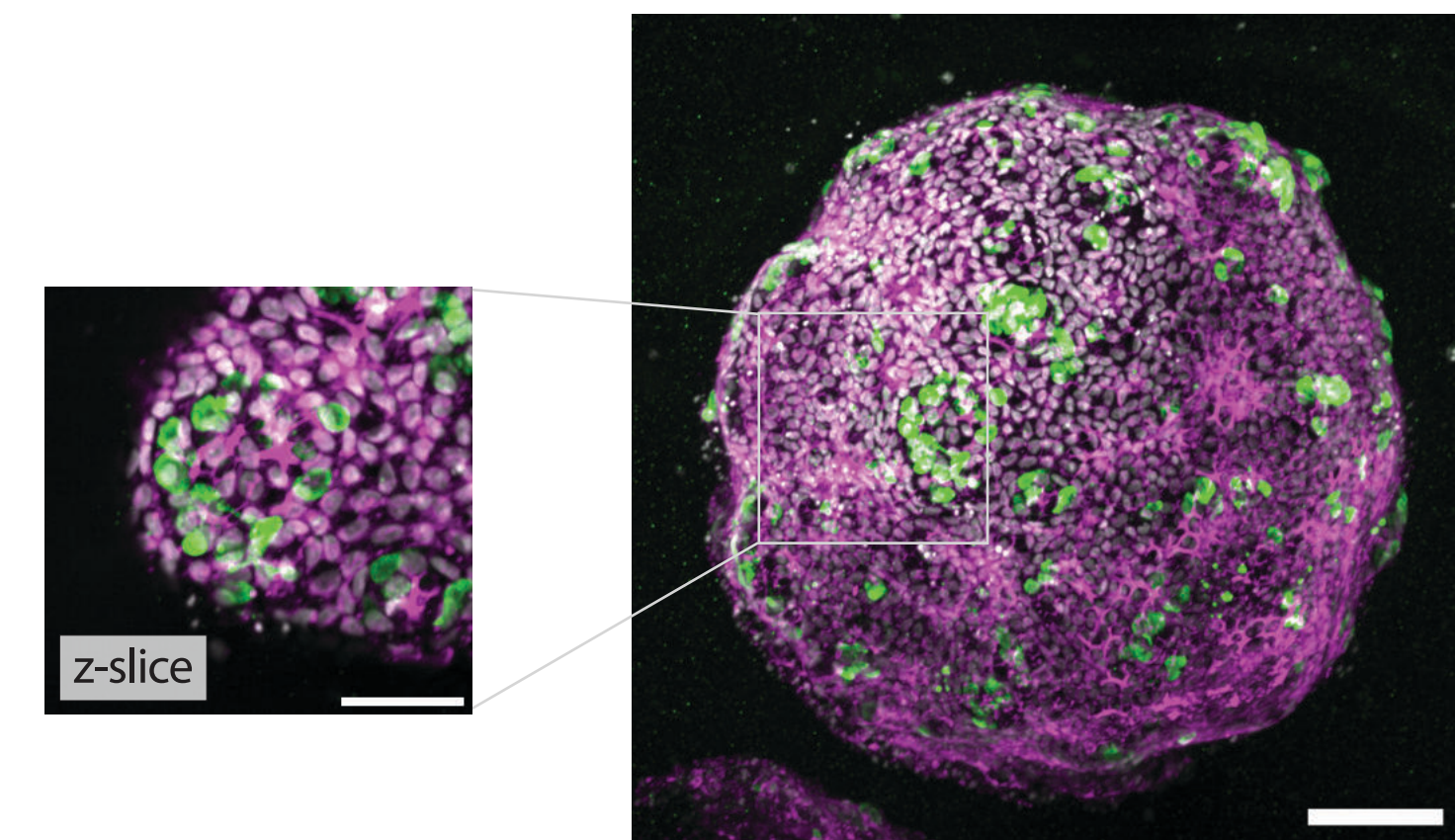
Candeli et al., 2026, biorxiv

NE differentiation media



NE cells ACTIN DAPI

NE expansion media:
NE cells and progenitors can
be expanded indefinitely



NE cells ACTIN DAPI

NE differentiation media:
generates large numbers of
mature NE cells

An organoid system for the study of human pulmonary NE cells

Candeli et al., 2026, biorxiv

NE differentiation media

How do (NE) cells respond to injury? *to chronic and acute environmental exposures?*

How do healthy (NE) cells become NE cancers? *How do environmental exposures contribute to this?*

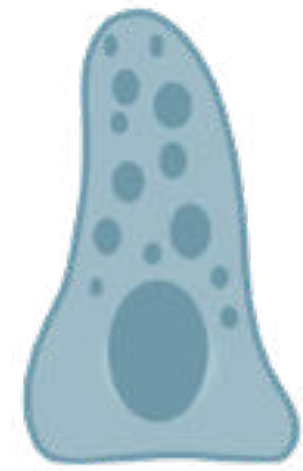
NE cells ACTIN DAPI

NE expansion media:
NE cells and progenitors can
be expanded indefinitely

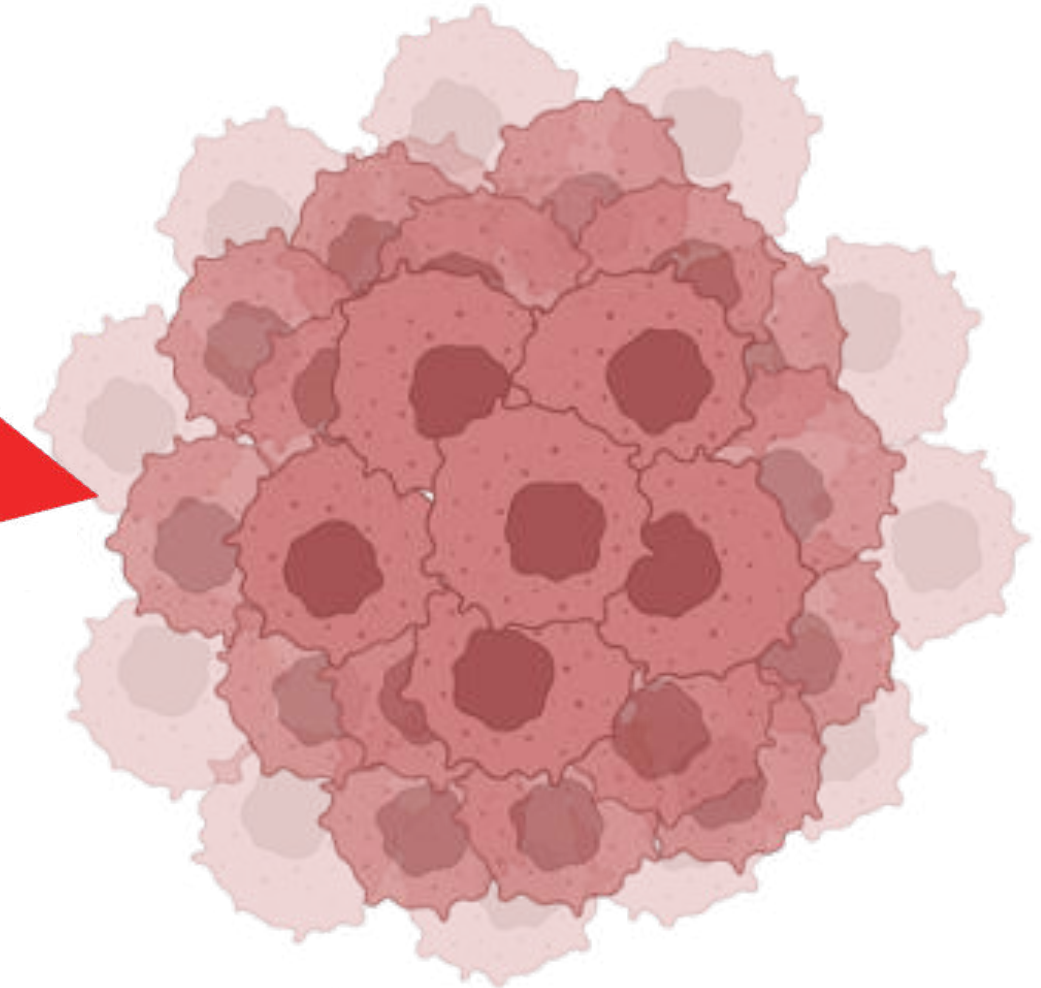
NE cells ACTIN DAPI

NE differentiation media:
generates large numbers of
mature NE cells

An organoid system for the study of tumor initiation in NE cells

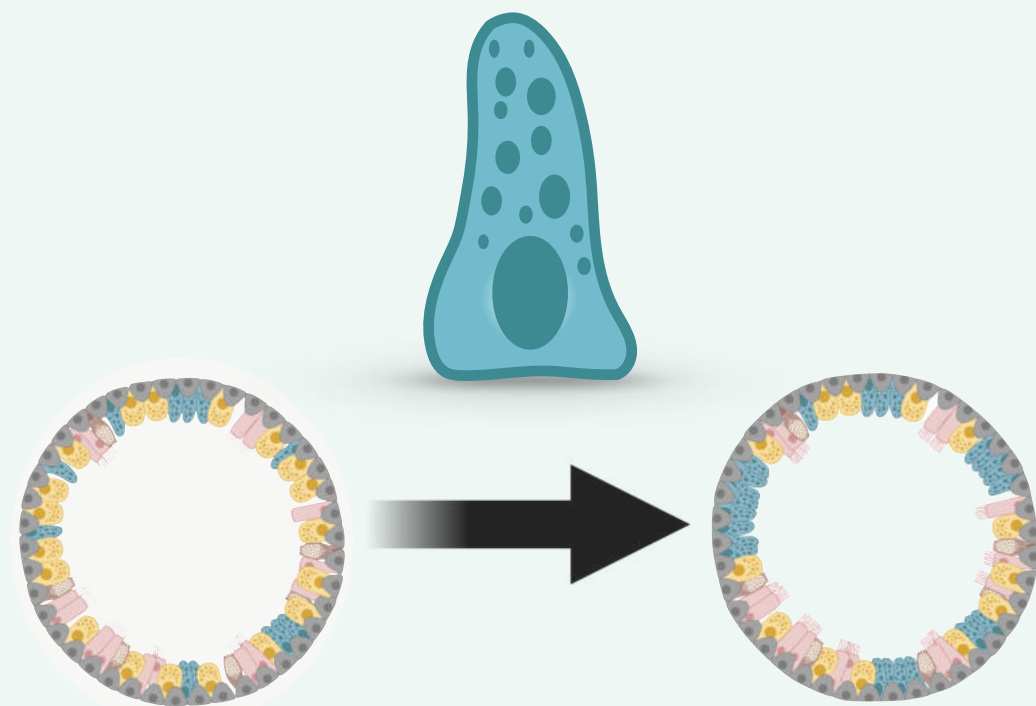


Healthy NE cells



NE cancers

NEr-fAOs



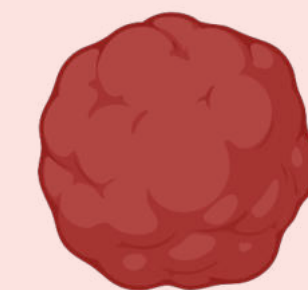
PDTOs of NE cancer



NETs

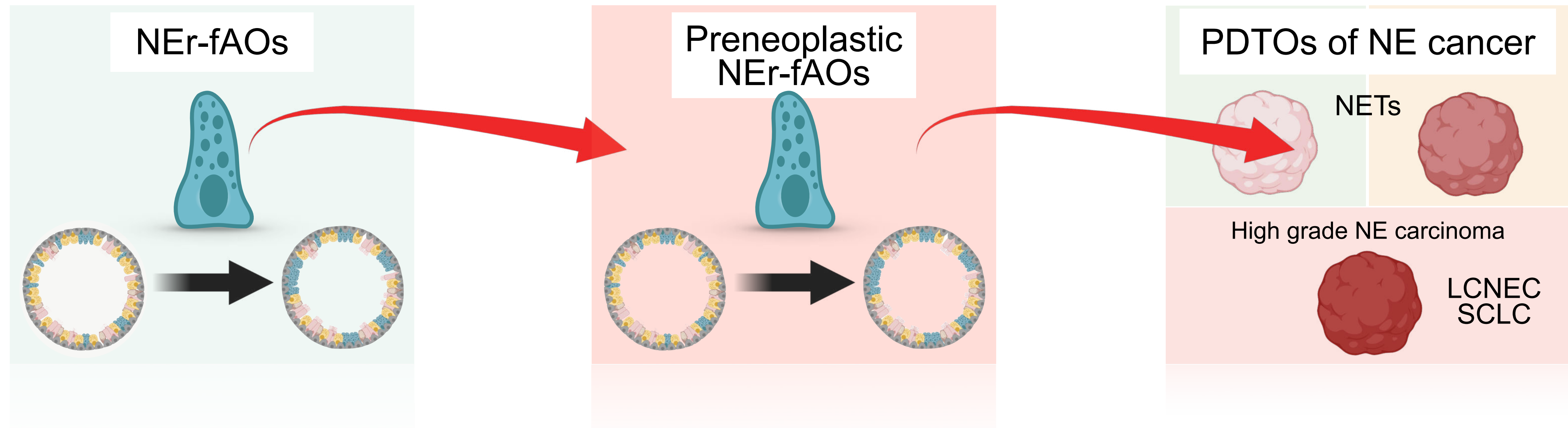
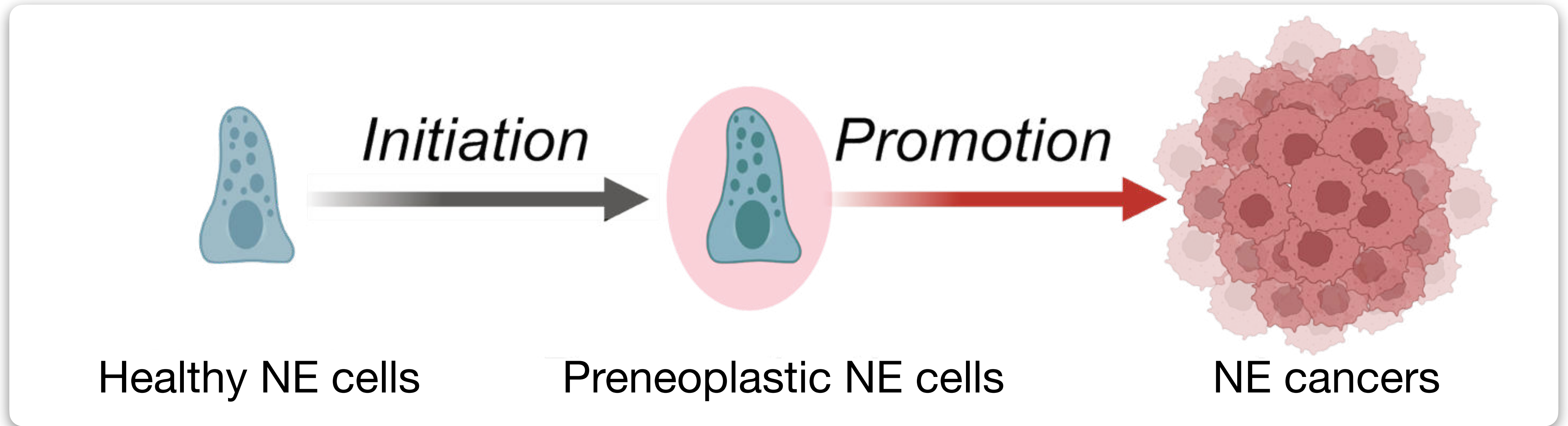


High grade NE carcinoma



LCNEC
SCLC

An organoid system for the study of tumor initiation in NE cells



An organoid system for the study of tumor initiation in NE cells



Andrés Marco

Small Cell Lung Cancer is an aggressive NE lung cancer with near universal loss of TP53 and RB1

***CHGA*^{2A}-tdTomato**

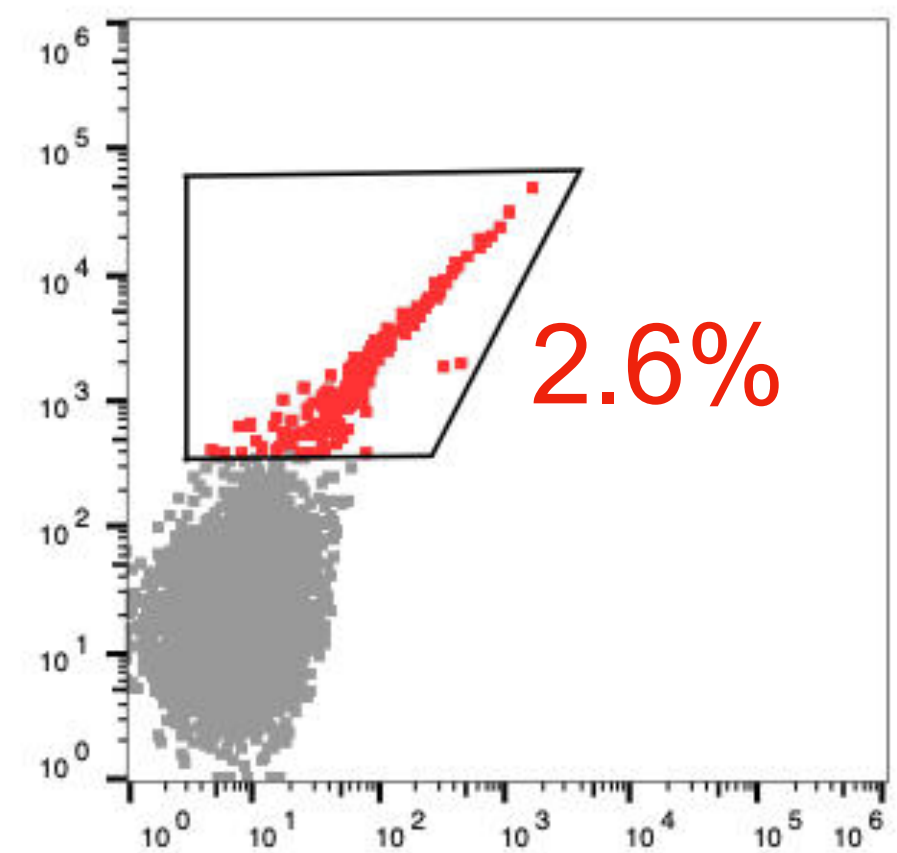


CRISPR/Cas9 induced
RB1 and *TP53* mutation

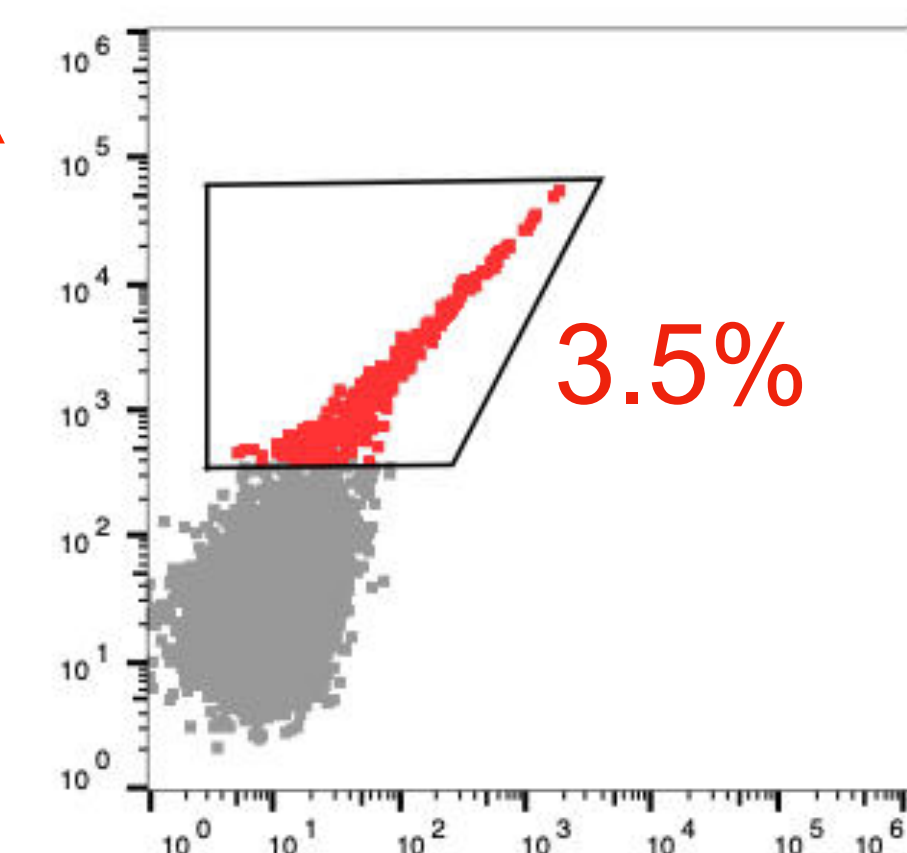
***CHGA*^{2A}-tdTomato; *RB1*^{-/-}; *TP53*^{-/-}**



tdTomato (NE cells)



tdTomato (NE cells)



CHGA = NE cell marker

Unpublished, do not post

An organoid system for the study of tumor initiation in NE cells



Andrés Marco

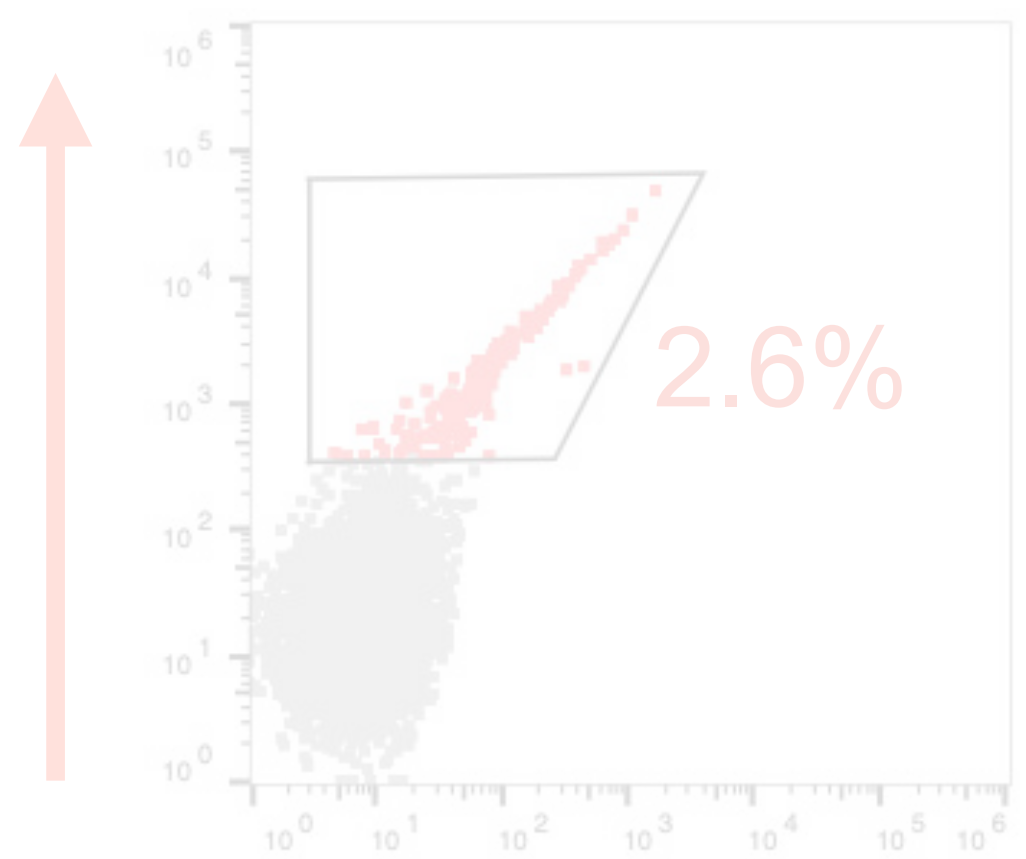
Small Cell Lung Cancer is an aggressive NE lung cancer with near universal loss of TP53 and RB1

CHGA^{2A}-tdTomato

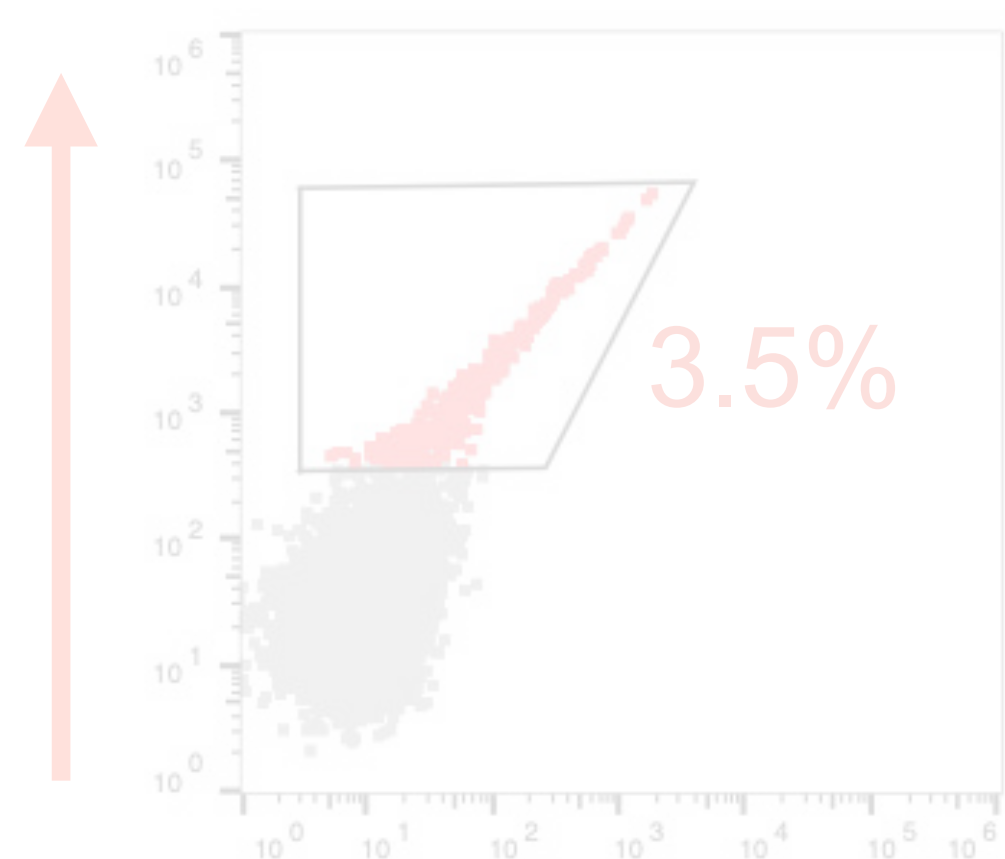
CHGA^{2A}-tdTomato; *RB1*^{-/-}; *TP53*^{-/-}

But mutation alone is not always sufficient to drive cancer...

tdTomato (NE cells)



tdTomato (NE cells)

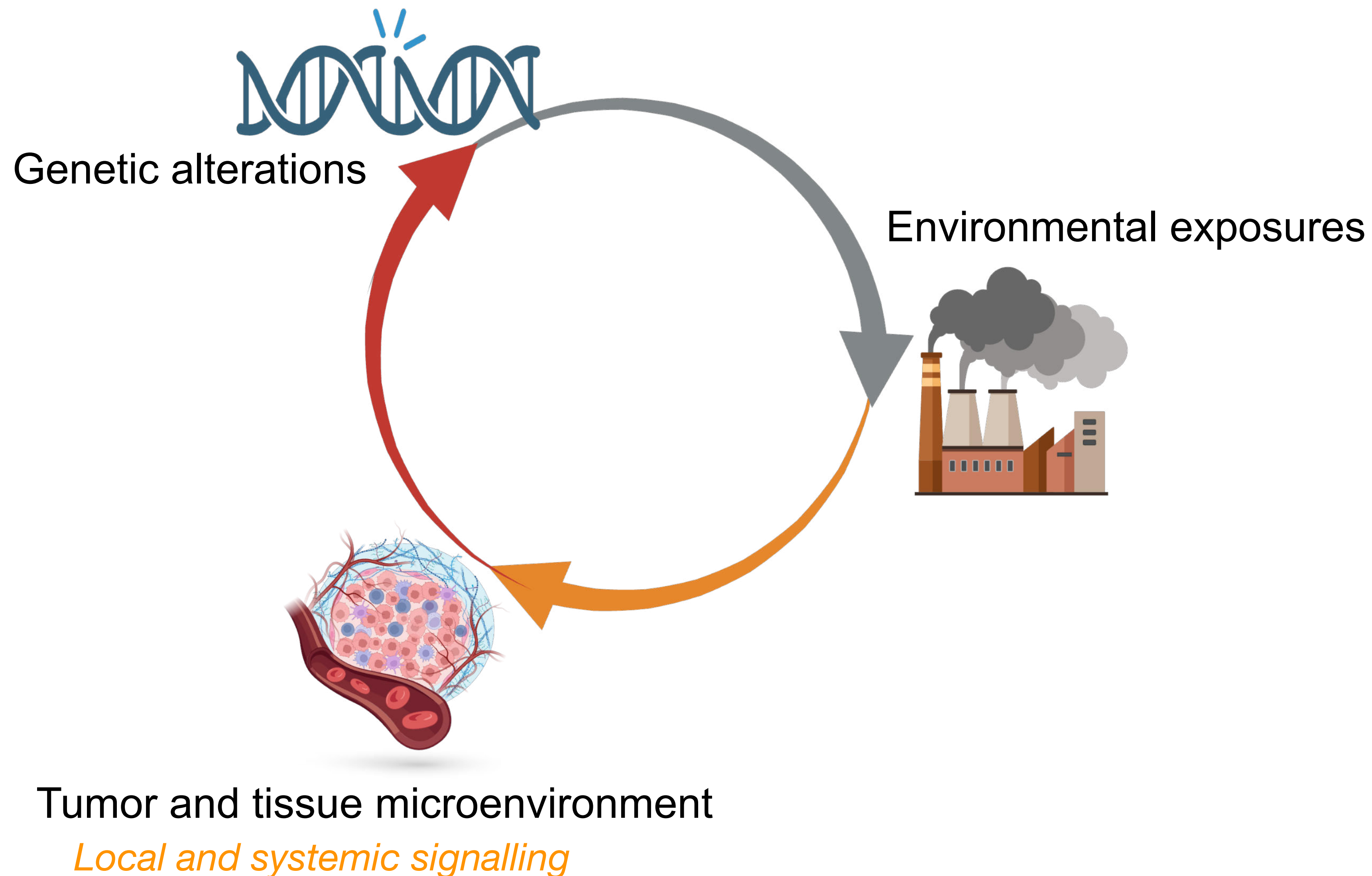


CHGA = NE cell marker

Unpublished, do not post



Cancer genesis and progression: a dynamic interplay between genetics, environmental exposures, and the microenvironment



Relevant environmental exposures to test in airway organoids

Personal exposome



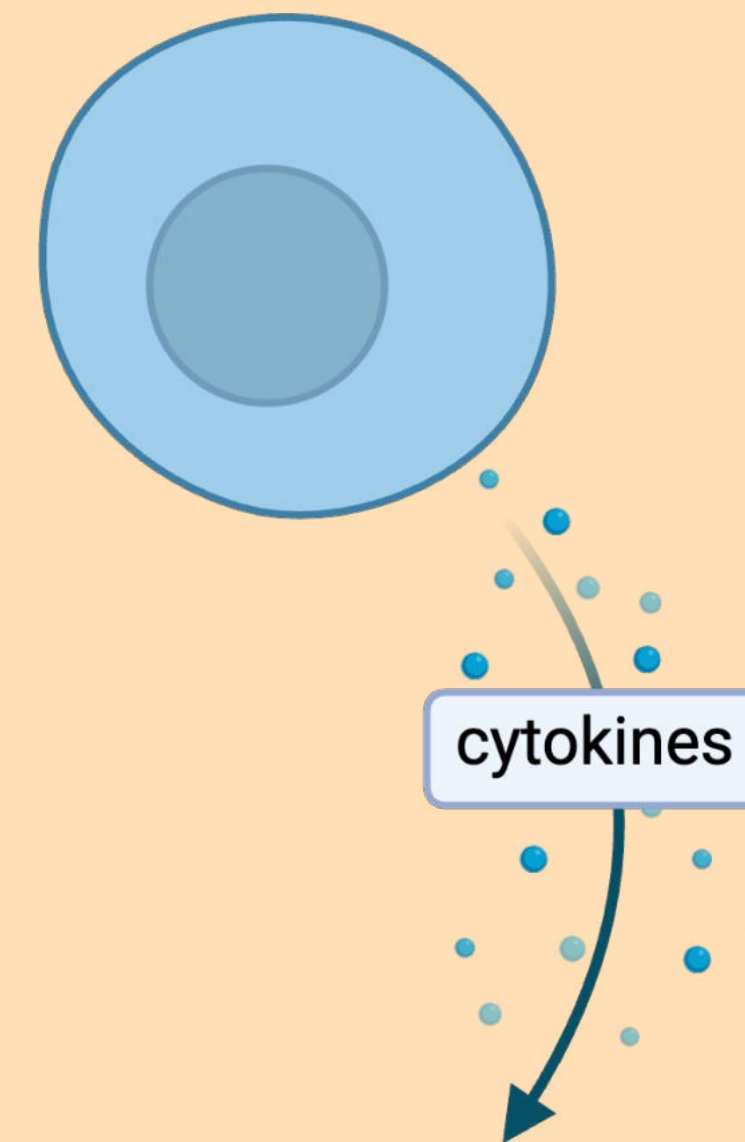
Nicotine & nicotine derivatives



Opioids

*with Mahdi Sheikh
(IARC - WHO)*

Internal exposome



Cytokines, neuropeptides,
Hormones, and metabolites

GxE cancer initiation studies in organoids — what is relevant?

What exposures are relevant and feasible?

Can the route of exposure be recapitulated?

Single exposure? Multiple exposures?

Are the relevant cell types represented in the organoid model?

What is your readout?

How will you know if what you generate in vitro represents a bona-fide pre-neoplastic state? A tumor?

Key messages

- Organoids can be grown from both healthy and diseased tissues
- Organoids can be used to model and study the full spectrum of tumorigenesis in a human cell based experimental system
- Organoids allow you to compare specific cell types in a given tissue
- Genetics and environment/exposure are important and they *also* impact cell state
- Organoids are a reductionist system — need to think about, what are the questions you can answer with this system?

Crosstalk between model systems is needed in Cancer Research

